

IIT Mandi B. Tech Curriculum

Batch-wise Curriculum Compilation

IIT Mandi

30th September 2026

*This compilation is organised **by batch** (2023 / 2024 / 2025 / 2026). Within each batch, every programme shows its credit distribution followed by the semester-wise course structure. Each semester table lists the Institute Core (IC / IC Basket / IKS / HSS) and Discipline Core (DC) courses fixed for that semester, with L-T-P-C from the course catalogue and a credit total. Discipline Electives and Free Electives are the student's own choice, so they are given as a remaining-credit note rather than named in the semester grid; each programme's approved Discipline-Elective list follows as a table. The 2026 batch includes the approved new programmes and their batch-specific curriculum structures.*

Contents

Batch 2023	4
Bachelor of Technology — Computer Science and Engineering	4
Bachelor of Technology — Data Science and Engineering	10
Bachelor of Technology — Electrical Engineering	17
Bachelor of Technology — Mechanical Engineering	24
Bachelor of Technology — Civil Engineering	31
Bachelor of Technology — Engineering Physics	38
Bachelor of Technology — Bioengineering	45
Bachelor of Technology — Mathematics and Computing	51
Bachelor of Technology — General Engineering - AI and Robotics	59
Bachelor of Technology — General Engineering - Mechatronics	64
Bachelor of Technology — General Engineering - Communication Technology	69
Bachelor of Technology — Materials Science and Engineering	74
Bachelor of Science — Chemical Sciences	79
Bachelor of Technology — Microelectronics and VLSI	86
Batch 2024	91
Bachelor of Technology — Computer Science and Engineering	91
Bachelor of Technology — Data Science and Engineering	97
Bachelor of Technology — Electrical Engineering	104
Bachelor of Technology — Mechanical Engineering	111
Bachelor of Technology — Civil Engineering	118
Bachelor of Technology — Engineering Physics	124
Bachelor of Technology — Bioengineering	130
Bachelor of Technology — Mathematics and Computing	136
Bachelor of Technology — General Engineering - AI and Robotics	144
Bachelor of Technology — General Engineering - Mechatronics	149
Bachelor of Technology — General Engineering - Communication Technology	154
Bachelor of Technology — General Engineering - Open Specialisation	159
Bachelor of Technology — General Engineering - Fintech	164
Bachelor of Technology — Materials Science and Engineering	169
Bachelor of Science — Chemical Sciences	174
Bachelor of Technology — Microelectronics and VLSI	181
Batch 2025	186
Bachelor of Technology — Computer Science and Engineering	186
Bachelor of Technology — Data Science and Artificial Intelligence	192
Bachelor of Technology — Electrical Engineering	199
Bachelor of Technology — Mechanical Engineering	206
Bachelor of Technology — Civil Engineering	213
Bachelor of Technology — Engineering Physics	219
Bachelor of Technology — Bioengineering	226
Bachelor of Technology — Mathematics and Computing	232
Bachelor of Technology — General Engineering - AI and Robotics	240
Bachelor of Technology — General Engineering - Mechatronics	245
Bachelor of Technology — General Engineering - Communication Technology	250
Bachelor of Technology — General Engineering - Open Specialisation	255
Bachelor of Technology — General Engineering - Data Analytics for Management and Finance	259
Bachelor of Technology — Materials Science and Engineering	264
Bachelor of Science — Chemical Sciences	269
Bachelor of Technology — Microelectronics and VLSI	276

Batch 2026	281
Bachelor of Technology — Computer Science and Engineering	281
Bachelor of Technology — Data Science and Artificial Intelligence	287
Bachelor of Technology — Electrical Engineering	293
Bachelor of Technology — Mechanical Engineering	300
Bachelor of Technology — Civil Engineering	307
Bachelor of Technology — Engineering Physics	313
Bachelor of Technology — Bioengineering	320
Bachelor of Technology — Mathematics and Computing	326
Bachelor of Technology — General Engineering - AI and Robotics	334
Bachelor of Technology — General Engineering - Mechatronics	339
Bachelor of Technology — General Engineering - Communication Technology	344
Bachelor of Technology — General Engineering - Open Specialisation	349
Bachelor of Technology — General Engineering - Data Analytics for Management and Finance	353
Bachelor of Technology — Materials Science and Engineering	358
Bachelor of Science — Chemical Sciences	363
Bachelor of Technology — Microelectronics and VLSI	370
Bachelor of Technology — Chemical Engineering with Data Analytics	375
Bachelor of Technology — Quantum Science and Engineering	379
Bachelor of Technology — Agricultural Engineering with Data Analytics	384

Batch 2023

Bachelor of Technology — Computer Science and Engineering

Table 1: Computer Science and Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	38
	Discipline Electives	28
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech Computer Science and Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
2	CS-212	Design of Algorithms	3-0-2-4	4
3	CS-213	Reverse Engineering	0-0-2-1	1
4	CS-214	Computer Organization	3-0-2-4	4
5	CS-304	Formal Language and Automata Theory	3-0-0-3	3
6	CS-309	Information Systems and Databases	3-0-2-4	4
7	CS-305	Artificial Intelligence	3-0-0-3	3
8	CS-312	Operating Systems	3-0-2-4	4
9	CS-313	Computer Networks	3-0-2-4	4
10	CS-302	Paradigms of Programming	3-0-2-4	4
11	CS-303	Software Engineering	2-0-2-3	3
Total				38

Table 3: B. Tech Computer Science and Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		HS-342	German I	3-0-0-3	
10		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Computer Science and Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
9	IKS + HSS (choose 0–3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0–3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22–25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Computer Science and Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	IC Basket (choose 1)	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
4		CS-214	Computer Organization	3-0-2-4	
5	DC	CS-212	Design of Algorithms	3-0-2-4	4
6	DC	CS-213	Reverse Engineering	0-0-2-1	1
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Computer Science and Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-304	Formal Language and Automata Theory	3-0-0-3	3
2	DC	CS-309	Information Systems and Databases	3-0-2-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Computer Science and Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-305	Artificial Intelligence	3-0-0-3	3
2	DC	CS-312	Operating Systems	3-0-2-4	4
3	DC	CS-313	Computer Networks	3-0-2-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Computer Science and Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-302	Paradigms of Programming	3-0-2-4	4
2	DC	CS-303	Software Engineering	2-0-2-3	3
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Computer Science and Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	CS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Computer Science and Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	CS-010	Internship		2
2	MTP	CS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 28 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Computer Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-501	Access Networks	3
2	CS-502	Compiler Design	4
3	CS-506	Cognitive Modeling	3
4	CS-507	Computer Architecture	4
5	CS-508	Introduction to Network Security	3
6	CS-510	Randomized Algorithms	4
7	CS-511	Applied Probability	2
8	CS-512	Matrix Theory	2
9	CS-513	Discrete Mathematics	4

Computer Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
10	CS-514	Advance Data Structure and Algorithm (CS-580)	4
11	CS-515	Advanced Computer Science Practicum	3
12	CS-516	Exploratory Project	3
13	CS-520	Introduction to Quantum Computing	3
14	CS-521	Introduction to Post Quantum Security	3
15	CS-522	Distributed Algorithms	3
16	CS-524	Computational Complexity Theory	4
17	CS-525	Computational Foundations of Cryptography	3
18	CS-526	Mathematical Foundation of Cryptography	3
19	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3
20	CS-541	IoT systems and the Cloud	3
21	CS-542	Design Patterns for Scalable Systems	3
22	CS-544	Formal Concept Analysis: Theory and Practice	3
23	CS-545	Software Design Patterns	3
24	CS-546	Design of Concurrent Software	3
25	CS-547	Network Management Systems	3
26	CS-548	Cloud Networking	3
27	CS-549	Performance Analysis of Computer Networks	3
28	CS-550	Computer Graphics and Geometric Design	3
29	CS-551	Human Computer Interaction	3
30	CS-560	Text Retrieval and Mining	3
31	CS-561	Map Reduce and Big Data	3
32	CS-563	Scalable Data Science	4
33	CS-570	Algorithms for Curves and Surface	3
34	CS-571	Programming Practicum	3
35	CS-591	Special Topics in Knowledge Discovery and Data	1
36	CS-592	Selected Topics in Artificial Intelligence	2
37	CS-601	Probability and Random Processes	4
38	CS-603	Managerial Decision Making	3
39	CS-606	Cognitive Modeling	3
40	CS-609	Speech processing	3
41	CS-611	Program Analysis	4
42	CS-621P	Post Graduate Project-1	12
43	CS-630	Speech Technology	4
44	CS-660	Data Mining for Decision Making	4
45	CS-661	Knowledge Representation and Reasoning	3
46	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
47	CS-670	Kernal Methods for Pattern Recognition	4
48	CS-671	Deep Learning and Applications	4
49	CS-672	Advanced Topics in Deep Learning	4

Computer Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	CS-673	Advanced Computer Vision	4
51	CS-677	Soft Computing	3
52	CS-683	Generative AI	4
53	CS-685	Natural Language Processing	3
54	CS-686	Large Language Models: Core Concepts to Custom	4
55	CS-693	Compressed Sensing and its applications	3
56	DS-201	Data handling and visualization	3
57	DS-301	Mathematical Foundations of Data Science	4
58	DS-314	Probability Theory and Statistics	4
59	DS-401	Optimization for data science	3
60	DS-402	Matrix Computations for Data Science	3
61	DS-403	Introduction to Statistical Learning	3
62	DS-404	Information Security and Privacy	3
63	DS-412	Matrix Computations for Data Science	4
64	DS-415	Optimization for Machine Learning	4
65	DS-416	Fundamentals of Machine Learning (CS-669)	4
66	EE-511	Computer Vision	4
67	EE-608	Digital Image Processing	4
68	MA-221	Numerical Analysis	4
69	MA-222	Applied Linear Programming	4
70	MA-322	Applied Graph Theory	4
71	MA-546	Introduction to Quantitative Finance	4
72	MA-552	Number Theory	3
73	MA-605	Statistical Data Analysis	3
<i>Compulsory project components — counted as DE, not selectable</i>			
74	CS-498P	Major Technical Project - I (compulsory; counted as DE)	4
75	CS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Data Science and Engineering

Table 12: Data Science and Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	33
	Discipline Electives	33
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 13: B. Tech Data Science and Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-213	Reverse Engineering	0-0-2-1	1
2	DS-201	Data Handling and Vizualisation	2-0-2-3	3
3	DS-301	Mathematical Foundations of Data Science	3-1-0-4	4
4	DS-302	Computing Systems for Data Processing	2-0-2-3	3
5	DS-313	Statistical Foundations of Data Science	3-1-0-4	4
6	DS-404	Information Security and Privacy	3-0-0-3	3
7	DS-412	Matrix Computations for Data Science	3-0-2-4	4
8	CS-305	Artificial Intelligence	3-0-0-3	3
9	DS-413	Introduction to Statistical Learning	3-1-0-4	4
10	DS-411	Optimization for Data Science	3-1-0-4	4
Total				33

Table 14: B. Tech Data Science and Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		HS-342	German I	3-0-0-3	
10		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 15: B. Tech Data Science and Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
9	IKS + HSS (choose 0–3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0–3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22–25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 16: B. Tech Data Science and Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	CS-213	Reverse Engineering	0-0-2-1	1
3	DC	DS-201	Data Handling and Vizualisation	2-0-2-3	3
4	DC	DS-301	Mathematical Foundations of Data Science	3-1-0-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 17: B. Tech Data Science and Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	DS-302	Computing Systems for Data Processing	2-0-2-3	3
3	DC	DS-313	Statistical Foundations of Data Science	3-1-0-4	4
4	DC	DS-404	Information Security and Privacy	3-0-0-3	3
5	DC	DS-412	Matrix Computations for Data Science	3-0-2-4	4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 18: B. Tech Data Science and Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-305	Artificial Intelligence	3-0-0-3	3
2	DC	DS-413	Introduction to Statistical Learning	3-1-0-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 19: B. Tech Data Science and Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-411	Optimization for Data Science	3-1-0-4	4
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					8

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 20: B. Tech Data Science and Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	DS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 21: B. Tech Data Science and Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	DS-010	Internship		2
2	MTP	DS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 33 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 22: Data Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-507	Probabilistic Robotics	3
2	AR-509	Deep Learning for Robotics	4
3	AR-516	Introduction to Blockchain and Web3	3
4	AR-517	Introduction to Cyber security	3
5	BE-303	Applied Biostatistics	4
6	BE-304	Bioinformatics	4
7	BE-502	Design and Analysis of Bioalgorithms	3
8	BE-505	Computational Biology	3
9	CS-212	Design and Analysis of Algorithms	4
10	CS-214	Computer Organization	4

Data Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	CS-241	Introduction to Cryptography	3
12	CS-302	Paradigms of Programming	4
13	CS-303	Software Engineering	3
14	CS-304	Formal Languages and Automata Theory	3
15	CS-306	Operating Systems	3
16	CS-307	Systems Practicum	2
17	CS-308	Large Application Practicum	2
18	CS-308P	Large Application Practicum	2
19	CS-309	Information and Database Systems	4
20	CS-310	Introduction to Communication and Distributed Processes	4
21	CS-313	Computer Networks	4
22	CS-347	Software Engineering	4
23	CS-350	Computer Graphics	3
24	CS-401	Architecture of High Performance Computers	3
25	CS-402	Compilers	2
26	CS-403	Algorithm Design and Analysis I	3
27	CS-405	Verification of Reactive Systems	3
28	CS-406	Computer Networks	4
29	CS-451	Computer Graphics and Game Design	3
30	CS-456	Distributed Databases	3
31	CS-501	Access Networks	3
32	CS-502	Compiler Design	4
33	CS-506	Cognitive Modeling	3
34	CS-507	Computer Architecture	4
35	CS-508	Introduction to Network Security	3
36	CS-510	Randomized Algorithms	4
37	CS-511	Applied Probability	2
38	CS-512	Matrix Theory	2
39	CS-513	Discrete Mathematics	4
40	CS-514	Advance Data Structure and Algorithm (CS-580)	4
41	CS-515	Advanced Computer Science Practicum	3
42	CS-516	Exploratory Project	3
43	CS-520	Introduction to Quantum Computing	3
44	CS-521	Introduction to Post Quantum Security	3
45	CS-522	Distributed Algorithms	3
46	CS-524	Computational Complexity Theory	4
47	CS-525	Computational Foundations of Cryptography	3
48	CS-526	Mathematical Foundation of Cryptography	3
49	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3

Data Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	CS-541	IoT systems and the Cloud	3
51	CS-542	Design Patterns for Scalable Systems	3
52	CS-544	Formal Concept Analysis: Theory and Practice	3
53	CS-545	Software Design Patterns	3
54	CS-546	Design of Concurrent Software	3
55	CS-547	Network Management Systems	3
56	CS-548	Cloud Networking	3
57	CS-549	Performance Analysis of Computer Networks	3
58	CS-550	Computer Graphics and Geometric Design	3
59	CS-551	Human Computer Interaction	3
60	CS-560	Text Retrieval and Mining	3
61	CS-561	Map Reduce and Big Data	3
62	CS-563	Scalable Data Science	4
63	CS-570	Algorithms for Curves and Surface	3
64	CS-571	Programming Practicum	3
65	CS-591	Special Topics in Knowledge Discovery and Data	1
66	CS-592	Selected Topics in Artificial Intelligence	2
67	CS-601	Probability and Random Processes	4
68	CS-603	Managerial Decision Making	3
69	CS-606	Cognitive Modeling	3
70	CS-609	Speech processing	3
71	CS-611	Program Analysis	4
72	CS-630	Speech Technology	4
73	CS-660	Data Mining for Decision Making	4
74	CS-661	Knowledge Representation and Reasoning	3
75	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
76	CS-670	Kernel Methods for Pattern Recognition	4
77	CS-671	Deep Learning and Applications	4
78	CS-672	Advanced Topics in Deep Learning	4
79	CS-673	Advanced Computer Vision	4
80	CS-677	Soft Computing	3
81	CS-683	Generative AI	4
82	CS-685	Natural Language Processing	3
83	CS-686	Large Language Models: Core Concepts to Custom	4
84	CS-693	Compressed Sensing and its applications	3
85	EE-203	Network Theory	3
86	EE-304	Communication Theory	4
87	EE-305	Digital Signal Processing	4
88	EE-511	Computer Vision	4
89	EE-522	Matrix Theory for Engineers	3

Data Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
90	EE-530	Optimization theory	3
91	EE-531	Estimation and Detection Theory	3
92	EE-534	Probability and Random Processes	3
93	EE-541	Tensors: Techniques, Algorithms, Applications for Signal	4
94	EE-543	Vision and Learning Based Control	3
95	EE-574	Biomedical Signal and Image Analysis	3
96	EE-575	Applied Statistics for Data and Signals	4
97	EE-608	Digital Image Processing	4
98	EE-620	Advanced Digital Signal Processing	3
99	EP-301	Engineering Mathematics - 2	4
100	MA-530	Graph Theory	4
101	MA-605	Statistical Data Analysis	3
102	ME-522	High-Performance Scientific Computing	3
103	ME-526	Programming Paradigm for Open-Source Software	1
104	PH-302	Introduction to Statistical Mechanics	3
105	PH-522	Statistical Mechanics	4
<i>Compulsory project components — counted as DE, not selectable</i>			
106	DS-498P	Major Technical Project - I (compulsory; counted as DE)	4
107	DS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Electrical Engineering

Table 23: Electrical Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	52
	Discipline Electives	20
Electives	Free Electives	17
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		161

Table 24: B. Tech Electrical Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-210	Digital Systems Design	3-0-0-3	3
3	EE-210P	Digital Systems Design Lab	0-0-2-1	1
4	EE-260	Signals and Systems	2.5-0.5-0-3	3
5	EE-261	Electrical Systems Around Us	3-0-0-3	3
6	EE-261P	Electrical Systems Around Us Lab	0-0-3-2	2
7	EE-311	Device Electronics	3-0-0-3	3
8	EE-201	Electromechanics	2.5-0.5-0-3	3
9	EE-201P	Electromechanics Lab	0-0-2-1	1
10	EE-202	Electromagnetics & Wave Propagation	3-0-0-3	3
11	EE-211	Analog Circuit Design	2-0-2-3	3
12	EE-223P	Reverse Engineering	1-0-0-1	1
13	EE-304	Communication Systems	3-0-1-4	4
14	EE-231	Measurement and Instrumentation	2-0-2-3	3
15	EE-302	Control Systems	3-0-2-4	4
16	EE-303	Power System	3-1-0-4	4
17	EE-314	Digital Signal Processing	3-0-2-4	4
18	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
Total				52

Table 25: B. Tech Electrical Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		HS-342	German I	3-0-0-3	
10		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 26: B. Tech Electrical Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket (choose 1)	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9		IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	
10		IC-241	Material Science for Engineers	3-0-0-3	
11		IC-253	Data Structures and Algorithms	2-0-2-3	
12	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
13		HS-151	Introduction to English Literature	3-0-0-3	
14		HS-357	French Language and Culture	2-0-0-2	
15		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 27: B. Tech Electrical Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	EE-203	Network Theory	3-0-0-3	3
3	DC	EE-210	Digital Systems Design	3-0-0-3	3
4	DC	EE-210P	Digital Systems Design Lab	0-0-2-1	1
5	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
6	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
7	DC	EE-261P	Electrical Systems Around Us Lab	0-0-3-2	2
8	DC	EE-311	Device Electronics	3-0-0-3	3
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 28: B. Tech Electrical Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
3	DC	EE-201P	Electromechanics Lab	0-0-2-1	1
4	DC	EE-202	Electromagnetics & Wave Propagation	3-0-0-3	3
5	DC	EE-211	Analog Circuit Design	2-0-2-3	3
6	DC	EE-223P	Reverse Engineering	1-0-0-1	1
7	DC	EE-304	Communication Systems	3-0-1-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 29: B. Tech Electrical Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	EE-302	Control Systems	3-0-2-4	4
3	DC	EE-303	Power System	3-1-0-4	4
4	DC	EE-314	Digital Signal Processing	3-0-2-4	4
5	DC	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 30: B. Tech Electrical Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 31: B. Tech Electrical Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	EE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 32: B. Tech Electrical Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	EE-010	Internship		2
2	MTP	EE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 20 credits of **Discipline Electives** and 17 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 33: Electrical Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BE-302	Bioelectric Systems Modeling	4
2	BE-304	Bioinformatics	4
3	CS-549	Performance Analysis of Computer Networks	3
4	CS-671	Deep Learning and Applications	4
5	EE-500	Network Control System	3
6	EE-501	Power System Operation and Control	3
7	EE-502	Analog System Design Laboratory	3
8	EE-503	Advance Communication Theory	3
9	EE-504	Switch Mode Power Conversion	3
10	EE-505	Materials for Semiconductor Industry	3
11	EE-506	Solar Photovoltaic Energy Systems	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	EE-507	Transmission Lines and Basic Microwave Engineering	4
13	EE-508	Fundamentals of Electric Drives	3
14	EE-508P	Practicum on fundamentals of Electrical Drives	2
15	EE-509	Linear Dynamical Systems	3
16	EE-511	Computer Vision	4
17	EE-512	CMOS Analog IC Design	4
18	EE-513	Special Electrical Machines	3
19	EE-514	Robust Control Systems	3
20	EE-515	Nonlinear Stability and Control	4
21	EE-516	Biomedical Systems	4
22	EE-517	Wireless Communications and Networking	3
23	EE-518	Information Theory	3
24	EE-519	CMOS Digital IC design Practicum	2
25	EE-519P	CMOS Digital IC Design Practicum	2
26	EE-520	Microelectronic Devices and Modelling	3
27	EE-522	Matrix Theory for Engineers	3
28	EE-523	Digital VLSI Architecture Design	4
29	EE-524	Digital MOS LSI Circuits	3
30	EE-526	Power Semiconductor Devices	3
31	EE-527	Analysis and Design of Power Electronic Converters	3
32	EE-527P	Practicum on Analysis and Design of Power Electronic Converters	2
33	EE-528	Modeling and Analysis of Electric Machines	3
34	EE-529	Embedded Systems	4
35	EE-530	Optimization theory	3
36	EE-531	Estimation and Detection Theory	3
37	EE-532	Supervised Research Exposure	3
38	EE-533	Access Network	4
39	EE-534	Probability and Random Processes	3
40	EE-535	Communication and Signal Processing Systems Design	3
41	EE-536	IoT Systems	3
42	EE-537	Power Quality Problems and Mitigation Techniques in Microgrids	3
43	EE-538	Data-driven Control	3
44	EE-540	Wide Band Gap Devices in Power Electronics Applications	3
45	EE-541	Sensors: Techniques, Algorithms, Applications for Signal	4
46	EE-542	Modelling, Simulation and Control of Hybrid Electric Vehicle	3
47	EE-543	Vision and Learning Based Control	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
48	EE-550	Optimal Control and Estimation	3
49	EE-551	Applied Photonics for Scientists and Engineers	3
50	EE-552	Power and Energy Systems	4
51	EE-553	Foundations of Intelligent Communication Systems-I	4
52	EE-554	Low Power VLSI Design	3
53	EE-555	Intelligent Control System	3
54	EE-556	Nuclear Reactor Control	1
55	EE-557	Adaptive Control	3
56	EE-560	Reconfigurable Computing	4
57	EE-570	Advanced Communication Theory	4
58	EE-574	Biomedical Signal and Image Analysis	3
59	EE-575	Applied Statistics for Data and Signals	4
60	EE-580	Network Systems: Modelling and Analysis	3
61	EE-581	Applied Statistics for Data and Signals	4
62	EE-582	Nonlinear Analysis and Control of Power Electronic Converters	3
63	EE-583	Smart Grids	3
64	EE-584	Power System Protection	3
65	EE-585	Theory, Modelling and Characterization of Photo-voltaic Devices	3
66	EE-593	Low Power VLSI Circuits	3
67	EE-597	Intellegent Control System	3
68	EE-601	Solid State Devices	3
69	EE-602	Control System Applications	3
70	EE-603	Renewable Energy and Smart Grid	3
71	EE-604	Practicum on Advanced Electric Drives	3
72	EE-604P	Practicum on Advanced Electric Drives	1
73	EE-605	Information Theory	3
74	EE-606	Introduction to High Voltage Engineering and Dielectric	3
75	EE-607	Optical Communication Systems	3
76	EE-608	Digital Image Processing	4
77	EE-609	Network Information Theory	3
78	EE-611	VLSI Technology	3
79	EE-611P	VLSI Fabrication Practicum	2
80	EE-612	OFDM For Optical Communications	3
81	EE-613	Wireless Communication	3
82	EE-614	Optical Communication Systems	3
83	EE-615	Nanoelectronics and Micronanofabrication	3
84	EE-616	Microwave engineering	2
85	EE-618	Industrial Process Control	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
86	EE-619	Mixed Signal VLSI Design	4
87	EE-620	Advanced Digital Signal Processing	3
88	EE-621	Radiating systems	4
89	EE-622	Microwave Integrated Circuits	3
90	EE-623P	Practicum on Digital Control of Power Electronics and Drives	3
91	EE-626P	Post Graduate Project - I	10
92	EE-630	HVDC Transmission and Flexible AC Transmission Systems	3
93	EE-641	Advance Wireless Technologies	4
94	EE-642	Research Study	3
95	EE-677	Analog Circuit Design	3
96	EE-678	Digital Circuit Design	3
97	EE-691	Advanced Topics in Dielectric Breakdown	1
98	EE-694	Photonics Computing	3
<i>Compulsory project components — counted as DE, not selectable</i>			
99	EE-498P	Major Technical Project - I (compulsory; counted as DE)	4
100	EE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Mechanical Engineering

Table 34: Mechanical Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	50
	Discipline Electives	16
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 35: B. Tech Mechanical Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-261	Electrical Systems Around Us	3-0-0-3	3
2	IC-241	Material Science for Engineers	3-0-0-3	3
3	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	ME-210	Fluid Mechanics	2.5-0.5-0-3	3
5	ME-212	Product Manufacturing Technologies	2-0-3-3	3
6	ME-213	Engineering Thermodynamics	3-1-0-4	4
7	ME-100	Reverse Engineering	0-0-2-1	1
8	ME-205	Machine Drawing	1-0-4-3	3
9	ME-210P	Fluid Mechanics Lab	0-0-2-1	1
10	ME-215	Manufacturing Engineering 1	3-0-0-3	3
11	ME-303	Heat Transfer	2.5-0.5-0-3	3
12	ME-307	Energy Conversion Devices	3-0-0-3	3
13	ME-303P	Heat Transfer Lab	0-0-3-1	1
14	ME-305	Design of Machine Elements	3-1-0-4	4
15	ME-309	Theory of Machines	3-1-0-4	4
16	ME-310	System Dynamics and Control	3-0-0-3	3
17	ME-311P	Design Lab 1	0-0-2-1	1
18	ME-315	Manufacturing Engineering 2	3-0-0-3	3
19	ME-312P	Design Lab 2	0-0-2-1	1
Total				50

Table 36: B. Tech Mechanical Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		HS-342	German I	3-0-0-3	
10		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 37: B. Tech Mechanical Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	IKS + HSS (choose 0–3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0–3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22–25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 38: B. Tech Mechanical Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
3	DC	IC-241	Material Science for Engineers	3-0-0-3	3
4	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
5	DC	ME-210	Fluid Mechanics	2.5-0.5-0-3	3
6	DC	ME-212	Product Manufacturing Technologies	2-0-3-3	3
7	DC	ME-213	Engineering Thermodynamics	3-1-0-4	4
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 39: B. Tech Mechanical Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	ME-100	Reverse Engineering	0-0-2-1	1
3	DC	ME-205	Machine Drawing	1-0-4-3	3
4	DC	ME-210P	Fluid Mechanics Lab	0-0-2-1	1
5	DC	ME-215	Manufacturing Engineering 1	3-0-0-3	3
6	DC	ME-303	Heat Transfer	2.5-0.5-0-3	3
7	DC	ME-307	Energy Conversion Devices	3-0-0-3	3
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 40: B. Tech Mechanical Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-303P	Heat Transfer Lab	0-0-3-1	1
2	DC	ME-305	Design of Machine Elements	3-1-0-4	4
3	DC	ME-309	Theory of Machines	3-1-0-4	4
4	DC	ME-310	System Dynamics and Control	3-0-0-3	3
5	DC	ME-311P	Design Lab 1	0-0-2-1	1
6	DC	ME-315	Manufacturing Engineering 2	3-0-0-3	3
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 41: B. Tech Mechanical Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-312P	Design Lab 2	0-0-2-1	1
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					5

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 42: B. Tech Mechanical Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	ME-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 43: B. Tech Mechanical Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	ME-010	Internship		2
2	MTP	ME-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 16 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 44: Mechanical Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	ME-451	Refrigeration and Air Conditioning	3
2	ME-452	Robot Kinematics, Dynamics and Control	4
3	ME-501	Nanomanufacturing	3
4	ME-502	Functional Materials	3
5	ME-503	Micro/Nano Scale heat transfer	3
6	ME-504	Numerical Methods for Engineering Computation	3
7	ME-505	Applied Finite Element Method	3
8	ME-506	Fundamentals of Fracture Mechanics	3
9	ME-507	Micro and Nanoscale Fluid Mechanics	3
10	ME-508	Fundamentals of project management	3

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	ME-509	Nano-manufacturing	3
12	ME-510	Advanced Manufacturing Processes	3
13	ME-511	Manufacturing of composites	3
14	ME-513	Finite Element Methods in Engineering	4
15	ME-514	Fundamentals of Multiphase Flow	3
16	ME-516	Polymer Technology for Engineers	3
17	ME-517	Advanced Analytical Techniques for Engineers	4
18	ME-518	Conduction and Radiation	3
19	ME-519	Technical communication for Engineers	2
20	ME-520	Microwave based Manufacturing Processes	3
21	ME-521P	Laboratory course on Vehicle Design and Control	1
22	ME-522	High-Performance Scientific Computing	3
23	ME-523	Product Design	3
24	ME-524	Additive Manufacturing	3
25	ME-526	Programming Paradigm for Open-Source Software	1
26	ME-528	Thermal Management of Electronics	3
27	ME-530	Continuum Mechanics	3
28	ME-595	Experiential Learning in Composite Design, Manufacturing, Testing and Recycling	3
29	ME-600	Research Methodology	1
30	ME-601	Advanced Finite Element Methods	3
31	ME-602	Mechanical Vibration	3
32	ME-603	Advanced Fluid Mechanics	3
33	ME-604	Experimental Methods in Thermal Engineering	3
34	ME-605	Air Conditioning and Ventilation	3
35	ME-606	Advanced Solid Mechanics	3
36	ME-607	Materials Science For Failure Analysis	3
37	ME-608	Thin Films And Devices	3
38	ME-609	Functional Materials	3
39	ME-610	Advanced Thermodynamics	4
40	ME-611	Design and Optimization of Thermal Systems	3
41	ME-612	Introduction to Bio-materials	3
42	ME-613	Thermal Radiation	3
43	ME-614	Compressible Flow and Gas Dynamics	3
44	ME-615	Applied Computational Fluid Dynamics	3
45	ME-616	Dielectrics and Related Materials	3
46	ME-617	Mechanics of Composite Materials	3
47	ME-618	Stealth Technology: Infrared Signatures	3
48	ME-619	Experiments in Materials Science	3
49	ME-620	Modeling and Simulation	3
50	ME-621	Aircraft Propulsion	3

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
51	ME-622	Biomechanics of Musculoskeletal System	3
52	ME-625	Introduction to Turbulence and its Modeling	3
53	ME-626	Acoustics	3
54	ME-627	Mesh Independent Computational Techniques	3
55	ME-628	Impact Mechanics	3
56	ME-630	Machine learning for engineers	3
57	ME-631	Heat Transfer and Fluid Flow in Energy Systems	3
58	ME-632	Mechanics for Energy Systems	3
59	ME-633	Design of Energy Systems	3
60	ME-634	Thermodynamics for Energy Systems	3
61	ME-635	Manufacturing for Energy Systems	3
62	ME-636	Combustion Technology	3
63	ME-637	Wind Power Plant	3
64	ME-638	Solar Thermal Power Plant	3
65	ME-639	Thermal Power Plant Engineering	3
66	ME-640	Solar Power Utilization	3
67	ME-641	Finite Element Method	3
68	MT-201	Physics of Solids	3
69	MT-202	Applied Quantum Mechanics	3
70	MT-203	Materials Synthesis & Characterization	4
71	MT-204	Thermodynamics and Kinetics of Materials	3
72	MT-205	Functional Properties of Materials	4
73	MT-206	Extraction and Materials Processing	4
74	MT-301	Phase Transformations	3
75	MT-302	Transport Phenomena	3
76	MT-303	Computational Materials Science	4
77	MT-304	Mechanical Behavior of Materials	4
78	MT-501	Energy Conversion and Storage Technologies	3
79	MT-502	Recycling and Circular Economy	3
80	MT-503	Semiconductor Materials and Devices	3
81	MT-504	Powder metallurgical processing of Materials	3
82	MT-505	Thin Film Technology	3
83	MT-506	Biomaterials	3
84	MT-507	Modelling and Simulations in Material Science	3
85	MT-508	Iron and Steel Making	3
86	MT-509	Hydrogen Energy	3
87	MT-510	Colloids and Interfaces	3
88	MT-511	Sensor Materials and Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
89	ME-498P	Major Technical Project - I (compulsory; counted as DE)	4

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
90	ME-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Civil Engineering

Table 45: Civil Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	49
	Discipline Electives	17
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 46: B. Tech Civil Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CE-202	Introduction to Civil Engineering Profession	1-0-0-1	1
2	CE-203	Construction Materials	3-0-0-3	3
3	CE-203P / CE-354P	Building (Constructions) Materials Lab	0-0-2-1	1
4	CE-252	Geology and Geomorphology	2-0-2-3	3
5	CE-301 / CE-310	Strength of Materials and Structures	3-0-1-3	3
6	CE-301P / CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
7	CE-201	Surveying Traditional and Digital	2-0-4-4	4
8	CE-251	Hydraulics Engineering	3-0-0-3	3
9	CE-302 / CE-311	Geotechnical Engineering I	3-0-0-3	3
10	CE-302P / CE-311P	Geotechnical Engineering Lab		1
11	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
12	CE-404	Analysis of Structures	3-0-0-3	3
13	CE-303	Water Resources Engineering	3-0-0-3	3
14	CE-351	Design of Reinforced Concrete Structures	2-1-0-3	3
15	CE-352	Transportation Engineering	3-0-0-3	3
16	CE-352P	Transportation Engineering Lab	0-0-2-1	1
17	CE-353P	Civil Engineering Drawing	0-0-2-1	1
18	CE-402	Geotechnical Engineering II	2-1-0-3	3
19	CE-305P	Environmental Engineering Lab	0-0-2-1	1
20	CE-356	Reverse Engineering	1-0-0-1	1
21	CE-401	Design of Steel Structures	2-1-0-3	3
22	CE-403	Water and Wastewater Engineering	3-0-0-3	3
Total				49

Table 47: B. Tech Civil Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 48: B. Tech Civil Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 49: B. Tech Civil Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CE-202	Introduction to Civil Engineering Profession	1-0-0-1	1
4	DC	CE-203	Construction Materials	3-0-0-3	3
5	DC	CE-203P / CE-354P	Building (Constructions) Materials Lab	0-0-2-1	1
6	DC	CE-252	Geology and Geomorphology	2-0-2-3	3
7	DC	CE-301 / CE-310	Strength of Materials and Structures	3-0-1-3	3
8	DC	CE-301P / CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 50: B. Tech Civil Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-201	Surveying Traditional and Digital	2-0-4-4	4
2	DC	CE-251	Hydraulics Engineering	3-0-0-3	3
3	DC	CE-302 / CE-311	Geotechnical Engineering I	3-0-0-3	3
4	DC	CE-302P / CE-311P	Geotechnical Engineering Lab		1
5	DC	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
6	DC	CE-404	Analysis of Structures	3-0-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 51: B. Tech Civil Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-303	Water Resources Engineering	3-0-0-3	3
2	DC	CE-351	Design of Reinforced Concrete Structures	2-1-0-3	3
3	DC	CE-352	Transportation Engineering	3-0-0-3	3
4	DC	CE-352P	Transportation Engineering Lab	0-0-2-1	1
5	DC	CE-353P	Civil Engineering Drawing	0-0-2-1	1
6	DC	CE-402	Geotechnical Engineering II	2-1-0-3	3
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 52: B. Tech Civil Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-305P	Environmental Engineering Lab	0-0-2-1	1
2	DC	CE-356	Reverse Engineering	1-0-0-1	1
3	DC	CE-401	Design of Steel Structures	2-1-0-3	3
4	DC	CE-403	Water and Wastewater Engineering	3-0-0-3	3
5	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 53: B. Tech Civil Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	CE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 54: B. Tech Civil Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	CE-010	Internship		2
2	MTP	CE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Beyond the core courses above, students choose 17 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).

Table 55: Civil Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CD-591	Special Topics in Disaster Resilient Hill Roads	1
2	CE-304	Hydraulics Engineering Lab	3
3	CE-305	Environment and Earth Science Lab	3
4	CE-306	Environmental Engineering Lab	3
5	CE-353	Civil Engineering Drawing	3
6	CE-354	Building and Pavement Materials Laboratory	3
7	CE-355	Hydrology	3
8	CE-405	Water and Wastewater Engineering	3
9	CE-451	Irrigation Engineering and Hydraulic Structures	3
10	CE-452	Open Channel Flow	3
11	CE-501	Remote Sensing	3
12	CE-502	Groundwater Flow and Contaminant Transport	3
13	CE-503	Fundamentals of Project Management	3
14	CE-504	Slope Stability and Retaining Structures	3
15	CE-505	Engineering of Ground Modification	3
16	CE-506	Analysis of Indeterminate Structures	3
17	CE-507	Advanced Concrete Science	3
18	CE-508	Photogeology and Photogrammetry	3
19	CE-509	Bridge Engineering	3
20	CE-510	Modelling and Simulation in Water Resources Engineering	3
21	CE-511	Structural Dynamics with Application to Earthquake Engineering	3
22	CE-512	Advanced Soil Mechanics	3
23	CE-513	Advanced Foundation Engineering	3
24	CE-514	Rock Mechanics	3
25	CE-515	Environmental Impact Assessment	3
26	CE-516	Uncertainty Analysis in Civil Engineering	3
27	CE-517	Hydroinformatics	4
28	CE-518	Structural Reliability and Risk Assessment	3
29	CE-519	Chemistry of Natural Waters	3
30	CE-520	Environmental Reaction Modeling	3
31	CE-521	Ecology and Environment Microbiology	4
32	CE-522	Matrix Methods for Structural Analysis	3
33	CE-523	Building Science	3
34	CE-524	Applied Hydroclimatology	3

Civil Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
35	CE-524P	Computational Hydroclimatology Lab	1
36	CE-526	Critical Zone Science	3
37	CE-551	Geosynthetics and their applications	3
38	CE-552	Concrete Technology	3
39	CE-554	Prestressed Concrete Structures	3
40	CE-555	Advanced Design of Structures	3
41	CE-556	Structural Engineering Laboratory	3
42	CE-556P	Structural Engineering laboratory	2
43	CE-557	Solid Mechanics in Structural Engineering	3
44	CE-558	Air pollution and its mitigation	3
45	CE-559	Biological Wastewater Treatment	3
46	CE-560	Soil Dynamics	3
47	CE-561	The science of climate change	3
48	CE-562	Numerical Methods for Engineering Computation	3
49	CE-563	Finite Element Methods in Engineering	4
50	CE-564	Fundamentals of Fracture Mechanics	3
51	CE-567	Highway Safety and Modeling	3
52	CE-568P	Experimental Structural Dynamics	1
53	CE-586	Mini Project	3
54	CE-586P	Mini Project	3
55	CE-587	Industrial / Academic Internship	3
56	CE-591	Special Topics in Civil and Mechanical Engineering	3
57	CE-600	Research Methodology for Civil Engineering	1
58	CE-601	Geo-Informatics	3
59	CE-602	Blast Engineering	3
60	CE-604	Theory of Plasticity	3
61	CE-605	Engineering Seismology and Seismic Hazard Assessment	3
62	CE-606	Constitutive Modeling of Frictional Material	3
63	CE-610	Analysis and Design for Earthquake Resistant Structures	3
64	CE-611	Structural Health Monitoring	3
65	CE-612	Theory of Plates and Shells	3
66	CE-613	Mechanics of Unsaturated Soils	3
67	CE-617	Mechanics of Composite Materials	3
68	CE-688	Post Graduate Project - I	3
69	CE-689	Post Graduate Project - II	3
<i>Compulsory project components — counted as DE, not selectable</i>			
70	CE-498P	Major Technical Project - I (compulsory; counted as DE)	4

Civil Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
71	CE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Engineering Physics

Table 56: Engineering Physics — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	37
	Discipline Electives	29
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 57: B. Tech Engineering Physics — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EP-301	Engineering Mathematics 2	3-1-0-4	4
2	EP-321	Foundations of Electrodynamics	3-0-0-3	3
3	PH-301	Quantum Mechanics and Applications	2.5-0.5-0-3	3
4	EE-223P	Reverse Engineering	1-0-0-1	1
5	EP-403	Physics of Atoms and Molecules	3-0-0-3	3
6	PH-302	Introduction to Statistical Mechanics	2.5-0.5-0-3	3
7	PH-501	Solid State Physics	3	3
8	EE-311	Device Electronics for Integrated Circuits	3-0-0-3	3
9	EP-302	Computational Methods for Engineering	2-0-2-3	3
10	EP-402P	Engineering Physics Practicum	1-0-5-4	4
11	EP-401P	Engineering of Instrumentation	1-0-5-4	4
12	PH-502	Photonics	3-0-0-3	3
Total				37

Table 58: B. Tech Engineering Physics — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 59: B. Tech Engineering Physics — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9	IKS + HSS (choose 0–3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0–3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22–25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 60: B. Tech Engineering Physics — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	EP-301	Engineering Mathematics 2	3-1-0-4	4
3	DC	EP-321	Foundations of Electrodynamics	3-0-0-3	3
4	DC	PH-301	Quantum Mechanics and Applications	2.5-0.5-0-3	3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 61: B. Tech Engineering Physics — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	EE-223P	Reverse Engineering	1-0-0-1	1
3	DC	EP-403	Physics of Atoms and Molecules	3-0-0-3	3
4	DC	PH-302	Introduction to Statistical Mechanics	2.5-0.5-0-3	3
5	DC	PH-501	Solid State Physics	3	3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 62: B. Tech Engineering Physics — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-311	Device Electronics for Integrated Circuits	3-0-0-3	3
2	DC	EP-302	Computational Methods for Engineering	2-0-2-3	3
3	DC	EP-402P	Engineering Physics Practicum	1-0-5-4	4
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 63: B. Tech Engineering Physics — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EP-401P	Engineering of Instrumentation	1-0-5-4	4
2	DC	PH-502	Photonics	3-0-0-3	3
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 64: B. Tech Engineering Physics — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	EP-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 65: B. Tech Engineering Physics — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	EP-010	Internship		2
2	MTP	EP-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 29 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 66: Engineering Physics — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-208	Mathematical Foundations of Computer Science	4
2	CS-241	Introduction to Cryptography	3
3	CS-304	Formal Languages and Automata Theory	3
4	CS-309	Information and Database Systems	4
5	CS-511	Applied Probability	2
6	CS-671	Deep Learning and Applications	4
7	CS-672	Advanced Topics in Deep Learning	4
8	DS-201	Data handling and visualization	3
9	DS-301	Mathematical Foundations of Data Science	4

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
10	DS-401	Optimization for data science	3
11	DS-404	Information Security and Privacy	3
12	DS-416	Fundamentals of Machine Learning (CS-669)	4
13	EE-203	Network Theory	3
14	EE-211	Analog Circuit Design	3
15	EE-307	Theory of Measurement	3
16	EE-511	Computer Vision	4
17	EE-512	CMOS Analog IC Design	4
18	EE-519P	CMOS Digital IC Design Practicum	2
19	EE-520	Microelectronic Devices and Modelling	3
20	EE-524	Digital MOS LSI Circuits	3
21	EE-534	Probability and Random Processes	3
22	EE-551	Applied Photonics for Scientists and Engineers	3
23	EE-593	Low Power VLSI Circuits	3
24	EE-611	VLSI Technology	3
25	EE-614	Optical Communication Systems	3
26	EE-621	Radiating systems	4
27	EN-502	Emerging Energy Sources	3
28	EP-201	Reverse Engineering	1
29	EP-401	Engineering of Instrumentation	3
30	EP-402	Engineering Physics Practicum	3
31	EP-502	Informatics for Materials Design	3
32	IC-253	Programming and Data Structures	3
33	MA-210	Real and Complex Analysis	3
34	MA-511	Real Analysis	4
35	MA-512	Linear Algebra	4
36	MA-513	Ordinary Differential Equation	4
37	MA-516	Topology	4
38	MA-521	Topology and Functional Analysis	4
39	MA-522	Partial Differential Equations	4
40	MA-560	Nonlinear Dynamics and Chaos	3
41	ME-210	Fluid Mechanics	3
42	ME-303	Heat Transfer	3
43	ME-307	Energy Conversion Devices	3
44	ME-308	Manufacturing Engineering	3
45	ME-503	Micro/Nano Scale heat transfer	3
46	ME-509	Nano-manufacturing	3
47	ME-510	Advanced Manufacturing Processes	3
48	ME-603	Advanced Fluid Mechanics	3
49	ME-615	Applied Computational Fluid Dynamics	3

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	PH-503	Laser and Applications	3
51	PH-504	Organic Optoelectronics	3
52	PH-507	X-rays as a Probe to Study Material Properties	3
53	PH-508	Magnetism and Magnetic Materials	3
54	PH-511	Mathematical Physics	4
55	PH-512	Classical Mechanics	4
56	PH-513	Quantum Mechanics	3
57	PH-514	Electronics	3
58	PH-515P	Physics Laboratory	3
59	PH-516	Research Project I	2
60	PH-521	Electromagnetic Theory	4
61	PH-525P	Electronics Lab	3
62	PH-528	Introduction to General Relativity	3
63	PH-530	Cosmology-I	3
64	PH-550	Introduction to Quantum Optics	3
65	PH-579	Quantum Computation and Information	3
66	PH-600	Research Methodology	1
67	PH-601	Mesoscopic Physics and Quantum Transport	3
68	PH-603	Advanced Condensed Matter Physics	3
69	PH-604	Optical Properties of Solids	3
70	PH-605	Superconductivity	3
71	PH-606	Quantum Field Theory	3
72	PH-607	Physics of Ultracold Quantum Gases	3
73	PH-608	Computer assisted quantum mechanics	3
74	PH-609	Theory of quantum collision and spectroscopy	3
75	PH-612	Nuclear and Particle Physics	3
76	PH-613	Special Topics in Quantum Mechanics	3
77	PH-615P	Mini Thesis I	3
78	PH-621	Computational Methods for Physicists	4
79	PH-625	Data Analysis in Particle Physics	4
80	PH-626	Elementary Theoretical Particle Physics	3
81	PH-627	Topological Quantum Matter	3
82	PH-701	Introduction to Molecular Simulations	4
83	PH-706	Introduction to Stochastic Problems in Physics	3
84	QS-501P	Experiments in Quantum Optics	3
85	QT-301	Survey of Quantum Technologies and Applications	3
86	QT-406	Introduction to Quantum communication	3
87	QT-407	Introduction to Quantum Sensing	3
88	QT-509	Engineering Foundations of Quantum Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
89	EP-498P	Major Technical Project - I (compulsory; counted as DE)	4
90	EP-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Bioengineering

Table 67: Bioengineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	42
	Discipline Electives	24
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 68: B. Tech Bioengineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	BE-201	Cell Biology	3-0-2-4	4
2	BE-202	Biochemistry and Molecular Biology	3-0-2-4	4
3	BE-308	Introduction to Biomanufacturing	3-0-2-4	4
4	BE-309	Biosensing & Bioinstrumentation	3-0-2-4	4
5	BE-203	Enzymology and Bioprocessing	3-0-2-4	4
6	BE-301	Biomechanics	3-0-2-4	4
7	BE-304	Bioinformatics	3-0-2-4	4
8	BE-101P	Reverse Engineering	0-0-2-1	1
9	BE-303	Applied Biostatistics	3-0-2-4	4
10	BE-305	Bioethics and Regulatory Affairs	1-0-0-1	1
11	BE-306	Fundamentals of Genetic Engineering	3-1-0-4	4
12	BE-310	Biomaterials	3-0-2-4	4
Total				42

Table 69: B. Tech Bioengineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 70: B. Tech Bioengineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 71: B. Tech Bioengineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	BE-201	Cell Biology	3-0-2-4	4
3	DC	BE-202	Biochemistry and Molecular Biology	3-0-2-4	4
4	DC	BE-308	Introduction to Biomanufacturing	3-0-2-4	4
5	DC	BE-309	Biosensing & Bioinstrumentation	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 72: B. Tech Bioengineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	BE-203	Enzymology and Bioprocessing	3-0-2-4	4
3	DC	BE-301	Biomechanics	3-0-2-4	4
4	DC	BE-304	Bioinformatics	3-0-2-4	4
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 73: B. Tech Bioengineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	BE-101P	Reverse Engineering	0-0-2-1	1
2	DC	BE-303	Applied Biostatistics	3-0-2-4	4
3	DC	BE-305	Bioethics and Regulatory Affairs	1-0-0-1	1
4	DC	BE-306	Fundamentals of Genetic Engineering	3-1-0-4	4
5	DC	BE-310	Biomaterials	3-0-2-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 74: B. Tech Bioengineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 75: B. Tech Bioengineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	BE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 76: B. Tech Bioengineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	BE-010	Internship		2
2	MTP	BE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 24 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 77: Bioengineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-506	Cognitive Robotics	3
2	BE-101	Reverse Engineering for Bioengineers	3
3	BE-302	Bioelectric Systems Modeling	4
4	BE-307	Reverse Engineering for Bioengineers	3
5	BE-401	Bioengineering Mini Project, Term Paper and Seminar	4
6	BE-501	Anatomy and Physiology	3
7	BE-502	Design and Analysis of Bioalgorithms	3
8	BE-503	Biosensing & Bioinstrumentation	4
9	BE-504	Biomaterials	4
10	BE-505	Computational Biology	3
11	BE-506	Biological Modelling and Simulation	3

Bioengineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	BE-507	Environmental Impact Assessment	3
13	BE-598	Thesis Project Part I	3
14	BE-599	Thesis Project Part II	3
15	BE-698	Thesis Project Part I	3
16	BE-699	Thesis Project Part II	17
17	BY-501	DNA Nanotechnology	3
18	BY-502	Biophysics and Protein Engineering	3
19	BY-503	Cellular Fuel and Cellular Communication	3
20	BY-504	Metabolic Systems Biology	4
21	BY-506	Advanced Immunology	3
22	BY-507	Genetic Engineering	4
23	BY-508	Practical Metabolomics	3
24	BY-509	Practical OMICs	3
25	BY-510	Advanced Cell Biology	3
26	BY-511	Cell Physiology in health and disease	3
27	BY-512	Quantitative and Computational Biology	3
28	BY-514	Analytical Biotechniques	3
29	BY-515	Molecular Biotechnology	3
30	BY-516	Introduction to OMICS and Systems Analysis	3
31	BY-517	Proteomics	3
32	BY-518	Disease Biology	3
33	BY-519	Protein Sciences in Therapeutics	3
34	BY-520	Cell Biology and Physiology Lab	3
35	BY-520P	Cell Biology and Physiology	1
36	BY-521	Computational Biology and Cellular Bioprocess Technology Lab	3
37	BY-521P	Computational Biology and Cellular Bioprocess Technology Lab	1
38	BY-522	Analytical and Molecular Biotechnology Lab	3
39	BY-523	Systems Biology Lab	3
40	BY-524	Medical and NanoBiotechnology Lab	3
41	BY-527	Gene silencing and genome editing: principles and applications	3
42	BY-528	Sensory Biology	3
43	BY-529	Mechanobiology of the Cell (MBoC)	3
44	BY-531	Quantitative Biology and Data Analytics	3
45	BY-532	Immunotechnology	3
46	BY-535P	Analytical Biotechniques Lab	1
47	BY-536P	Immunotechnology Lab	1
48	BY-537	Computational Biology-1	4
49	BY-606	Bioinformatics Applications for Systems	3

Bioengineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	BY-613	Metagenomics and Next Generation Sequencing Technologies	3
51	CE-521	Ecology and Environment Microbiology	4
52	CE-559	Biological Wastewater Treatment	3
53	CS-309	Information and Database Systems	4
54	CS-606	Cognitive Modeling	3
55	CS-671	Deep Learning and Applications	4
56	CY-344	Food Chemistry: Processing, Preservation and Storage	3
57	CY-550	Bioinspired Materials	3
58	CY-642	Molecular- and Bio-electronics	3
59	CY-643	Advanced Analytical Techniques	3
60	CY-644	Bioinorganic chemistry	3
61	CY-670	Fluorescence spectroscopy, microscopy and applications	3
62	DS-303	Statistical Foundations of Data Science	3
63	DS-314	Probability Theory and Statistics	4
64	EE-305	Digital Signal Processing	4
65	EE-314	Digital Signal Processing	4
66	EE-516	Biomedical Systems	4
67	EE-574	Biomedical Signal and Image Analysis	3
68	EE-608	Digital Image Processing	4
69	MA-621	Modeling Population Dynamics	3
70	MA-650	Mathematical Models for Infectious Diseases	3
71	ME-303	Heat Transfer	3
72	ME-308	Manufacturing Engineering	3
73	ME-527	Biofluid Dynamics	3
74	ME-612	Introduction to Bio-materials	3
75	ME-622	Biomechanics of Musculoskeletal System	3
76	MT-506	Biomaterials	3
<i>Compulsory project components — counted as DE, not selectable</i>			
77	BE-498P	Major Technical Project - I (compulsory; counted as DE)	4
78	BE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Mathematics and Computing

Table 78: Mathematics and Computing — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	51
	Discipline Electives	15
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 79: B. Tech Mathematics and Computing — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
2	CS-214	Computer Organisation	3-0-2-4	4
3	MA-210	Real and Complex Analysis	2.5-0.5-0-3	3
4	MA-211	Ordinary Differential Equations	3-1-0-4	4
5	CS-304	Formal Language and Automata Theory	3-0-0-3	3
6	MA-220	Partial Differential Equations	3-1-0-4	4
7	MA-221	Numerical Analysis	3-1-0-4	4
8	MA-222	Applied Linear Programming	3-1-0-4	4
9	CS-212	Design of Algorithms	3-0-2-4	4
10	MA-310	Matrix Computation and Lab	3-0-2-4	4
11	MA-311	Mathematical Modelling	3-0-0-3	3
12	MA-322	Applied Graph Theory		4
13	MA-321	Numerics of Differential Equation		4
14	MA-323P	Applied Databases Practicum	0-0-3-2	2
Total				51

Table 80: B. Tech Mathematics and Computing — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 81: B. Tech Mathematics and Computing — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 82: B. Tech Mathematics and Computing — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
4	DC	CS-214	Computer Organisation	3-0-2-4	4
5	DC	MA-210	Real and Complex Analysis	2.5-0.5-0-3	3
6	DC	MA-211	Ordinary Differential Equations	3-1-0-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 83: B. Tech Mathematics and Computing — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-304	Formal Language and Automata Theory	3-0-0-3	3
2	DC	MA-220	Partial Differential Equations	3-1-0-4	4
3	DC	MA-221	Numerical Analysis	3-1-0-4	4
4	DC	MA-222	Applied Linear Programming	3-1-0-4	4
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 84: B. Tech Mathematics and Computing — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-212	Design of Algorithms	3-0-2-4	4
2	DC	MA-310	Matrix Computation and Lab	3-0-2-4	4
3	DC	MA-311	Mathematical Modelling	3-0-0-3	3
4	DC	MA-322	Applied Graph Theory		4
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 85: B. Tech Mathematics and Computing — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	MA-321	Numerics of Differential Equation	3-0-2-4	4
2	DC	MA-323P	Applied Databases Practicum	0-0-3-2	2
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
4	DC	MA-322	Applied Graph Theory	3-0-2-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 86: B. Tech Mathematics and Computing — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	MC-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 87: B. Tech Mathematics and Computing — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	MC-010	Internship		2
2	MTP	MC-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 15 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

This programme groups its Discipline Electives into the named baskets below; the remaining approved electives follow as a general list.

Table 88: Mathematics and Computing — Discipline Elective Basket I — Foundation Module (choose from these)

Sl. No	Course Code	Course Name	Credits
1	MA-251	Abstract Algebra	3
2	MA-252	Functional Analysis	3
3	MA-253	Measure Theory	3
4	MA-254	Topology	3
5	MA-255	Number Theory	3

Table 89: Mathematics and Computing — Discipline Elective Basket II — Advanced Modelling Module (choose from these)

Sl. No	Course Code	Course Name	Credits
1	MA-351	Climate Modelling	3
2	MA-352	Computational Financial Modelling & Lab	3
3	MA-353	Modelling of Infectious Disease	3
4	MA-354	Mathematical Image Processing	3
5	MA-355	Mathematical Control Theory	3
6	MA-356	Modelling and Simulation	3
7	MA-357	Modelling Population Dynamics	3
8	MA-546	Introduction to Quantitative Finance	4
9	MA-570	Data Driven Dynamical Systems	3

Table 90: Mathematics and Computing — Other Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BE-303	Applied Biostatistics	4
2	BE-304	Bioinformatics	4
3	CE-251	Hydraulics Engineering	3
4	CS-241	Introduction to Cryptography	3
5	CS-302	Paradigms of Programming	4
6	CS-306	Operating Systems	3
7	CS-309	Information and Database Systems	4
8	CS-313	Computer Networks	4
9	CS-315	Artificial Intelligence (CS-362, CS-562)	4
10	CS-502	Compiler Design	4
11	CS-514	Advance Data Structure and Algorithm (CS-580)	4
12	CS-563	Scalable Data Science	4
13	CS-606	Cognitive Modeling	3
14	CS-609	Speech processing	3
15	CS-677	Soft Computing	3
16	DS-201	Data handling and visualization	3
17	DS-312	Computing Systems for Data Processing	4
18	DS-416	Fundamentals of Machine Learning (CS-669)	4
19	EE-203	Network Theory	3
20	EE-260	Signals and Systems	3
21	EE-301	Control Systems	3
22	EE-511	Computer Vision	4
23	EE-530	Optimization theory	3
24	EE-608	Digital Image Processing	4
25	MA-001	Preparatory Mathematics - 1	4
26	MA-002	Preparatory Mathematics - 2	4

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
27	MA-101	Calculus I	4
28	MA-102	Linear Algebra	4
29	MA-120	Introduction to Computing Systems and Databases	4
30	MA-201	Probability and Statistics	4
31	MA-460	Nonlinear Dynamics and Chaos	3
32	MA-465	Ordinary Differential Equations	3
33	MA-510	Climate Change Analysis	3
34	MA-516	Topology	4
35	MA-521	Topology and Functional Analysis	4
36	MA-522	Partial Differential Equations	4
37	MA-523	Numerical Analysis	4
38	MA-524	Probability and Statistics	4
39	MA-525	Heuristic Optimization	3
40	MA-526	An Introduction to Wavelets	3
41	MA-527	Field and Galois Theory	3
42	MA-528	Measure Theory and Integration	4
43	MA-529	Statistical Inference	4
44	MA-530	Graph Theory	4
45	MA-546	Introduction to Quantitative Finance	4
46	MA-548	Abstract Algebra	4
47	MA-549	Abstract Algebra	4
48	MA-550	Statistical Data Analysis	3
49	MA-551	Numerical Analysis	3
50	MA-552	Number Theory	3
51	MA-553	Mathematical Foundations of Financial Engineering	3
52	MA-555	Introduction to Partial Differential Equations for Engineers	3
53	MA-560	Nonlinear Dynamics and Chaos	3
54	MA-563	Statistical Computation and Simulation	4
55	MA-565	Numerical Methods in Quantitative Finance	3
56	MA-568	Real Analysis	3
57	MA-570	Data-driven Dynamical Systems	3
58	MA-571	Introduction to Approximation algorithms	4
59	MA-572	Applied Multivariate Statistical Analysis	4
60	MA-581P	Post Graduate Project I	6
61	MA-582P	Post Graduate Project-2	8
62	MA-588	Mathematical Control Theory	4
63	MA-595	Calculus as Ganita	3
64	MA-601	Real and Functional Analysis	3
65	MA-603	Advanced Partial Differential Equations	3
66	MA-604	Introduction To Game Theory	3

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
67	MA-605	Statistical Data Analysis	3
68	MA-607	Numerical Analysis	3
69	MA-608	Computational Fluid Dynamics	3
70	MA-609	Numerics of Partial Differential Equations	3
71	MA-610	Mathematical Modeling	3
72	MA-611	Statistical tools and Computing	4
73	MA-612	Operator Theory	4
74	MA-621	Modeling Population Dynamics	3
75	MA-641	Operations Research	3
76	MA-644	Dynamical Systems	3
77	MA-650	Mathematical Models for Infectious Diseases	3
78	MA-651	Optimization Techniques	3
79	MA-652	Stability Theory of Differential Equations	3
80	MA-653	Computational Financial Modelling	3
81	MA-653P	Computational Financial Modelling Lab	1
82	MA-654	Financial Engineering	3
83	MA-655	Fixed Income Securities	3
84	MA-656	Stochastic Calculus for Financial Engineering	3
85	MA-665	Semigroup Of Bounded Linear Operators	3
86	MA-704	Dynamical System	3
87	MA-709	Numerical Linear Algebra	3
88	MA-765	Fractional Differential Equations	4
89	MA-780	Topics on Semigroup Theory	3
90	ME-210	Fluid Mechanics	3
91	ME-213	Engineering Thermodynamics	4
92	ME-303	Heat Transfer	3
93	ME-308	Manufacturing Engineering	3
94		Time Series Analysis	3
95		Mathematical Method for Signal Processing	4
96	CS-669	Pattern Recognition	4
97	BE-301	Biomechanics	4
98	BY-507	Genetic Engineering	4
99	CS-352	Transportation Engineering	3
100	CS-352P	Transporting Engg. Lab	1
101		Information Systems and Databases	4
102		Times Series Analysis and Applications / Bayesian Data Analysis and Applications	3
103		Big Data: Management and Analytics	4
104	EP-302	Computational Methods for Engineering	3
105	ME-310	System Dynamics and Control	3
106	ME-210P	Fluid Mechanics Lab	1

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
107	ME-303P	Heat Transfer Lab	1
108		Cellular Automata	3
109	CS-606	Computational Modeling of Social Systems	3
110	MA-562	An Introduction to Bayesian Analysis	4
111	MA-561	Fluidics and Microfluidics	4
112	MA-573	Advanced Mathematical Foundations of Machine Learning	4
113	MA-574	An Introduction to Probabilistic Machine Learning	4
114	MA-576	Nonlinear Optimization	4
<i>Compulsory project components — counted as DE, not selectable</i>			
115	MC-498P	Major Technical Project - I (compulsory; counted as DE)	4
116	MC-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - AI and Robotics

Table 91: General Engineering - AI and Robotics — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	37
	Discipline Electives	29
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 92: B. Tech General Engineering - AI and Robotics — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	DS-201	Data Handling and Vizualization	2-0-2-3	3
2	EE-261	Electrical System Around Us	3-0-0-3	3
3	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	EE-201	Electromechanics	2.5-0.5-0-3	3
5	IC-253	Programming and Data Structures	2-0-2-3	3
6	ME-100	Reverse Engineering	0-0-2-1	1
7	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
8	AR-521	Control of Robotic Systems		3
9	AR-523	Robot Manipulators		3
10	EE-302	Control Systems	3-0-2-4	4
11	ME-305	Design of Machine Elements	3-1-0-4	4
12	ME-309	Theory of Machines	3-1-0-4	4
Total				37

Table 93: B. Tech General Engineering - AI and Robotics — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 94: B. Tech General Engineering - AI and Robotics — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 95: B. Tech General Engineering - AI and Robotics — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	DS-201	Data Handling and Vizualization	2-0-2-3	3
4	DC	EE-261	Electrical System Around Us	3-0-0-3	3
5	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 96: B. Tech General Engineering - AI and Robotics — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
2	DC	IC-253	Programming and Data Structures	2-0-2-3	3
3	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 97: B. Tech General Engineering - AI and Robotics — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
2	DC	AR-521	Control of Robotic Systems		3
3	DC	AR-523	Robot Manipulators		3
4	DC	EE-302	Control Systems	3-0-2-4	4
5	DC	ME-305	Design of Machine Elements	3-1-0-4	4
6	DC	ME-309	Theory of Machines	3-1-0-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 98: B. Tech General Engineering - AI and Robotics — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 99: B. Tech General Engineering - AI and Robotics — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 100: B. Tech General Engineering - AI and Robotics — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 29 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 101: General Engineering - AI and Robotics — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-502	Advanced Design Practicum	4
2	AR-505	Principles of Robot Autonomy	3
3	AR-508	Marine Robotics	3
4	AR-509	Deep Learning for Robotics	4
5	AR-510	Underactuated Robotics	3
6	AR-511	Autonomous Mobile Robots	3
7	AR-513	Unmanned Aerial Systems (UAS)	3
8	AR-514	Vision and Learning Based Control	3
9	AR-519	Robot Manipulators	3
10	AR-522	Robot Vision	3
11	AR-524	Robot Simulators	3

General Engineering - AI and Robotics — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
12	AR-525	Reinforcement Learning for Robotics	3
13	AR-526	Multimodal Learning	3
14	CE-251	Hydraulics Engineering	3
15	CS-212	Design and Analysis of Algorithms	4
16	CS-514	Advance Data Structure and Algorithm (CS-580)	4
17	DS-314	Probability Theory and Statistics	4
18	EE-201P	Electromechanics Lab	1
19	EE-203	Network Theory	3
20	EE-211	Analog Circuit Design	3
21	EE-260	Signals and Systems	3
22	EE-529	Embedded Systems	4
23	EE-604	Practicum on Advanced Electric Drives	3
24	EE-608	Digital Image Processing	4
25	IC-241	Materials Science for Engineers	3
26	ME-210	Fluid Mechanics	3
27	ME-212	Product Manufacturing Technology	3
28	ME-213	Engineering Thermodynamics	4
29	ME-510	Advanced Manufacturing Processes	3
<i>Compulsory project components — counted as DE, not selectable</i>			
30	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
31	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Mechatronics

Table 102: General Engineering - Mechatronics — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	37
	Discipline Electives	29
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 103: B. Tech General Engineering - Mechatronics — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-260	Signals and Systems	2.5-0.5-0-3	3
3	EE-261	Electrical System Around Us	3-0-0-3	3
4	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
5	EE-201	Electromechanics	2.5-0.5-0-3	3
6	EE-211	Analog Circuit Design	2-0-2-3	3
7	ME-100	Reverse Engineering	0-0-2-1	1
8	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
9	EE-231	Measurement and Instrumentation	2-0-2-3	3
10	EE-302	Control Systems	3-0-2-4	4
11	ME-305	Design of Machine Elements	3-1-0-4	4
12	ME-309	Theory of Machines	3-1-0-4	4
Total				37

Table 104: B. Tech General Engineering - Mechatronics — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 105: B. Tech General Engineering - Mechatronics — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 106: B. Tech General Engineering - Mechatronics — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-203	Network Theory	3-0-0-3	3
4	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
5	DC	EE-261	Electrical System Around Us	3-0-0-3	3
6	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 107: B. Tech General Engineering - Mechatronics — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
2	DC	EE-211	Analog Circuit Design	2-0-2-3	3
3	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 108: B. Tech General Engineering - Mechatronics — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
2	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
3	DC	EE-302	Control Systems	3-0-2-4	4
4	DC	ME-305	Design of Machine Elements	3-1-0-4	4
5	DC	ME-309	Theory of Machines	3-1-0-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 109: B. Tech General Engineering - Mechatronics — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 110: B. Tech General Engineering - Mechatronics — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 111: B. Tech General Engineering - Mechatronics — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 29 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 112: General Engineering - Mechatronics — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-502	Advanced Design Practicum	4
2	AR-505	Principles of Robot Autonomy	3
3	AR-508	Marine Robotics	3
4	AR-509	Deep Learning for Robotics	4
5	AR-510	Underactuated Robotics	3
6	AR-511	Autonomous Mobile Robots	3
7	AR-519	Robot Manipulators	3
8	AR-521	Control of Robotic Systems	3
9	AR-522	Robot Vision	3
10	AR-523	Robot Manipulators	3
11	AR-524	Robot Simulators	3

General Engineering - Mechatronics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	AR-526	Multimodal Learning	3
13	CS-212	Design and Analysis of Algorithms	4
14	CS-313	Computer Networks	4
15	CS-315	Artificial Intelligence (CS-362, CS-562)	4
16	EE-201P	Electromechanics Lab	1
17	EE-210	Digital Systems Design	3
18	EE-210P	Digital Systems Design Lab	1
19	EE-311	Device Electronics for Integrated Circuits	3
20	EE-314	Digital Signal Processing	4
21	EE-326	Computer Organization and Processor Architecture Design	4
22	EE-529	Embedded Systems	4
23	EE-536	IoT Systems	3
24	IC-121	Mechanics of Particles and waves	3
25	IC-241	Materials Science for Engineers	3
26	ME-210	Fluid Mechanics	3
27	ME-212	Product Manufacturing Technology	3
28	ME-213	Engineering Thermodynamics	4
29	ME-215	Manufacturing Engineering-1	3
30	ME-315	Manufacturing Engineering-II	3
31	ME-452	Robot Kinematics, Dynamics and Control	4
32	ME-504	Numerical Methods for Engineering Computation	3
33	ME-510	Advanced Manufacturing Processes	3
34	ME-513	Finite Element Methods in Engineering	4
35	ME-527	Biofluid Dynamics	3
36	ME-630	Machine learning for engineers	3
<i>Compulsory project components — counted as DE, not selectable</i>			
37	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
38	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Communication Technology

Table 113: General Engineering - Communication Technology — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	40
	Discipline Electives	26
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 114: B. Tech General Engineering - Communication Technology — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-313	Computer Networks	3-0-2-4	4
2	EE-203	Network Theory	3-0-0-3	3
3	EE-260	Signals and Systems	2.5-0.5-0-3	3
4	EE-261	Electrical System Around Us	3-0-0-3	3
5	DS-404	Information Security and Privacy	3-0-0-3	3
6	EE-201	Electromechanics	2.5-0.5-0-3	3
7	EE-202	Electromagnetic Theory	3-0-0-3	3
8	EE-316	Communication Systems	3-0-2-4	3
9	IC-253	Programming and Data Structures	2-0-2-3	3
10	ME-100	Reverse Engineering	0-0-2-1	1
11	EE-231	Measurement and Instrumentation	2-0-2-3	3
12	EE-302	Control Systems	3-0-2-4	4
13	EE-314	Digital Signal Processing	3-0-2-4	4
Total				40

Table 115: B. Tech General Engineering - Communication Technology — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 116: B. Tech General Engineering - Communication Technology — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 117: B. Tech General Engineering - Communication Technology — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-313	Computer Networks	3-0-2-4	4
4	DC	EE-203	Network Theory	3-0-0-3	3
5	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
6	DC	EE-261	Electrical System Around Us	3-0-0-3	3
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 118: B. Tech General Engineering - Communication Technology — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-404	Information Security and Privacy	3-0-0-3	3
2	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
3	DC	EE-202	Electromagnetic Theory	3-0-0-3	3
4	DC	EE-316	Communication Systems	3-0-2-4	3
5	DC	IC-253	Programming and Data Structures	2-0-2-3	3
6	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 119: B. Tech General Engineering - Communication Technology — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	EE-302	Control Systems	3-0-2-4	4
3	DC	EE-314	Digital Signal Processing	3-0-2-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 120: B. Tech General Engineering - Communication Technology — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 121: B. Tech General Engineering - Communication Technology — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 122: B. Tech General Engineering - Communication Technology — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 26 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 123: General Engineering - Communication Technology — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-549	Performance Analysis of Computer Networks	3
2	DS-201	Data handling and visualization	3
3	DS-314	Probability Theory and Statistics	4
4	EE-201P	Electromechanics Lab	1
5	EE-211	Analog Circuit Design	3
6	EE-503	Advance Communication Theory	3
7	EE-517	Wireless Communications and Networking	3
8	EE-518	Information Theory	3
9	EE-530	Optimization theory	3
10	EE-533	Access Network	4

General Engineering - Communication Technology — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	EE-536	IoT Systems	3
12	EE-541	Tensors: Techniques, Algorithms, Applications for Signal	4
13	EE-553	Foundations of Intelligent Communication Systems-I	4
14	EE-574	Biomedical Signal and Image Analysis	3
15	EE-581	Applied Statistics for Data and Signals	4
16	EE-607	Optical Communication Systems	3
17	EE-608	Digital Image Processing	4
18	EE-641	Advance Wireless Technologies	4
19	IC-241	Materials Science for Engineers	3
20	ME-212	Product Manufacturing Technology	3
21	VL-402	Rfic Design	3
<i>Compulsory project components — counted as DE, not selectable</i>			
22	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
23	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Materials Science and Engineering

Table 124: Materials Science and Engineering — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	45
	Discipline Electives	21
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 125: B. Tech Materials Science and Engineering — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
2	MT-201	Physics of Solids		3
3	MT-202	Quantum Mechanics and Applications		3
4	MT-203	Material Synthesis and Characterisation		4
5	ME-100	Reverse Engineering	0-0-2-1	1
6	MT-204	Thermodynamics and Kinetics and Materials	3	3
7	MT-205	Functional Properties of Materials	4	4
8	MT-206	Extraction and Materials Processing	4	4
9	ME-212	Product Realization (Manufacturing) Technology	2-0-3-3	3
10	MT-301	Phase Transformations		3
11	MT-302	Transport Phenomena		3
12	MT-303	Computational Materials Science		4
13	MT-304	Mechanical Behaviour of Materials		4
14	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total				45

Table 126: B. Tech Materials Science and Engineering — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 127: B. Tech Materials Science and Engineering — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-241	Material Science for Engineers	3-0-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 128: B. Tech Materials Science and Engineering — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	DC	MT-201	Physics of Solids		3
5	DC	MT-202	Quantum Mechanics and Applications		3
6	DC	MT-203	Material Synthesis and Characterisation		4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 129: B. Tech Materials Science and Engineering — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-100	Reverse Engineering	0-0-2-1	1
2	DC	MT-204	Thermodynamics and Kinetics and Materials	3	3
3	DC	MT-205	Functional Properties of Materials	4	4
4	DC	MT-206	Extraction and Materials Processing	4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 130: B. Tech Materials Science and Engineering — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-212	Product Realization (Manufacturing) Technology	2-0-3-3	3
2	DC	MT-301	Phase Transformations		3
3	DC	MT-302	Transport Phenomena		3
4	DC	MT-303	Computational Materials Science		4
5	DC	MT-304	Mechanical Behaviour of Materials		4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 131: B. Tech Materials Science and Engineering — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 132: B. Tech Materials Science and Engineering — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	MS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 133: B. Tech Materials Science and Engineering — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	MS-010	Internship		2
2	MTP	MS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 21 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 134: Materials Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	ME-303	Heat Transfer	3
2	ME-308	Manufacturing Engineering	3
3	ME-353	Electronic Materials and Their Applications	3
4	ME-515	Carbon Materials and Technology	3
5	MT-501	Energy Conversion and Storage Technologies	3
6	MT-502	Recycling and Circular Economy	3
7	MT-503	Semiconductor Materials and Devices	3
8	MT-504	Powder metallurgical processing of Materials	3
9	MT-505	Thin Film Technology	3
10	MT-506	Biomaterials	3

Materials Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	MT-507	Modelling and Simulations in Material Science	3
12	MT-511	Sensor Materials and Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
13	MS-498P	Major Technical Project - I (compulsory; counted as DE)	4
14	MS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Science — Chemical Sciences

Table 135: Chemical Sciences — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	31
	IC Basket	9
	IKS + HSS	12
Discipline Courses	Discipline Core	62
	Discipline Electives	20
Electives	Free Electives	15
	Research Lit Presentation (1+1)	2
	RP (equivalent as DE)	12
Total		163

Table 136: B. S. Chemical Sciences — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CY-201P	Physical Chemistry Laboratory	0-0-4-2	2
2	CY-301	Principles and Theories of Physical Chemistry	3-0-0-3	3
3	CY-302	Principles of Organic Chemistry	3-0-0-3	3
4	CY-303	Fundamentals of Inorganic Chemistry	3-0-0-3	3
5	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
6	CY-202P	Organic Chemistry Laboratory	0-0-4-2	2
7	CY-203P	Inorganic Chemistry Laboratory	0-0-4-2	2
8	CY-304	Fundamental Analytical Chemistry	3-0-0-3	3
9	CY-401	Introduction to Quantum Chemistry and Molecular Spectroscopy	3-0-0-3	3
10	CY-512	Advanced Quantum Chemistry	3-0-0-3	3
11	CY-512P	Physical Chemistry Laboratory		3
12	CY-531	Organic Reactions and Mechanisms		3
13	CY-533	Chemistry of Main Group Elements		3
14	CY-533P	Inorganic Chemistry Laboratory		3
15	CY-511	Group Theory and Spectroscopy	3-0-0-3	3
16	CY-531P	Organic Chemistry Laboratory		3
17	CY-532	Photochemistry and Pericyclic Reactions		3
18	CY-534	Chemistry of Transition Elements		3
19	CY-514	Chemical and Statistical Thermodynamics	3-0-0-3	3
20	CY-535	Introduction to Organometallic Chemistry		3
21	CY-504	Heterocyclic Chemistry	2-0-0-2	2
22	CY-513	Chemical Kinetics and Reaction Dynamics	3-0-0-3	3
Total				62

Table 137: B. S. Chemical Sciences — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 138: B. S. Chemical Sciences — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 139: B. S. Chemical Sciences — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-201P	Physical Chemistry Laboratory	0-0-4-2	2
2	DC	CY-301	Principles and Theories of Physical Chemistry	3-0-0-3	3
3	DC	CY-302	Principles of Organic Chemistry	3-0-0-3	3
4	DC	CY-303	Fundamentals of Inorganic Chemistry	3-0-0-3	3
5	DC	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 140: B. S. Chemical Sciences — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-202P	Organic Chemistry Laboratory	0-0-4-2	2
2	DC	CY-203P	Inorganic Chemistry Laboratory	0-0-4-2	2
3	DC	CY-304	Fundamental Analytical Chemistry	3-0-0-3	3
4	DC	CY-401	Introduction to Quantum Chemistry and Molecular Spectroscopy	3-0-0-3	3
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 141: B. S. Chemical Sciences — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-512	Advanced Quantum Chemistry	3-0-0-3	3
2	DC	CY-512P	Physical Chemistry Laboratory		3
3	DC	CY-531	Organic Reactions and Mechanisms		3
4	DC	CY-533	Chemistry of Main Group Elements		3
5	DC	CY-533P	Inorganic Chemistry Laboratory		3
6	RP	DP-403P	Research Literature Presentation 1		1
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 142: B. S. Chemical Sciences — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-511	Group Theory and Spectroscopy	3-0-0-3	3
2	DC	CY-531P	Organic Chemistry Laboratory		3
3	DC	CY-532	Photochemistry and Pericyclic Reactions		3
4	DC	CY-534	Chemistry of Transition Elements		3
5	RP	DP-404P	Research Literature Presentation 2		1
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 143: B. S. Chemical Sciences — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-514	Chemical and Statistical Thermodynamics	3-0-0-3	3
2	DC	CY-535	Introduction to Organometallic Chemistry		3
3	RP	DP-551P	Undergraduate Research Project 1		6
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 144: B. S. Chemical Sciences — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-504	Heterocyclic Chemistry	2-0-0-2	2
2	DC	CY-513	Chemical Kinetics and Reaction Dynamics	3-0-0-3	3
3	RP	DP-552P	Undergraduate Research Project 2		6
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 20 credits of **Discipline Electives** and 15 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 145: Chemical Sciences — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BY-505	Nanobiotechnology	3
2	BY-514	Analytical Biotechniques	3
3	CE-519	Chemistry of Natural Waters	3
4	CH-101	General Chemistry	4
5	CY-001	Preparatory Chemistry - 1	4
6	CY-002	Preparatory Chemistry - 2	4
7	CY-100	Preparatory Chemistry 1	3
8	CY-101	Chemistry	5
9	CY-200	Foundations and applications of Chemistry	3
10	CY-241	Nano-scale Science and Technology	3
11	CY-242	Introduction to Molecular Spectroscopy	3
12	CY-243	Engineering Chemistry	3
13	CY-247	Introduction to Molecular Thermodynamics	3
14	CY-248	Molecular Physical Chemistry for Engineers	3
15	CY-342	Understanding Small Systems	3
16	CY-344	Food Chemistry: Processing, Preservation and Storage	3
17	CY-402	Applied Materials Chemistry	3
18	CY-403	Numerical Methods and Data Analysis in Chemistry	3
19	CY-404	Fundamentals of Soft Matter Science and Applications	3
20	CY-405	Introduction to structural chemistry	3
21	CY-501	Organic Reaction and Mechanism	4
22	CY-501P	Organic Chemistry Laboratory	4
23	CY-502	Photochemistry and pericyclic reactions	4
24	CY-506	Chemistry of Transition Elements	4
25	CY-506P	Inorganic Chemistry Lab	4
26	CY-507	Chemistry of Main Group Elements	4
27	CY-508	Organometallic Chemistry	2
28	CY-511P	Physical Chemistry Laboratory	4
29	CY-515	Advanced Inorganic Spectroscopy	3
30	CY-521	Mathematics for Chemists	3
31	CY-522	Computational Chemistry	3
32	CY-523	Colloids and Interface Science and Technology	3
33	CY-524	Basic and Applied Electrochemistry	3
34	CY-541	Fundamentals of Organic Chemistry	3
35	CY-545	Reagents in organic synthesis	3
36	CY-547	Chemical Crystallography	3
37	CY-550	Bioinspired Materials	3
38	CY-552	Hydrogen Generation and Storage	3
39	CY-553	Organic Inorganic Spectroscopy	3
40	CY-554	Science and Technology of Nanomaterials	3

Chemical Sciences — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
41	CY-555	Introduction to Polymer Science & Technology	3
42	CY-556	Organic Spectroscopy	3
43	CY-560	Introduction to Supramolecular Chemistry	3
44	CY-561	Natural Products Synthesis	3
45	CY-562	Stereochemistry of organic compounds	3
46	CY-562P	Design Practicum - II	1
47	CY-598P	Post Graduate Project-1 (CH)	4
48	CY-599P	Post Graduate Project-2	9
49	CY-641	Polymer Synthesis	3
50	CY-642	Molecular- and Bio-electronics	3
51	CY-643	Advanced Analytical Techniques	3
52	CY-644	Bioinorganic chemistry	3
53	CY-645	Reagents in organic synthesis	3
54	CY-646	Advanced NMR Spectroscopy - A Problem Based Approach	3
55	CY-660	Photocatalysis: Concepts and Applications	3
56	CY-670	Fluorescence spectroscopy, microscopy and applications	3
57	CY-701	Advances Physical Methods in Chemistry Theory and	3
58	CY-702	Advanced Inorganic Chemistry: Theory and Applications	3
59	CY-703	Advanced Organic Chemistry	3
60	CY-704	Introduction to Theoretical Chemistry	3
61	CY-705	Modern Methods in Organic Synthesis	3
62	CY-746	Self Assembly of Surfactants and Polymers in Solution	3
63	EN-503	Energy Storage Technologies	3
64	ME-353	Electronic Materials and Their Applications	3
65	ME-509	Nanomanufacturing	3
66	ME-515	Carbon Materials and Technology	3
67	ME-516	Polymer Technology for Engineers	3
68	MT-201	Physics of Solids	3
69	MT-501	Energy Conversion and Storage Technologies	3
70	MT-502	Recycling and Circular Economy	3
71	MT-503	Semiconductor Materials and Devices	3
72	MT-505	Thin Film Technology	3
73	MT-506	Biomaterials	3
74	MT-509	Hydrogen Energy	3
75	MT-510	Colloids and Interfaces	3
76	PH-504	Organic Optoelectronics	3
<i>Compulsory project components — counted as DE, not selectable</i>			
77	DP-551P	Undergraduate Research Project 1 (compulsory; counted as DE)	6

Chemical Sciences — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
78	DP-552P	Undergraduate Research Project 2 (compulsory; counted as DE)	6

Bachelor of Technology — Microelectronics and VLSI

Table 146: Microelectronics and VLSI — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	54
	Discipline Electives	12
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 147: B. Tech Microelectronics and VLSI — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-210	Digital Systems Design	3-0-0-3	3
3	EE-210P	Digital Systems Design Lab	0-0-2-1	1
4	EE-260	Signals and Systems	2.5-0.5-0-3	3
5	EE-302	Control Systems	3-0-2-4	4
6	EE-311	Device Electronics	3-0-0-3	3
7	EE-202	Electromagnetic Theory and Transmission Lines	3-0-0-3	3
8	EE-211	Analog Circuit Design	2-0-2-3	4
9	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
10	VL-300	Reverse Engineering - E-waste management	0-0-2-1	1
11	VL-311	CMOS Processing and Practicum	3-0-2-4	4
12	VL-312	Electronic System Packaging	3-0-0-3	3
13	VL-401	RTL Design and Verification	2-0-2-3	3
14	VL-402	RF IC Design	3-0-0-3	3
15	VL-403	CMOS Digital IC Design	3-0-2-4	4
16	VL-404	CMOS Analog IC Design	3-0-2-4	4
17	VL-405	Design for Testability	3-0-2-4	4
Total				54

Table 148: B. Tech Microelectronics and VLSI — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		HS-342	German I	3-0-0-3	
10		HS-108	Basic English for Engineers	3-0-0-3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 149: B. Tech Microelectronics and VLSI — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket (choose 1)	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9		IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	
10		IC-241	Material Science for Engineers	3-0-0-3	
11		IC-253	Data Structures and Algorithms	2-0-2-3	
12	IKS + HSS (choose 0–3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0–3
13		HS-151	Introduction to English Literature	3-0-0-3	
14		HS-357	French Language and Culture	2-0-0-2	
15		HS-308	European Literature	2-0-0-2	
Total					22–25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 150: B. Tech Microelectronics and VLSI — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-203	Network Theory	3-0-0-3	3
4	DC	EE-210	Digital Systems Design	3-0-0-3	3
5	DC	EE-210P	Digital Systems Design Lab	0-0-2-1	1
6	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
7	DC	EE-302	Control Systems	3-0-2-4	4
8	DC	EE-311	Device Electronics	3-0-0-3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 151: B. Tech Microelectronics and VLSI — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-202	Electromagnetic Theory and Transmission Lines	3-0-0-3	3
2	DC	EE-211	Analog Circuit Design	2-0-2-3	4
3	DC	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
4	DC	VL-300	Reverse Engineering - E-waste management	0-0-2-1	1
5	DC	VL-311	CMOS Processing and Practicum	3-0-2-4	4
6	DC	VL-312	Electronic System Packaging	3-0-0-3	3
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 152: B. Tech Microelectronics and VLSI — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	VL-401	RTL Design and Verification	2-0-2-3	3
2	DC	VL-402	RF IC Design	3-0-0-3	3
3	DC	VL-403	CMOS Digital IC Design	3-0-2-4	4
4	DC	VL-404	CMOS Analog IC Design	3-0-2-4	4
5	DC	VL-405	Design for Testability	3-0-2-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 153: B. Tech Microelectronics and VLSI — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 154: B. Tech Microelectronics and VLSI — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	VL-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 155: B. Tech Microelectronics and VLSI — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	VL-010	Internship		2
2	MTP	VL-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 12 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 156: Microelectronics and VLSI — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	EE-507	Transmission Lines and Basic Microwave Engineering	4
2	EE-516	Biomedical Systems	4
3	EE-520	Microelectronic Devices and Modelling	3
4	EE-523	Digital VLSI Architecture Design	4
5	EE-529	Embedded Systems	4
6	EE-554	Low Power VLSI Design	3
7	EE-574	Biomedical Signal and Image Analysis	3
8	EE-585	Theory, Modelling and Characterization of Photo-voltaic Devices	3
9	EE-611	VLSI Technology	3
10	EE-615	Nanoelectronics and Micronanofabrication	3

Microelectronics and VLSI — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	EE-619	Mixed Signal VLSI Design	4
12	EE-621	Radiating systems	4
13	EE-622	Microwave Integrated Circuits	3
<i>Compulsory project components — counted as DE, not selectable</i>			
14	VL-498P	Major Technical Project - I (compulsory; counted as DE)	4
15	VL-499P	Major Technical Project - II (compulsory; counted as DE)	4

Batch 2024

Bachelor of Technology — Computer Science and Engineering

Table 157: Computer Science and Engineering — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	38
	Discipline Electives	28
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 158: B. Tech Computer Science and Engineering — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
2	CS-212	Design of Algorithms	3-0-2-4	4
3	CS-213	Reverse Engineering	0-0-2-1	1
4	CS-214	Computer Organization	3-0-2-4	4
5	CS-304	Formal Language and Automata Theory	3-0-0-3	3
6	CS-309	Information Systems and Databases	3-0-2-4	4
7	CS-305	Artificial Intelligence	3-0-0-3	3
8	CS-312	Operating Systems	3-0-2-4	4
9	CS-313	Computer Networks	3-0-2-4	4
10	CS-302	Paradigms of Programming	3-0-2-4	4
11	CS-303	Software Engineering	2-0-2-3	3
Total				38

Table 159: B. Tech Computer Science and Engineering — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-110	Japanese for Beginners	3	
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 160: B. Tech Computer Science and Engineering — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 161: B. Tech Computer Science and Engineering — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
4	DC	CS-212	Design of Algorithms	3-0-2-4	4
5	DC	CS-213	Reverse Engineering	0-0-2-1	1
6	DC	CS-214	Computer Organization	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 162: B. Tech Computer Science and Engineering — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	CS-304	Formal Language and Automata Theory	3-0-0-3	3
3	DC	CS-309	Information Systems and Databases	3-0-2-4	4
Total					9

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 163: B. Tech Computer Science and Engineering — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-305	Artificial Intelligence	3-0-0-3	3
2	DC	CS-312	Operating Systems	3-0-2-4	4
3	DC	CS-313	Computer Networks	3-0-2-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 164: B. Tech Computer Science and Engineering — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-302	Paradigms of Programming	3-0-2-4	4
2	DC	CS-303	Software Engineering	2-0-2-3	3
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 165: B. Tech Computer Science and Engineering — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	CS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 166: B. Tech Computer Science and Engineering — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	CS-010	Internship		2
2	MTP	CS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 28 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 167: Computer Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-501	Access Networks	3
2	CS-502	Compiler Design	4
3	CS-506	Cognitive Modeling	3
4	CS-507	Computer Architecture	4
5	CS-508	Introduction to Network Security	3
6	CS-510	Randomized Algorithms	4
7	CS-511	Applied Probability	2
8	CS-512	Matrix Theory	2
9	CS-513	Discrete Mathematics	4

Computer Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
10	CS-514	Advance Data Structure and Algorithm (CS-580)	4
11	CS-515	Advanced Computer Science Practicum	3
12	CS-516	Exploratory Project	3
13	CS-520	Introduction to Quantum Computing	3
14	CS-521	Introduction to Post Quantum Security	3
15	CS-522	Distributed Algorithms	3
16	CS-524	Computational Complexity Theory	4
17	CS-525	Computational Foundations of Cryptography	3
18	CS-526	Mathematical Foundation of Cryptography	3
19	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3
20	CS-541	IoT systems and the Cloud	3
21	CS-542	Design Patterns for Scalable Systems	3
22	CS-544	Formal Concept Analysis: Theory and Practice	3
23	CS-545	Software Design Patterns	3
24	CS-546	Design of Concurrent Software	3
25	CS-547	Network Management Systems	3
26	CS-548	Cloud Networking	3
27	CS-549	Performance Analysis of Computer Networks	3
28	CS-550	Computer Graphics and Geometric Design	3
29	CS-551	Human Computer Interaction	3
30	CS-560	Text Retrieval and Mining	3
31	CS-561	Map Reduce and Big Data	3
32	CS-563	Scalable Data Science	4
33	CS-570	Algorithms for Curves and Surface	3
34	CS-571	Programming Practicum	3
35	CS-591	Special Topics in Knowledge Discovery and Data	1
36	CS-592	Selected Topics in Artificial Intelligence	2
37	CS-601	Probability and Random Processes	4
38	CS-603	Managerial Decision Making	3
39	CS-606	Cognitive Modeling	3
40	CS-609	Speech processing	3
41	CS-611	Program Analysis	4
42	CS-621P	Post Graduate Project-1	12
43	CS-630	Speech Technology	4
44	CS-660	Data Mining for Decision Making	4
45	CS-661	Knowledge Representation and Reasoning	3
46	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
47	CS-670	Kernal Methods for Pattern Recognition	4
48	CS-671	Deep Learning and Applications	4
49	CS-672	Advanced Topics in Deep Learning	4

Computer Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	CS-673	Advanced Computer Vision	4
51	CS-677	Soft Computing	3
52	CS-683	Generative AI	4
53	CS-685	Natural Language Processing	3
54	CS-686	Large Language Models: Core Concepts to Custom	4
55	CS-693	Compressed Sensing and its applications	3
56	DS-201	Data handling and visualization	3
57	DS-301	Mathematical Foundations of Data Science	4
58	DS-314	Probability Theory and Statistics	4
59	DS-401	Optimization for data science	3
60	DS-402	Matrix Computations for Data Science	3
61	DS-403	Introduction to Statistical Learning	3
62	DS-404	Information Security and Privacy	3
63	DS-412	Matrix Computations for Data Science	4
64	DS-415	Optimization for Machine Learning	4
65	DS-416	Fundamentals of Machine Learning (CS-669)	4
66	EE-511	Computer Vision	4
<i>Compulsory project components — counted as DE, not selectable</i>			
67	CS-498P	Major Technical Project - I (compulsory; counted as DE)	4
68	CS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Data Science and Engineering

Table 168: Data Science and Engineering — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	33
	Discipline Electives	33
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 169: B. Tech Data Science and Engineering — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-213	Reverse Engineering	0-0-2-1	1
2	DS-201	Data Handling and Vizualisation	2-0-2-3	3
3	DS-301	Mathematical Foundations of Data Science	3-1-0-4	4
4	DS-302	Computing Systems for Data Processing	2-0-2-3	3
5	DS-313	Statistical Foundations of Data Science	3-1-0-4	4
6	DS-412	Matrix Computations for Data Science	3-0-2-4	4
7	CS-305	Artificial Intelligence	3-0-0-3	3
8	DS-404	Information Security and Privacy	3-0-0-3	3
9	DS-413	Introduction to Statistical Learning	3-1-0-4	4
10	DS-411	Optimization for Data Science	3-1-0-4	4
Total				33

Table 170: B. Tech Data Science and Engineering — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-110	Japanese for Beginners	3	
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 171: B. Tech Data Science and Engineering — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 172: B. Tech Data Science and Engineering — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-213	Reverse Engineering	0-0-2-1	1
4	DC	DS-201	Data Handling and Vizualisation	2-0-2-3	3
5	DC	DS-301	Mathematical Foundations of Data Science	3-1-0-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 173: B. Tech Data Science and Engineering — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	DS-302	Computing Systems for Data Processing	2-0-2-3	3
3	DC	DS-313	Statistical Foundations of Data Science	3-1-0-4	4
4	DC	DS-412	Matrix Computations for Data Science	3-0-2-4	4
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 174: B. Tech Data Science and Engineering — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-305	Artificial Intelligence	3-0-0-3	3
2	DC	DS-404	Information Security and Privacy	3-0-0-3	3
3	DC	DS-413	Introduction to Statistical Learning	3-1-0-4	4
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 175: B. Tech Data Science and Engineering — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-411	Optimization for Data Science	3-1-0-4	4
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					8

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 176: B. Tech Data Science and Engineering — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	DS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 177: B. Tech Data Science and Engineering — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	DS-010	Internship		2
2	MTP	DS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 33 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 178: Data Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-507	Probabilistic Robotics	3
2	AR-509	Deep Learning for Robotics	4
3	AR-516	Introduction to Blockchain and Web3	3
4	AR-517	Introduction to Cyber security	3
5	BE-303	Applied Biostatistics	4
6	BE-304	Bioinformatics	4
7	BE-502	Design and Analysis of Bioalgorithms	3
8	BE-505	Computational Biology	3
9	CS-212	Design and Analysis of Algorithms	4
10	CS-214	Computer Organization	4

Data Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	CS-241	Introduction to Cryptography	3
12	CS-302	Paradigms of Programming	4
13	CS-303	Software Engineering	3
14	CS-304	Formal Languages and Automata Theory	3
15	CS-306	Operating Systems	3
16	CS-307	Systems Practicum	2
17	CS-308	Large Application Practicum	2
18	CS-308P	Large Application Practicum	2
19	CS-309	Information and Database Systems	4
20	CS-310	Introduction to Communication and Distributed Processes	4
21	CS-313	Computer Networks	4
22	CS-347	Software Engineering	4
23	CS-350	Computer Graphics	3
24	CS-401	Architecture of High Performance Computers	3
25	CS-402	Compilers	2
26	CS-403	Algorithm Design and Analysis I	3
27	CS-405	Verification of Reactive Systems	3
28	CS-406	Computer Networks	4
29	CS-451	Computer Graphics and Game Design	3
30	CS-456	Distributed Databases	3
31	CS-501	Access Networks	3
32	CS-502	Compiler Design	4
33	CS-506	Cognitive Modeling	3
34	CS-507	Computer Architecture	4
35	CS-508	Introduction to Network Security	3
36	CS-510	Randomized Algorithms	4
37	CS-511	Applied Probability	2
38	CS-512	Matrix Theory	2
39	CS-513	Discrete Mathematics	4
40	CS-514	Advance Data Structure and Algorithm (CS-580)	4
41	CS-515	Advanced Computer Science Practicum	3
42	CS-516	Exploratory Project	3
43	CS-520	Introduction to Quantum Computing	3
44	CS-521	Introduction to Post Quantum Security	3
45	CS-522	Distributed Algorithms	3
46	CS-524	Computational Complexity Theory	4
47	CS-525	Computational Foundations of Cryptography	3
48	CS-526	Mathematical Foundation of Cryptography	3
49	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3

Data Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	CS-541	IoT systems and the Cloud	3
51	CS-542	Design Patterns for Scalable Systems	3
52	CS-544	Formal Concept Analysis: Theory and Practice	3
53	CS-545	Software Design Patterns	3
54	CS-546	Design of Concurrent Software	3
55	CS-547	Network Management Systems	3
56	CS-548	Cloud Networking	3
57	CS-549	Performance Analysis of Computer Networks	3
58	CS-550	Computer Graphics and Geometric Design	3
59	CS-551	Human Computer Interaction	3
60	CS-560	Text Retrieval and Mining	3
61	CS-561	Map Reduce and Big Data	3
62	CS-563	Scalable Data Science	4
63	CS-570	Algorithms for Curves and Surface	3
64	CS-571	Programming Practicum	3
65	CS-591	Special Topics in Knowledge Discovery and Data	1
66	CS-592	Selected Topics in Artificial Intelligence	2
67	CS-601	Probability and Random Processes	4
68	CS-603	Managerial Decision Making	3
69	CS-606	Cognitive Modeling	3
70	CS-609	Speech processing	3
71	CS-611	Program Analysis	4
72	CS-630	Speech Technology	4
73	CS-660	Data Mining for Decision Making	4
74	CS-661	Knowledge Representation and Reasoning	3
75	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
76	CS-670	Kernel Methods for Pattern Recognition	4
77	CS-671	Deep Learning and Applications	4
78	CS-672	Advanced Topics in Deep Learning	4
79	CS-673	Advanced Computer Vision	4
80	CS-677	Soft Computing	3
81	CS-683	Generative AI	4
82	CS-685	Natural Language Processing	3
83	CS-686	Large Language Models: Core Concepts to Custom	4
84	CS-693	Compressed Sensing and its applications	3
85	EE-203	Network Theory	3
86	EE-304	Communication Theory	4
87	EE-305	Digital Signal Processing	4
88	EE-511	Computer Vision	4
89	EE-522	Matrix Theory for Engineers	3

Data Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
90	EE-530	Optimization theory	3
91	EE-531	Estimation and Detection Theory	3
92	EE-534	Probability and Random Processes	3
93	EE-541	Tensors: Techniques, Algorithms, Applications for Signal	4
94	EE-543	Vision and Learning Based Control	3
95	EE-574	Biomedical Signal and Image Analysis	3
96	EE-575	Applied Statistics for Data and Signals	4
97	EE-608	Digital Image Processing	4
98	EE-620	Advanced Digital Signal Processing	3
99	EP-301	Engineering Mathematics - 2	4
100	MA-530	Graph Theory	4
101	MA-605	Statistical Data Analysis	3
102	ME-522	High-Performance Scientific Computing	3
103	ME-526	Programming Paradigm for Open-Source Software	1
104	PH-302	Introduction to Statistical Mechanics	3
105	PH-522	Statistical Mechanics	4
<i>Compulsory project components — counted as DE, not selectable</i>			
106	DS-498P	Major Technical Project - I (compulsory; counted as DE)	4
107	DS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Electrical Engineering

Table 179: Electrical Engineering — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	52
	Discipline Electives	20
Electives	Free Electives	17
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		161

Table 180: B. Tech Electrical Engineering — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-210	Digital Systems Design	3-0-0-3	3
3	EE-210P	Digital Systems Design Lab	0-0-2-1	1
4	EE-260	Signals and Systems	2.5-0.5-0-3	3
5	EE-261	Electrical Systems Around Us	3-0-0-3	3
6	EE-261P	Electrical Systems Around Us Lab	0-0-3-2	2
7	EE-311	Device Electronics	3-0-0-3	3
8	EE-201	Electromechanics	2.5-0.5-0-3	3
9	EE-201P	Electromechanics Lab	0-0-2-1	1
10	EE-205	Electromagnetics & Wave Propagation	3-0-0-3	3
11	EE-211	Analog Circuit Design	2-0-2-3	3
12	EE-223P	Reverse Engineering	1-0-0-1	1
13	EE-304	Communication Systems	3-0-1-4	4
14	EE-231	Measurement and Instrumentation	2-0-2-3	3
15	EE-302	Control Systems	3-0-2-4	4
16	EE-303	Power System	3-1-0-4	4
17	EE-314	Digital Signal Processing	3-0-2-4	4
18	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
Total				52

Table 181: B. Tech Electrical Engineering — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-110	Japanese for Beginners	3	
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 182: B. Tech Electrical Engineering — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket (choose 1)	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
8		IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	
9		IC-241	Material Science for Engineers	3-0-0-3	
10		IC-253	Data Structures and Algorithms	2-0-2-3	
11	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 183: B. Tech Electrical Engineering — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-203	Network Theory	3-0-0-3	3
4	DC	EE-210	Digital Systems Design	3-0-0-3	3
5	DC	EE-210P	Digital Systems Design Lab	0-0-2-1	1
6	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
7	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
8	DC	EE-261P	Electrical Systems Around Us Lab	0-0-3-2	2
9	DC	EE-311	Device Electronics	3-0-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 184: B. Tech Electrical Engineering — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
3	DC	EE-201P	Electromechanics Lab	0-0-2-1	1
4	DC	EE-205	Electromagnetics & Wave Propagation	3-0-0-3	3
5	DC	EE-211	Analog Circuit Design	2-0-2-3	3
6	DC	EE-223P	Reverse Engineering	1-0-0-1	1
7	DC	EE-304	Communication Systems	3-0-1-4	4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 185: B. Tech Electrical Engineering — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	EE-302	Control Systems	3-0-2-4	4
3	DC	EE-303	Power System	3-1-0-4	4
4	DC	EE-314	Digital Signal Processing	3-0-2-4	4
5	DC	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 186: B. Tech Electrical Engineering — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 187: B. Tech Electrical Engineering — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	EE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 188: B. Tech Electrical Engineering — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	EE-010	Internship		2
2	MTP	EE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 20 credits of **Discipline Electives** and 17 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 189: Electrical Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BE-302	Bioelectric Systems Modeling	4
2	BE-304	Bioinformatics	4
3	CS-549	Performance Analysis of Computer Networks	3
4	CS-671	Deep Learning and Applications	4
5	EE-500	Network Control System	3
6	EE-501	Power System Operation and Control	3
7	EE-502	Analog System Design Laboratory	3
8	EE-503	Advance Communication Theory	3
9	EE-504	Switch Mode Power Conversion	3
10	EE-505	Materials for Semiconductor Industry	3
11	EE-506	Solar Photovoltaic Energy Systems	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	EE-507	Transmission Lines and Basic Microwave Engineering	4
13	EE-508	Fundamentals of Electric Drives	3
14	EE-508P	Practicum on fundamentals of Electrical Drives	2
15	EE-509	Linear Dynamical Systems	3
16	EE-511	Computer Vision	4
17	EE-512	CMOS Analog IC Design	4
18	EE-513	Special Electrical Machines	3
19	EE-514	Robust Control Systems	3
20	EE-515	Nonlinear Stability and Control	4
21	EE-516	Biomedical Systems	4
22	EE-517	Wireless Communications and Networking	3
23	EE-518	Information Theory	3
24	EE-519	CMOS Digital IC design Practicum	2
25	EE-519P	CMOS Digital IC Design Practicum	2
26	EE-520	Microelectronic Devices and Modelling	3
27	EE-522	Matrix Theory for Engineers	3
28	EE-523	Digital VLSI Architecture Design	4
29	EE-524	Digital MOS LSI Circuits	3
30	EE-526	Power Semiconductor Devices	3
31	EE-527	Analysis and Design of Power Electronic Converters	3
32	EE-527P	Practicum on Analysis and Design of Power Electronic Converters	2
33	EE-528	Modeling and Analysis of Electric Machines	3
34	EE-529	Embedded Systems	4
35	EE-530	Optimization theory	3
36	EE-531	Estimation and Detection Theory	3
37	EE-532	Supervised Research Exposure	3
38	EE-533	Access Network	4
39	EE-534	Probability and Random Processes	3
40	EE-535	Communication and Signal Processing Systems Design	3
41	EE-536	IoT Systems	3
42	EE-537	Power Quality Problems and Mitigation Techniques in Microgrids	3
43	EE-538	Data-driven Control	3
44	EE-540	Wide Band Gap Devices in Power Electronics Applications	3
45	EE-541	Sensors: Techniques, Algorithms, Applications for Signal	4
46	EE-542	Modelling, Simulation and Control of Hybrid Electric Vehicle	3
47	EE-543	Vision and Learning Based Control	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
48	EE-550	Optimal Control and Estimation	3
49	EE-551	Applied Photonics for Scientists and Engineers	3
50	EE-552	Power and Energy Systems	4
51	EE-553	Foundations of Intelligent Communication Systems-I	4
52	EE-554	Low Power VLSI Design	3
53	EE-555	Intelligent Control System	3
54	EE-556	Nuclear Reactor Control	1
55	EE-557	Adaptive Control	3
56	EE-560	Reconfigurable Computing	4
57	EE-570	Advanced Communication Theory	4
58	EE-574	Biomedical Signal and Image Analysis	3
59	EE-575	Applied Statistics for Data and Signals	4
60	EE-580	Network Systems: Modelling and Analysis	3
61	EE-581	Applied Statistics for Data and Signals	4
62	EE-582	Nonlinear Analysis and Control of Power Electronic Converters	3
63	EE-583	Smart Grids	3
64	EE-584	Power System Protection	3
65	EE-585	Theory, Modelling and Characterization of Photo-voltaic Devices	3
66	EE-593	Low Power VLSI Circuits	3
67	EE-597	Intellegent Control System	3
68	EE-601	Solid State Devices	3
69	EE-602	Control System Applications	3
70	EE-603	Renewable Energy and Smart Grid	3
71	EE-604	Practicum on Advanced Electric Drives	3
72	EE-604P	Practicum on Advanced Electric Drives	1
73	EE-605	Information Theory	3
74	EE-606	Introduction to High Voltage Engineering and Dielectric	3
75	EE-607	Optical Communication Systems	3
76	EE-608	Digital Image Processing	4
77	EE-609	Network Information Theory	3
78	EE-611	VLSI Technology	3
79	EE-611P	VLSI Fabrication Practicum	2
80	EE-612	OFDM For Optical Communications	3
81	EE-613	Wireless Communication	3
82	EE-614	Optical Communication Systems	3
83	EE-615	Nanoelectronics and Micronanofabrication	3
84	EE-616	Microwave engineering	2
85	EE-618	Industrial Process Control	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
86	EE-619	Mixed Signal VLSI Design	4
87	EE-620	Advanced Digital Signal Processing	3
88	EE-621	Radiating systems	4
89	EE-622	Microwave Integrated Circuits	3
90	EE-623P	Practicum on Digital Control of Power Electronics and Drives	3
91	EE-626P	Post Graduate Project - I	10
92	EE-630	HVDC Transmission and Flexible AC Transmission Systems	3
93	EE-641	Advance Wireless Technologies	4
94	EE-642	Research Study	3
95	EE-677	Analog Circuit Design	3
96	EE-678	Digital Circuit Design	3
97	EE-691	Advanced Topics in Dielectric Breakdown	1
98	EE-694	Photonics Computing	3
<i>Compulsory project components — counted as DE, not selectable</i>			
99	EE-498P	Major Technical Project - I (compulsory; counted as DE)	4
100	EE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Mechanical Engineering

Table 190: Mechanical Engineering — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	50
	Discipline Electives	16
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 191: B. Tech Mechanical Engineering — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-261	Electrical Systems Around Us	3-0-0-3	3
2	IC-241	Material Science for Engineers	3-0-0-3	3
3	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	ME-210	Fluid Mechanics	2.5-0.5-0-3	3
5	ME-212	Product Manufacturing Technologies	2-0-3-3	3
6	ME-213	Engineering Thermodynamics	3-1-0-4	4
7	ME-100	Reverse Engineering	0-0-2-1	1
8	ME-205	Machine Drawing	1-0-4-3	3
9	ME-210P	Fluid Mechanics Lab	0-0-2-1	1
10	ME-215	Manufacturing Engineering 1	3-0-0-3	3
11	ME-303	Heat Transfer	2.5-0.5-0-3	3
12	ME-307	Energy Conversion Devices	3-0-0-3	3
13	ME-303P	Heat Transfer Lab	0-0-3-1	1
14	ME-305	Design of Machine Elements	3-1-0-4	4
15	ME-309	Theory of Machines	3-1-0-4	4
16	ME-310	System Dynamics and Control	3-0-0-3	3
17	ME-311P	Design Lab 1	0-0-2-1	1
18	ME-315	Manufacturing Engineering 2	3-0-0-3	3
19	ME-312P	Design Lab 2	0-0-2-1	1
Total				50

Table 192: B. Tech Mechanical Engineering — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-110	Japanese for Beginners	3	
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 193: B. Tech Mechanical Engineering — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 194: B. Tech Mechanical Engineering — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
4	DC	IC-241	Material Science for Engineers	3-0-0-3	3
5	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
6	DC	ME-210	Fluid Mechanics	2.5-0.5-0-3	3
7	DC	ME-212	Product Manufacturing Technologies	2-0-3-3	3
8	DC	ME-213	Engineering Thermodynamics	3-1-0-4	4
Total					25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 195: B. Tech Mechanical Engineering — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	ME-100	Reverse Engineering	0-0-2-1	1
3	DC	ME-205	Machine Drawing	1-0-4-3	3
4	DC	ME-210P	Fluid Mechanics Lab	0-0-2-1	1
5	DC	ME-215	Manufacturing Engineering 1	3-0-0-3	3
6	DC	ME-303	Heat Transfer	2.5-0.5-0-3	3
7	DC	ME-307	Energy Conversion Devices	3-0-0-3	3
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 196: B. Tech Mechanical Engineering — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-303P	Heat Transfer Lab	0-0-3-1	1
2	DC	ME-305	Design of Machine Elements	3-1-0-4	4
3	DC	ME-309	Theory of Machines	3-1-0-4	4
4	DC	ME-310	System Dynamics and Control	3-0-0-3	3
5	DC	ME-311P	Design Lab 1	0-0-2-1	1
6	DC	ME-315	Manufacturing Engineering 2	3-0-0-3	3
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 197: B. Tech Mechanical Engineering — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-312P	Design Lab 2	0-0-2-1	1
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					5

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 198: B. Tech Mechanical Engineering — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	ME-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 199: B. Tech Mechanical Engineering — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	ME-010	Internship		2
2	MTP	ME-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 16 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 200: Mechanical Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	ME-451	Refrigeration and Air Conditioning	3
2	ME-452	Robot Kinematics, Dynamics and Control	4
3	ME-501	Nanomanufacturing	3
4	ME-502	Functional Materials	3
5	ME-503	Micro/Nano Scale heat transfer	3
6	ME-504	Numerical Methods for Engineering Computation	3
7	ME-505	Applied Finite Element Method	3
8	ME-506	Fundamentals of Fracture Mechanics	3
9	ME-507	Micro and Nanoscale Fluid Mechanics	3
10	ME-508	Fundamentals of project management	3

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	ME-509	Nano-manufacturing	3
12	ME-510	Advanced Manufacturing Processes	3
13	ME-511	Manufacturing of composites	3
14	ME-513	Finite Element Methods in Engineering	4
15	ME-514	Fundamentals of Multiphase Flow	3
16	ME-516	Polymer Technology for Engineers	3
17	ME-517	Advanced Analytical Techniques for Engineers	4
18	ME-518	Conduction and Radiation	3
19	ME-519	Technical communication for Engineers	2
20	ME-520	Microwave based Manufacturing Processes	3
21	ME-521P	Laboratory course on Vehicle Design and Control	1
22	ME-522	High-Performance Scientific Computing	3
23	ME-523	Product Design	3
24	ME-524	Additive Manufacturing	3
25	ME-526	Programming Paradigm for Open-Source Software	1
26	ME-528	Thermal Management of Electronics	3
27	ME-530	Continuum Mechanics	3
28	ME-595	Experiential Learning in Composite Design, Manufacturing, Testing and Recycling	3
29	ME-600	Research Methodology	1
30	ME-601	Advanced Finite Element Methods	3
31	ME-602	Mechanical Vibration	3
32	ME-603	Advanced Fluid Mechanics	3
33	ME-604	Experimental Methods in Thermal Engineering	3
34	ME-605	Air Conditioning and Ventilation	3
35	ME-606	Advanced Solid Mechanics	3
36	ME-607	Materials Science For Failure Analysis	3
37	ME-608	Thin Films And Devices	3
38	ME-609	Functional Materials	3
39	ME-610	Advanced Thermodynamics	4
40	ME-611	Design and Optimization of Thermal Systems	3
41	ME-612	Introduction to Bio-materials	3
42	ME-613	Thermal Radiation	3
43	ME-614	Compressible Flow and Gas Dynamics	3
44	ME-615	Applied Computational Fluid Dynamics	3
45	ME-616	Dielectrics and Related Materials	3
46	ME-617	Mechanics of Composite Materials	3
47	ME-618	Stealth Technology: Infrared Signatures	3
48	ME-619	Experiments in Materials Science	3
49	ME-620	Modeling and Simulation	3
50	ME-621	Aircraft Propulsion	3

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
51	ME-622	Biomechanics of Musculoskeletal System	3
52	ME-625	Introduction to Turbulence and its Modeling	3
53	ME-626	Acoustics	3
54	ME-627	Mesh Independent Computational Techniques	3
55	ME-628	Impact Mechanics	3
56	ME-630	Machine learning for engineers	3
57	ME-631	Heat Transfer and Fluid Flow in Energy Systems	3
58	ME-632	Mechanics for Energy Systems	3
59	ME-633	Design of Energy Systems	3
60	ME-634	Thermodynamics for Energy Systems	3
61	ME-635	Manufacturing for Energy Systems	3
62	ME-636	Combustion Technology	3
63	ME-637	Wind Power Plant	3
64	ME-638	Solar Thermal Power Plant	3
65	ME-639	Thermal Power Plant Engineering	3
66	ME-640	Solar Power Utilization	3
67	ME-641	Finite Element Method	3
68	MT-201	Physics of Solids	3
69	MT-202	Applied Quantum Mechanics	3
70	MT-203	Materials Synthesis & Characterization	4
71	MT-204	Thermodynamics and Kinetics of Materials	3
72	MT-205	Functional Properties of Materials	4
73	MT-206	Extraction and Materials Processing	4
74	MT-301	Phase Transformations	3
75	MT-302	Transport Phenomena	3
76	MT-303	Computational Materials Science	4
77	MT-304	Mechanical Behavior of Materials	4
78	MT-501	Energy Conversion and Storage Technologies	3
79	MT-502	Recycling and Circular Economy	3
80	MT-503	Semiconductor Materials and Devices	3
81	MT-504	Powder metallurgical processing of Materials	3
82	MT-505	Thin Film Technology	3
83	MT-506	Biomaterials	3
84	MT-507	Modelling and Simulations in Material Science	3
85	MT-508	Iron and Steel Making	3
86	MT-509	Hydrogen Energy	3
87	MT-510	Colloids and Interfaces	3
88	MT-511	Sensor Materials and Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
89	ME-498P	Major Technical Project - I (compulsory; counted as DE)	4

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
90	ME-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Civil Engineering

Table 201: Civil Engineering — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	32
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	49
	Discipline Electives	21
Electives	Free Electives	25
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 202: B. Tech Civil Engineering — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CE-202	Introduction to Civil Engineering Profession	1-0-0-1	1
2	CE-203	Construction Materials	3-0-0-3	3
3	CE-203P	Building (Constructions) Materials Lab	0-0-2-1	1
4	CE-252	Geology and Geomorphology	2-0-2-3	3
5	CE-310	Strength of Materials and Structures	3-0-0-3	3
6	CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
7	CE-201	Surveying Traditional and Digital	2-0-4-4	4
8	CE-251	Hydraulics Engineering	3-0-0-3	3
9	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
10	CE-311	Geotechnical Engineering I	3-0-0-3	3
11	CE-311P	Geotechnical Engineering Lab	0-0-2-1	1
12	CE-404	Analysis of Structures	3-0-0-3	3
13	CE-303	Water Resources Engineering	3-0-0-3	3
14	CE-351	Design of Reinforced Concrete Structures	2-1-0-3	3
15	CE-352	Transportation Engineering	3-0-0-3	3
16	CE-352P	Transportation Engineering Lab	0-0-2-1	1
17	CE-353P	Civil Engineering Drawing	0-0-2-1	1
18	CE-402	Geotechnical Engineering II	2-1-0-3	3
19	CE-305P	Environmental Engineering Lab	0-0-2-1	1
20	CE-356	Reverse Engineering	1-0-0-1	1
21	CE-401	Design of Steel Structures	2-1-0-3	3
22	CE-403	Water and Wastewater Engineering	3-0-0-3	3
Total				49

Table 203: B. Tech Civil Engineering — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-110	Japanese for Beginners	3	
7		HS-151	Introduction to English Literature	3-0-0-3	
8		HS-256	Introduction to Spanish	3	
9	IKS + HSS	IC-181	Introduction to Consciousness and Holistic Wellbeing	2-0-2-3	3
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 204: B. Tech Civil Engineering — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
8	DC	CE-202	Introduction to Civil Engineering Profession	1-0-0-1	1
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 205: B. Tech Civil Engineering — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	CE-203	Construction Materials	3-0-0-3	3
3	DC	CE-203P	Building (Constructions) Materials Lab	0-0-2-1	1
4	DC	CE-252	Geology and Geomorphology	2-0-2-3	3
5	DC	CE-310	Strength of Materials and Structures	3-0-0-3	3
6	DC	CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 206: B. Tech Civil Engineering — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	CE-201	Surveying Traditional and Digital	2-0-4-4	4
3	DC	CE-251	Hydraulics Engineering	3-0-0-3	3
4	DC	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
5	DC	CE-311	Geotechnical Engineering I	3-0-0-3	3
6	DC	CE-311P	Geotechnical Engineering Lab	0-0-2-1	1
7	DC	CE-404	Analysis of Structures	3-0-0-3	3
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 207: B. Tech Civil Engineering — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-303	Water Resources Engineering	3-0-0-3	3
2	DC	CE-351	Design of Reinforced Concrete Structures	2-1-0-3	3
3	DC	CE-352	Transportation Engineering	3-0-0-3	3
4	DC	CE-352P	Transportation Engineering Lab	0-0-2-1	1
5	DC	CE-353P	Civil Engineering Drawing	0-0-2-1	1
6	DC	CE-402	Geotechnical Engineering II	2-1-0-3	3
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 208: B. Tech Civil Engineering — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-305P	Environmental Engineering Lab	0-0-2-1	1
2	DC	CE-356	Reverse Engineering	1-0-0-1	1
3	DC	CE-401	Design of Steel Structures	2-1-0-3	3
4	DC	CE-403	Water and Wastewater Engineering	3-0-0-3	3
5	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 209: B. Tech Civil Engineering — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	CE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 210: B. Tech Civil Engineering — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	CE-010	Internship		2
2	MTP	CE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 21 credits of **Discipline Electives** and 25 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 211: Civil Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CD-591	Special Topics in Disaster Resilient Hill Roads	1
2	CE-304	Hydraulics Engineering Lab	3
3	CE-305	Environment and Earth Science Lab	3
4	CE-306	Environmental Engineering Lab	3
5	CE-353	Civil Engineering Drawing	3
6	CE-354	Building and Pavement Materials Laboratory	3

Civil Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
7	CE-355	Hydrology	3
8	CE-405	Water and Wastewater Engineering	3
9	CE-451	Irrigation Engineering and Hydraulic Structures	3
10	CE-452	Open Channel Flow	3
11	CE-501	Remote Sensing	3
12	CE-502	Groundwater Flow and Contaminant Transport	3
13	CE-503	Fundamentals of Project Management	3
14	CE-504	Slope Stability and Retaining Structures	3
15	CE-505	Engineering of Ground Modification	3
16	CE-506	Analysis of Indeterminate Structures	3
17	CE-507	Advanced Concrete Science	3
18	CE-508	Photogeology and Photogrammetry	3
19	CE-509	Bridge Engineering	3
20	CE-510	Modelling and Simulation in Water Resources Engineering	3
21	CE-511	Structural Dynamics with Application to Earthquake Engineering	3
22	CE-512	Advanced Soil Mechanics	3
23	CE-513	Advanced Foundation Engineering	3
24	CE-514	Rock Mechanics	3
25	CE-515	Environmental Impact Assessment	3
26	CE-516	Uncertainty Analysis in Civil Engineering	3
27	CE-517	Hydroinformatics	4
28	CE-518	Structural Reliability and Risk Assessment	3
29	CE-519	Chemistry of Natural Waters	3
30	CE-520	Environmental Reaction Modeling	3
31	CE-521	Ecology and Environment Microbiology	4
32	CE-522	Matrix Methods for Structural Analysis	3
33	CE-523	Building Science	3
34	CE-524	Applied Hydroclimatology	3
35	CE-524P	Computational Hydroclimatology Lab	1
36	CE-526	Critical Zone Science	3
37	CE-551	Geosynthetics and their applications	3
38	CE-552	Concrete Technology	3
39	CE-554	Prestressed Concrete Structures	3
40	CE-555	Advanced Design of Structures	3
41	CE-556	Structural Engineering Laboratory	3
42	CE-556P	Structural Engineering laboratory	2
43	CE-557	Solid Mechanics in Structural Engineering	3
44	CE-558	Air pollution and its mitigation	3
45	CE-559	Biological Wastewater Treatment	3

Civil Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
46	CE-560	Soil Dynamics	3
47	CE-561	The science of climate change	3
48	CE-562	Numerical Methods for Engineering Computation	3
49	CE-563	Finite Element Methods in Engineering	4
50	CE-564	Fundamentals of Fracture Mechanics	3
51	CE-567	Highway Safety and Modeling	3
52	CE-568P	Experimental Structural Dynamics	1
53	CE-586	Mini Project	3
54	CE-586P	Mini Project	3
55	CE-587	Industrial / Academic Internship	3
56	CE-591	Special Topics in Civil and Mechanical Engineering	3
57	CE-600	Research Methodology for Civil Engineering	1
58	CE-601	Geo-Informatics	3
59	CE-602	Blast Engineering	3
60	CE-604	Theory of Plasticity	3
61	CE-605	Engineering Seismology and Seismic Hazard Assessment	3
62	CE-606	Constitutive Modeling of Frictional Material	3
63	CE-610	Analysis and Design for Earthquake Resistant Structures	3
64	CE-611	Structural Health Monitoring	3
65	CE-612	Theory of Plates and Shells	3
66	CE-613	Mechanics of Unsaturated Soils	3
67	CE-617	Mechanics of Composite Materials	3
68	CE-688	Post Graduate Project - I	3
69	CE-689	Post Graduate Project - II	3
<i>Compulsory project components — counted as DE, not selectable</i>			
70	CE-498P	Major Technical Project - I (compulsory; counted as DE)	4
71	CE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Engineering Physics

Table 212: Engineering Physics — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	32
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	37
	Discipline Electives	33
Electives	Free Electives	25
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 213: B. Tech Engineering Physics — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EP-301	Engineering Mathematics 2	3-1-0-4	4
2	EP-321	Foundations of Electrodynamics	3-0-0-3	3
3	PH-301	Quantum Mechanics and Applications	2.5-0.5-0-3	3
4	EE-223P	Reverse Engineering	1-0-0-1	1
5	EP-403	Physics of Atoms and Molecules	3-0-0-3	3
6	PH-302	Introduction to Statistical Mechanics	2.5-0.5-0-3	3
7	PH-501	Solid State Physics	3	3
8	EE-311	Device Electronics for Integrated Circuits	3-0-0-3	3
9	EP-302	Computational Methods for Engineering	2-0-2-3	3
10	EP-402P	Engineering Physics Practicum	1-0-5-4	4
11	EP-401P	Engineering of Instrumentation	1-0-5-4	4
12	PH-502	Photonics	3-0-0-3	3
Total				37

Table 214: B. Tech Engineering Physics — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-110	Japanese for Beginners	3	
7		HS-151	Introduction to English Literature	3-0-0-3	
8		HS-256	Introduction to Spanish	3	
9	IKS + HSS	IC-181	Introduction to Consciousness and Holistic Wellbeing	2-0-2-3	3
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 215: B. Tech Engineering Physics — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
Total					20

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 216: B. Tech Engineering Physics — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	EP-301	Engineering Mathematics 2	3-1-0-4	4
3	DC	EP-321	Foundations of Electrodynamics	3-0-0-3	3
4	DC	PH-301	Quantum Mechanics and Applications	2.5-0.5-0-3	3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 217: B. Tech Engineering Physics — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	EE-223P	Reverse Engineering	1-0-0-1	1
3	DC	EP-403	Physics of Atoms and Molecules	3-0-0-3	3
4	DC	PH-302	Introduction to Statistical Mechanics	2.5-0.5-0-3	3
5	DC	PH-501	Solid State Physics	3	3
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 218: B. Tech Engineering Physics — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-311	Device Electronics for Integrated Circuits	3-0-0-3	3
2	DC	EP-302	Computational Methods for Engineering	2-0-2-3	3
3	DC	EP-402P	Engineering Physics Practicum	1-0-5-4	4
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 219: B. Tech Engineering Physics — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EP-401P	Engineering of Instrumentation	1-0-5-4	4
2	DC	PH-502	Photonics	3-0-0-3	3
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 220: B. Tech Engineering Physics — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	EP-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 221: B. Tech Engineering Physics — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	EP-010	Internship		2
2	MTP	EP-499P	Major Technical Project - II	0-0-6-4	4
				Total	6

*Aim for **20–25 credits this semester** — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

*Beyond the core courses above, students choose 33 credits of **Discipline Electives** and 25 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 222: Engineering Physics — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-208	Mathematical Foundations of Computer Science	4
2	CS-241	Introduction to Cryptography	3
3	CS-304	Formal Languages and Automata Theory	3
4	CS-309	Information and Database Systems	4
5	CS-511	Applied Probability	2
6	CS-671	Deep Learning and Applications	4
7	CS-672	Advanced Topics in Deep Learning	4
8	DS-201	Data handling and visualization	3
9	DS-301	Mathematical Foundations of Data Science	4
10	DS-401	Optimization for data science	3
11	DS-404	Information Security and Privacy	3
12	DS-416	Fundamentals of Machine Learning (CS-669)	4
13	EE-203	Network Theory	3
14	EE-211	Analog Circuit Design	3
15	EE-307	Theory of Measurement	3
16	EE-511	Computer Vision	4
17	EE-512	CMOS Analog IC Design	4
18	EE-519P	CMOS Digital IC Design Practicum	2
19	EE-520	Microelectronic Devices and Modelling	3
20	EE-524	Digital MOS LSI Circuits	3
21	EE-534	Probability and Random Processes	3
22	EE-551	Applied Photonics for Scientists and Engineers	3
23	EE-593	Low Power VLSI Circuits	3
24	EE-611	VLSI Technology	3
25	EE-614	Optical Communication Systems	3
26	EE-621	Radiating systems	4
27	EN-502	Emerging Energy Sources	3

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
28	EP-201	Reverse Engineering	1
29	EP-401	Engineering of Instrumentation	3
30	EP-402	Engineering Physics Practicum	3
31	EP-502	Informatics for Materials Design	3
32	IC-253	Programming and Data Structures	3
33	MA-210	Real and Complex Analysis	3
34	MA-511	Real Analysis	4
35	MA-512	Linear Algebra	4
36	MA-513	Ordinary Differential Equation	4
37	MA-516	Topology	4
38	MA-521	Topology and Functional Analysis	4
39	MA-522	Partial Differential Equations	4
40	MA-560	Nonlinear Dynamics and Chaos	3
41	ME-210	Fluid Mechanics	3
42	ME-303	Heat Transfer	3
43	ME-307	Energy Conversion Devices	3
44	ME-308	Manufacturing Engineering	3
45	ME-503	Micro/Nano Scale heat transfer	3
46	ME-509	Nano-manufacturing	3
47	ME-510	Advanced Manufacturing Processes	3
48	ME-603	Advanced Fluid Mechanics	3
49	ME-615	Applied Computational Fluid Dynamics	3
50	PH-503	Laser and Applications	3
51	PH-504	Organic Optoelectronics	3
52	PH-507	X-rays as a Probe to Study Material Properties	3
53	PH-508	Magnetism and Magnetic Materials	3
54	PH-511	Mathematical Physics	4
55	PH-512	Classical Mechanics	4
56	PH-513	Quantum Mechanics	3
57	PH-514	Electronics	3
58	PH-515P	Physics Laboratory	3
59	PH-516	Research Project I	2
60	PH-521	Electromagnetic Theory	4
61	PH-525P	Electronics Lab	3
62	PH-528	Introduction to General Relativity	3
63	PH-530	Cosmology-I	3
64	PH-550	Introduction to Quantum Optics	3
65	PH-579	Quantum Computation and Information	3
66	PH-600	Research Methodology	1
67	PH-601	Mesoscopic Physics and Quantum Transport	3

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
68	PH-603	Advanced Condensed Matter Physics	3
69	PH-604	Optical Properties of Solids	3
70	PH-605	Superconductivity	3
71	PH-606	Quantum Field Theory	3
72	PH-607	Physics of Ultracold Quantum Gases	3
73	PH-608	Computer assisted quantum mechanics	3
74	PH-609	Theory of quantum collision and spectroscopy	3
75	PH-612	Nuclear and Particle Physics	3
76	PH-613	Special Topics in Quantum Mechanics	3
77	PH-615P	Mini Thesis I	3
78	PH-621	Computational Methods for Physicists	4
79	PH-625	Data Analysis in Particle Physics	4
80	PH-626	Elementary Theoretical Particle Physics	3
81	PH-627	Topological Quantum Matter	3
82	PH-701	Introduction to Molecular Simulations	4
83	PH-706	Introduction to Stochastic Problems in Physics	3
84	QS-501P	Experiments in Quantum Optics	3
85	QT-301	Survey of Quantum Technologies and Applications	3
86	QT-406	Introduction to Quantum communication	3
87	QT-407	Introduction to Quantum Sensing	3
88	QT-509	Engineering Foundations of Quantum Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
89	EP-498P	Major Technical Project - I (compulsory; counted as DE)	4
90	EP-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Bioengineering

Table 223: Bioengineering — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	32
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	42
	Discipline Electives	28
Electives	Free Electives	25
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 224: B. Tech Bioengineering — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	BE-101P	Reverse Engineering	0-0-2-1	1
2	BE-201	Cell Biology	3-0-2-4	4
3	BE-202	Biochemistry and Molecular Biology	3-0-2-4	4
4	BE-308	Introduction to Biomanufacturing	3-0-2-4	4
5	BE-309	Biosensing & Bioinstrumentation	3-0-2-4	4
6	BE-203	Enzymology and Bioprocessing	3-0-2-4	4
7	BE-301	Biomechanics	3-0-2-4	4
8	BE-304	Bioinformatics	3-0-2-4	4
9	BE-303	Applied Biostatistics	3-0-2-4	4
10	BE-305	Bioethics and Regulatory Affairs	1-0-0-1	1
11	BE-306	Fundamentals of Genetic Engineering	3-1-0-4	4
12	BE-310	Biomaterials	3-0-2-4	4
Total				42

Table 225: B. Tech Bioengineering — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-110	Japanese for Beginners	3	
7		HS-151	Introduction to English Literature	3-0-0-3	
8		HS-256	Introduction to Spanish	3	
9	IKS + HSS	IC-181	Introduction to Consciousness and Holistic Wellbeing	2-0-2-3	3
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 226: B. Tech Bioengineering — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
8	DC	BE-101P	Reverse Engineering	0-0-2-1	1
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 227: B. Tech Bioengineering — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	BE-201	Cell Biology	3-0-2-4	4
3	DC	BE-202	Biochemistry and Molecular Biology	3-0-2-4	4
4	DC	BE-308	Introduction to Biomanufacturing	3-0-2-4	4
5	DC	BE-309	Biosensing & Bioinstrumentation	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 228: B. Tech Bioengineering — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	BE-203	Enzymology and Bioprocessing	3-0-2-4	4
3	DC	BE-301	Biomechanics	3-0-2-4	4
4	DC	BE-304	Bioinformatics	3-0-2-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 229: B. Tech Bioengineering — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	BE-303	Applied Biostatistics	3-0-2-4	4
2	DC	BE-305	Bioethics and Regulatory Affairs	1-0-0-1	1
3	DC	BE-306	Fundamentals of Genetic Engineering	3-1-0-4	4
4	DC	BE-310	Biomaterials	3-0-2-4	4
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 230: B. Tech Bioengineering — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 231: B. Tech Bioengineering — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	BE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 232: B. Tech Bioengineering — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	BE-010	Internship		2
2	MTP	BE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 28 credits of **Discipline Electives** and 25 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 233: Bioengineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-506	Cognitive Robotics	3
2	BE-101	Reverse Engineering for Bioengineers	3
3	BE-302	Bioelectric Systems Modeling	4
4	BE-307	Reverse Engineering for Bioengineers	3
5	BE-401	Bioengineering Mini Project, Term Paper and Seminar	4
6	BE-501	Anatomy and Physiology	3
7	BE-502	Design and Analysis of Bioalgorithms	3
8	BE-503	Biosensing & Bioinstrumentation	4
9	BE-504	Biomaterials	4
10	BE-505	Computational Biology	3
11	BE-506	Biological Modelling and Simulation	3
12	BE-507	Environmental Impact Assessment	3
13	BE-598	Thesis Project Part I	3
14	BE-599	Thesis Project Part II	3
15	BE-698	Thesis Project Part I	3
16	BE-699	Thesis Project Part II	17
17	BY-501	DNA Nanotechnology	3
18	BY-502	Biophysics and Protein Engineering	3
19	BY-503	Cellular Fuel and Cellular Communication	3

Bioengineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
20	BY-504	Metabolic Systems Biology	4
21	BY-506	Advanced Immunology	3
22	BY-507	Genetic Engineering	4
23	BY-508	Practical Metabolomics	3
24	BY-509	Practical OMICs	3
25	BY-510	Advanced Cell Biology	3
26	BY-511	Cell Physiology in health and disease	3
27	BY-512	Quantitative and Computational Biology	3
28	BY-514	Analytical Biotechniques	3
29	BY-515	Molecular Biotechnology	3
30	BY-516	Introduction to OMICS and Systems Analysis	3
31	BY-517	Proteomics	3
32	BY-518	Disease Biology	3
33	BY-519	Protein Sciences in Therapeutics	3
34	BY-520	Cell Biology and Physiology Lab	3
35	BY-520P	Cell Biology and Physiology	1
36	BY-521	Computational Biology and Cellular Bioprocess Technology Lab	3
37	BY-521P	Computational Biology and Cellular Bioprocess Technology Lab	1
38	BY-522	Analytical and Molecular Biotechnology Lab	3
39	BY-523	Systems Biology Lab	3
40	BY-524	Medical and NanoBiotechnology Lab	3
41	BY-527	Gene silencing and genome editing: principles and applications	3
42	BY-528	Sensory Biology	3
43	BY-529	Mechanobiology of the Cell (MBoC)	3
44	BY-531	Quantitative Biology and Data Analytics	3
45	BY-532	Immunotechnology	3
46	BY-535P	Analytical Biotechniques Lab	1
47	BY-536P	Immunotechnology Lab	1
48	BY-537	Computational Biology-1	4
49	BY-606	Bioinformatics Applications for Systems	3
50	BY-613	Metagenomics and Next Generation Sequencing Technologies	3
51	CE-521	Ecology and Environment Microbiology	4
52	CE-559	Biological Wastewater Treatment	3
53	CS-309	Information and Database Systems	4
54	CS-606	Cognitive Modeling	3
55	CS-671	Deep Learning and Applications	4
56	CY-344	Food Chemistry: Processing, Preservation and Storage	3

Bioengineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
57	CY-550	Bioinspired Materials	3
58	CY-642	Molecular- and Bio-electronics	3
59	CY-643	Advanced Analytical Techniques	3
60	CY-644	Bioinorganic chemistry	3
61	CY-670	Fluorescence spectroscopy, microscopy and applications	3
62	DS-303	Statistical Foundations of Data Science	3
63	DS-314	Probability Theory and Statistics	4
64	EE-305	Digital Signal Processing	4
65	EE-314	Digital Signal Processing	4
66	EE-516	Biomedical Systems	4
67	EE-574	Biomedical Signal and Image Analysis	3
68	EE-608	Digital Image Processing	4
69	MA-621	Modeling Population Dynamics	3
70	MA-650	Mathematical Models for Infectious Diseases	3
71	ME-303	Heat Transfer	3
72	ME-308	Manufacturing Engineering	3
73	ME-527	Biofluid Dynamics	3
74	ME-612	Introduction to Bio-materials	3
75	ME-622	Biomechanics of Musculoskeletal System	3
76	MT-506	Biomaterials	3
<i>Compulsory project components — counted as DE, not selectable</i>			
77	BE-498P	Major Technical Project - I (compulsory; counted as DE)	4
78	BE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Mathematics and Computing

Table 234: Mathematics and Computing — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	35
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	55
	Discipline Electives	15
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 235: B. Tech Mathematics and Computing — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	MA-120	Introduction to Computing Systems and Databases	4	4
2	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
3	MA-210	Real and Complex Analysis	2.5-0.5-0-3	3
4	MA-211	Ordinary Differential Equations	3-1-0-4	4
5	CS-201	Computer Organization	3-0-0-3	3
6	CS-201P	Computer Organization Laboratory	0-0-2-1	1
7	MA-220	Partial Differential Equations	3-1-0-4	4
8	MA-221	Numerical Analysis	3-1-0-4	4
9	MA-222	Applied Linear Programming	3-1-0-4	4
10	CS-304	Formal Language and Automata Theory	3-0-0-3	3
11	MA-310	Matrix Computation and Lab	3-0-2-4	4
12	MA-311	Mathematical Modelling	3-0-0-3	3
13	MA-312	Design of Algorithms	3-0-2-4	4
14	MA-322	Applied Graph Theory		4
15	MA-321	Numerics of Differential Equation		4
16	MA-323P	Applied Databases Practicum	0-0-3-2	2
Total				55

Table 236: B. Tech Mathematics and Computing — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-110	Japanese for Beginners	3	
7		HS-151	Introduction to English Literature	3-0-0-3	
8		HS-256	Introduction to Spanish	3	
9	IKS + HSS	IC-181	Introduction to Consciousness and Holistic Wellbeing	2-0-2-3	3
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 237: B. Tech Mathematics and Computing — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
8	DC	MA-120	Introduction to Computing Systems and Databases	4	4
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 238: B. Tech Mathematics and Computing — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
4	DC	MA-210	Real and Complex Analysis	2.5-0.5-0-3	3
5	DC	MA-211	Ordinary Differential Equations	3-1-0-4	4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 239: B. Tech Mathematics and Computing — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	CS-201	Computer Organization	3-0-0-3	3
3	DC	CS-201P	Computer Organization Laboratory	0-0-2-1	1
4	DC	MA-220	Partial Differential Equations	3-1-0-4	4
5	DC	MA-221	Numerical Analysis	3-1-0-4	4
6	DC	MA-222	Applied Linear Programming	3-1-0-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 240: B. Tech Mathematics and Computing — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-304	Formal Language and Automata Theory	3-0-0-3	3
2	DC	MA-310	Matrix Computation and Lab	3-0-2-4	4
3	DC	MA-311	Mathematical Modelling	3-0-0-3	3
4	DC	MA-312	Design of Algorithms	3-0-2-4	4
5	DC	MA-322	Applied Graph Theory		4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 241: B. Tech Mathematics and Computing — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	MA-321	Numerics of Differential Equation	3-0-2-4	4
2	DC	MA-323P	Applied Databases Practicum	0-0-3-2	2
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
4	DC	MA-322	Applied Graph Theory	3-0-2-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 242: B. Tech Mathematics and Computing — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	MC-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 243: B. Tech Mathematics and Computing — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	MC-010	Internship		2
2	MTP	MC-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 15 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

This programme groups its Discipline Electives into the named baskets below; the remaining approved electives follow as a general list.

Table 244: Mathematics and Computing — Discipline Elective Basket I — Foundation Module (choose from these)

Sl. No	Course Code	Course Name	Credits
1	MA-251	Abstract Algebra	3
2	MA-252	Functional Analysis	3
3	MA-253	Measure Theory	3
4	MA-254	Topology	3
5	MA-255	Number Theory	3

Table 245: Mathematics and Computing — Discipline Elective Basket II — Advanced Modelling Module (choose from these)

Sl. No	Course Code	Course Name	Credits
1	MA-351	Climate Modelling	3
2	MA-352	Computational Financial Modelling & Lab	3
3	MA-353	Modelling of Infectious Disease	3
4	MA-354	Mathematical Image Processing	3
5	MA-355	Mathematical Control Theory	3
6	MA-356	Modelling and Simulation	3
7	MA-357	Modelling Population Dynamics	3
8	MA-546	Introduction to Quantitative Finance	4
9	MA-570	Data Driven Dynamical Systems	3

Table 246: Mathematics and Computing — Other Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BE-303	Applied Biostatistics	4
2	BE-304	Bioinformatics	4
3	CE-251	Hydraulics Engineering	3
4	CS-241	Introduction to Cryptography	3
5	CS-302	Paradigms of Programming	4
6	CS-306	Operating Systems	3
7	CS-309	Information and Database Systems	4
8	CS-313	Computer Networks	4
9	CS-315	Artificial Intelligence (CS-362, CS-562)	4
10	CS-502	Compiler Design	4
11	CS-514	Advance Data Structure and Algorithm (CS-580)	4
12	CS-563	Scalable Data Science	4
13	CS-606	Cognitive Modeling	3
14	CS-609	Speech processing	3
15	CS-677	Soft Computing	3
16	DS-201	Data handling and visualization	3
17	DS-312	Computing Systems for Data Processing	4
18	DS-416	Fundamentals of Machine Learning (CS-669)	4
19	EE-203	Network Theory	3
20	EE-260	Signals and Systems	3
21	EE-301	Control Systems	3
22	EE-511	Computer Vision	4
23	EE-530	Optimization theory	3
24	EE-608	Digital Image Processing	4
25	MA-001	Preparatory Mathematics - 1	4
26	MA-002	Preparatory Mathematics - 2	4

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
27	MA-101	Calculus I	4
28	MA-102	Linear Algebra	4
29	MA-201	Probability and Statistics	4
30	MA-460	Nonlinear Dynamics and Chaos	3
31	MA-465	Ordinary Differential Equations	3
32	MA-510	Climate Change Analysis	3
33	MA-516	Topology	4
34	MA-521	Topology and Functional Analysis	4
35	MA-522	Partial Differential Equations	4
36	MA-523	Numerical Analysis	4
37	MA-524	Probability and Statistics	4
38	MA-525	Heuristic Optimization	3
39	MA-526	An Introduction to Wavelets	3
40	MA-527	Field and Galois Theory	3
41	MA-528	Measure Theory and Integration	4
42	MA-529	Statistical Inference	4
43	MA-530	Graph Theory	4
44	MA-546	Introduction to Quantitative Finance	4
45	MA-548	Abstract Algebra	4
46	MA-549	Abstract Algebra	4
47	MA-550	Statistical Data Analysis	3
48	MA-551	Numerical Analysis	3
49	MA-552	Number Theory	3
50	MA-553	Mathematical Foundations of Financial Engineering	3
51	MA-555	Introduction to Partial Differential Equations for Engineers	3
52	MA-560	Nonlinear Dynamics and Chaos	3
53	MA-563	Statistical Computation and Simulation	4
54	MA-565	Numerical Methods in Quantitative Finance	3
55	MA-568	Real Analysis	3
56	MA-570	Data-driven Dynamical Systems	3
57	MA-571	Introduction to Approximation algorithms	4
58	MA-572	Applied Multivariate Statistical Analysis	4
59	MA-581P	Post Graduate Project I	6
60	MA-582P	Post Graduate Project-2	8
61	MA-588	Mathematical Control Theory	4
62	MA-595	Calculus as Ganita	3
63	MA-601	Real and Functional Analysis	3
64	MA-603	Advanced Partial Differential Equations	3
65	MA-604	Introduction To Game Theory	3
66	MA-605	Statistical Data Analysis	3

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
67	MA-607	Numerical Analysis	3
68	MA-608	Computational Fluid Dynamics	3
69	MA-609	Numerics of Partial Differential Equations	3
70	MA-610	Mathematical Modeling	3
71	MA-611	Statistical tools and Computing	4
72	MA-612	Operator Theory	4
73	MA-621	Modeling Population Dynamics	3
74	MA-641	Operations Research	3
75	MA-644	Dynamical Systems	3
76	MA-650	Mathematical Models for Infectious Diseases	3
77	MA-651	Optimization Techniques	3
78	MA-652	Stability Theory of Differential Equations	3
79	MA-653	Computational Financial Modelling	3
80	MA-653P	Computational Financial Modelling Lab	1
81	MA-654	Financial Engineering	3
82	MA-655	Fixed Income Securities	3
83	MA-656	Stochastic Calculus for Financial Engineering	3
84	MA-665	Semigroup Of Bounded Linear Operators	3
85	MA-704	Dynamical System	3
86	MA-709	Numerical Linear Algebra	3
87	MA-765	Fractional Differential Equations	4
88	MA-780	Topics on Semigroup Theory	3
89	ME-210	Fluid Mechanics	3
90	ME-213	Engineering Thermodynamics	4
91	ME-303	Heat Transfer	3
92	ME-308	Manufacturing Engineering	3
93		Time Series Analysis	3
94		Mathematical Method for Signal Processing	4
95	CS-669	Pattern Recognition	4
96	BE-301	Biomechanics	4
97	BY-507	Genetic Engineering	4
98	CS-352	Transportation Engineering	3
99	CS-352P	Transporting Engg. Lab	1
100		Information Systems and Databases	4
101		Times Series Analysis and Applications / Bayesian Data Analysis and Applications	3
102		Big Data: Management and Analytics	4
103	EP-302	Computational Methods for Engineering	3
104	ME-310	System Dynamics and Control	3
105	ME-210P	Fluid Mechanics Lab	1
106	ME-303P	Heat Transfer Lab	1

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
107		Cellular Automata	3
108	CS-606	Computational Modeling of Social Systems	3
109	MA-562	An Introduction to Bayesian Analysis	4
110	MA-561	Fluidics and Microfluidics	4
111	MA-573	Advanced Mathematical Foundations of Machine Learning	4
112	MA-574	An Introduction to Probabilistic Machine Learning	4
113	MA-576	Nonlinear Optimization	4
<i>Compulsory project components — counted as DE, not selectable</i>			
114	MC-498P	Major Technical Project - I (compulsory; counted as DE)	4
115	MC-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - AI and Robotics

Table 247: General Engineering - AI and Robotics — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	37
	Discipline Electives	29
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 248: B. Tech General Engineering - AI and Robotics — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	DS-201	Data Handling and Vizualization	2-0-2-3	3
2	EE-261	Electrical System Around Us	3-0-0-3	3
3	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	EE-201	Electromechanics	2.5-0.5-0-3	3
5	IC-253	Programming and Data Structures	2-0-2-3	3
6	ME-100	Reverse Engineering	0-0-2-1	1
7	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
8	AR-521	Control of Robotic Systems		3
9	AR-523	Robot Manipulators		3
10	EE-302	Control Systems	3-0-2-4	4
11	ME-305	Design of Machine Elements	3-1-0-4	4
12	ME-309	Theory of Machines	3-1-0-4	4
Total				37

Table 249: B. Tech General Engineering - AI and Robotics — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
7		HS-110	Japanese for Beginners	3	
8		HS-151	Introduction to English Literature	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 250: B. Tech General Engineering - AI and Robotics — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 251: B. Tech General Engineering - AI and Robotics — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	DS-201	Data Handling and Vizualization	2-0-2-3	3
4	DC	EE-261	Electrical System Around Us	3-0-0-3	3
5	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 252: B. Tech General Engineering - AI and Robotics — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
3	DC	IC-253	Programming and Data Structures	2-0-2-3	3
4	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					9

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 253: B. Tech General Engineering - AI and Robotics — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
2	DC	AR-521	Control of Robotic Systems		3
3	DC	AR-523	Robot Manipulators		3
4	DC	EE-302	Control Systems	3-0-2-4	4
5	DC	ME-305	Design of Machine Elements	3-1-0-4	4
6	DC	ME-309	Theory of Machines	3-1-0-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 254: B. Tech General Engineering - AI and Robotics — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 255: B. Tech General Engineering - AI and Robotics — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 256: B. Tech General Engineering - AI and Robotics — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 29 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 257: General Engineering - AI and Robotics — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-502	Advanced Design Practicum	4
2	AR-505	Principles of Robot Autonomy	3
3	AR-508	Marine Robotics	3
4	AR-509	Deep Learning for Robotics	4
5	AR-510	Underactuated Robotics	3
6	AR-511	Autonomous Mobile Robots	3
7	AR-513	Unmanned Aerial Systems (UAS)	3
8	AR-514	Vision and Learning Based Control	3
9	AR-519	Robot Manipulators	3
10	AR-522	Robot Vision	3
11	AR-524	Robot Simulators	3

General Engineering - AI and Robotics — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
12	AR-525	Reinforcement Learning for Robotics	3
13	AR-526	Multimodal Learning	3
14	CE-251	Hydraulics Engineering	3
15	CS-212	Design and Analysis of Algorithms	4
16	CS-514	Advance Data Structure and Algorithm (CS-580)	4
17	DS-314	Probability Theory and Statistics	4
18	EE-201P	Electromechanics Lab	1
19	EE-203	Network Theory	3
20	EE-211	Analog Circuit Design	3
21	EE-260	Signals and Systems	3
22	EE-529	Embedded Systems	4
23	EE-604	Practicum on Advanced Electric Drives	3
24	EE-608	Digital Image Processing	4
25	IC-241	Materials Science for Engineers	3
26	ME-210	Fluid Mechanics	3
27	ME-212	Product Manufacturing Technology	3
28	ME-213	Engineering Thermodynamics	4
29	ME-510	Advanced Manufacturing Processes	3
<i>Compulsory project components — counted as DE, not selectable</i>			
30	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
31	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Mechatronics

Table 258: General Engineering - Mechatronics — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	37
	Discipline Electives	29
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 259: B. Tech General Engineering - Mechatronics — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-260	Signals and Systems	2.5-0.5-0-3	3
3	EE-261	Electrical System Around Us	3-0-0-3	3
4	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
5	EE-201	Electromechanics	2.5-0.5-0-3	3
6	EE-211	Analog Circuit Design	2-0-2-3	3
7	ME-100	Reverse Engineering	0-0-2-1	1
8	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
9	EE-231	Measurement and Instrumentation	2-0-2-3	3
10	EE-302	Control Systems	3-0-2-4	4
11	ME-305	Design of Machine Elements	3-1-0-4	4
12	ME-309	Theory of Machines	3-1-0-4	4
Total				37

Table 260: B. Tech General Engineering - Mechatronics — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
7		HS-110	Japanese for Beginners	3	
8		HS-151	Introduction to English Literature	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 261: B. Tech General Engineering - Mechatronics — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 262: B. Tech General Engineering - Mechatronics — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-203	Network Theory	3-0-0-3	3
4	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
5	DC	EE-261	Electrical System Around Us	3-0-0-3	3
6	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 263: B. Tech General Engineering - Mechatronics — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
3	DC	EE-211	Analog Circuit Design	2-0-2-3	3
4	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					9

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 264: B. Tech General Engineering - Mechatronics — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
2	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
3	DC	EE-302	Control Systems	3-0-2-4	4
4	DC	ME-305	Design of Machine Elements	3-1-0-4	4
5	DC	ME-309	Theory of Machines	3-1-0-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 265: B. Tech General Engineering - Mechatronics — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 266: B. Tech General Engineering - Mechatronics — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 267: B. Tech General Engineering - Mechatronics — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 29 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 268: General Engineering - Mechatronics — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-502	Advanced Design Practicum	4
2	AR-505	Principles of Robot Autonomy	3
3	AR-508	Marine Robotics	3
4	AR-509	Deep Learning for Robotics	4
5	AR-510	Underactuated Robotics	3
6	AR-511	Autonomous Mobile Robots	3
7	AR-519	Robot Manipulators	3
8	AR-521	Control of Robotic Systems	3
9	AR-522	Robot Vision	3
10	AR-523	Robot Manipulators	3
11	AR-524	Robot Simulators	3

General Engineering - Mechatronics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	AR-526	Multimodal Learning	3
13	CS-212	Design and Analysis of Algorithms	4
14	CS-313	Computer Networks	4
15	CS-315	Artificial Intelligence (CS-362, CS-562)	4
16	EE-201P	Electromechanics Lab	1
17	EE-210	Digital Systems Design	3
18	EE-210P	Digital Systems Design Lab	1
19	EE-311	Device Electronics for Integrated Circuits	3
20	EE-314	Digital Signal Processing	4
21	EE-326	Computer Organization and Processor Architecture Design	4
22	EE-529	Embedded Systems	4
23	EE-536	IoT Systems	3
24	IC-121	Mechanics of Particles and waves	3
25	IC-241	Materials Science for Engineers	3
26	ME-210	Fluid Mechanics	3
27	ME-212	Product Manufacturing Technology	3
28	ME-213	Engineering Thermodynamics	4
29	ME-215	Manufacturing Engineering-1	3
30	ME-315	Manufacturing Engineering-II	3
31	ME-452	Robot Kinematics, Dynamics and Control	4
32	ME-504	Numerical Methods for Engineering Computation	3
33	ME-510	Advanced Manufacturing Processes	3
34	ME-513	Finite Element Methods in Engineering	4
35	ME-527	Biofluid Dynamics	3
36	ME-630	Machine learning for engineers	3
<i>Compulsory project components — counted as DE, not selectable</i>			
37	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
38	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Communication Technology

Table 269: General Engineering - Communication Technology — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	40
	Discipline Electives	26
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 270: B. Tech General Engineering - Communication Technology — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-313	Computer Networks	3-0-2-4	4
2	EE-203	Network Theory	3-0-0-3	3
3	EE-260	Signals and Systems	2.5-0.5-0-3	3
4	EE-261	Electrical System Around Us	3-0-0-3	3
5	DS-404	Information Security and Privacy	3-0-0-3	3
6	EE-201	Electromechanics	2.5-0.5-0-3	3
7	EE-202	Electromagnetic Theory	3-0-0-3	3
8	EE-316	Communication Systems	3-0-2-4	3
9	IC-253	Programming and Data Structures	2-0-2-3	3
10	ME-100	Reverse Engineering	0-0-2-1	1
11	EE-231	Measurement and Instrumentation	2-0-2-3	3
12	EE-302	Control Systems	3-0-2-4	4
13	EE-314	Digital Signal Processing	3-0-2-4	4
Total				40

Table 271: B. Tech General Engineering - Communication Technology — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
7		HS-110	Japanese for Beginners	3	
8		HS-151	Introduction to English Literature	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 272: B. Tech General Engineering - Communication Technology — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 273: B. Tech General Engineering - Communication Technology — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-313	Computer Networks	3-0-2-4	4
4	DC	EE-203	Network Theory	3-0-0-3	3
5	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
6	DC	EE-261	Electrical System Around Us	3-0-0-3	3
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 274: B. Tech General Engineering - Communication Technology — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	DS-404	Information Security and Privacy	3-0-0-3	3
3	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
4	DC	EE-202	Electromagnetic Theory	3-0-0-3	3
5	DC	EE-316	Communication Systems	3-0-2-4	3
6	DC	IC-253	Programming and Data Structures	2-0-2-3	3
7	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 275: B. Tech General Engineering - Communication Technology — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	EE-302	Control Systems	3-0-2-4	4
3	DC	EE-314	Digital Signal Processing	3-0-2-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 276: B. Tech General Engineering - Communication Technology — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 277: B. Tech General Engineering - Communication Technology — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 278: B. Tech General Engineering - Communication Technology — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 26 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 279: General Engineering - Communication Technology — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-549	Performance Analysis of Computer Networks	3
2	DS-201	Data handling and visualization	3
3	DS-314	Probability Theory and Statistics	4
4	EE-201P	Electromechanics Lab	1
5	EE-211	Analog Circuit Design	3
6	EE-503	Advance Communication Theory	3
7	EE-517	Wireless Communications and Networking	3
8	EE-518	Information Theory	3
9	EE-530	Optimization theory	3
10	EE-533	Acess Network	4

General Engineering - Communication Technology — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	EE-536	IoT Systems	3
12	EE-541	Tensors: Techniques, Algorithms, Applications for Signal	4
13	EE-553	Foundations of Intelligent Communication Systems-I	4
14	EE-574	Biomedical Signal and Image Analysis	3
15	EE-581	Applied Statistics for Data and Signals	4
16	EE-607	Optical Communication Systems	3
17	EE-608	Digital Image Processing	4
18	EE-641	Advance Wireless Technologies	4
19	IC-241	Materials Science for Engineers	3
20	ME-212	Product Manufacturing Technology	3
21	VL-402	Rfic Design	3
<i>Compulsory project components — counted as DE, not selectable</i>			
22	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
23	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Open Specialisation

Table 280: General Engineering - Open Specialisation — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	40
	Discipline Electives	26
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 281: B. Tech General Engineering - Open Specialisation — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-261	Electrical Systems Around Us	3-0-0-3	3
2	IC-241	Material Science for Engineers	3-0-0-3	3
3	ME-212	Product Manufacturing Technology	2-0-3-3	3
4	ME-213	Thermodynamics	3-1-0-4	3
5	EE-201	Electromechanics	2.5-0.5-0-3	3
6	EE-211	Analog Circuit Design	2-0-2-3	3
7	HS-307	Macroeconomics	3-0-0-3	3
8	IC-253	Programming and Data Structures	2-0-2-3	3
9	ME-100	Reverse Engineering	0-0-2-1	1
10	DS-201	Data Handling and Visualisation	2-0-2-3	3
11	EE-231	Measurement and Instrumentation	2-0-2-3	3
12	EE-302	Control Systems	3-0-2-4	4
13	ME-305	Design of Machine Elements	3-1-0-4	4
14	HS-541	Technical Communication	1-0-0-1	1
Total				40

Table 282: B. Tech General Engineering - Open Specialisation — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
7		HS-110	Japanese for Beginners	3	
8		HS-151	Introduction to English Literature	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 283: B. Tech General Engineering - Open Specialisation — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 284: B. Tech General Engineering - Open Specialisation — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
4	DC	IC-241	Material Science for Engineers	3-0-0-3	3
5	DC	ME-212	Product Manufacturing Technology	2-0-3-3	3
6	DC	ME-213	Thermodynamics	3-1-0-4	3
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 285: B. Tech General Engineering - Open Specialisation — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
3	DC	EE-211	Analog Circuit Design	2-0-2-3	3
4	DC	HS-307	Macroeconomics	3-0-0-3	3
5	DC	IC-253	Programming and Data Structures	2-0-2-3	3
6	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 286: B. Tech General Engineering - Open Specialisation — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-201	Data Handling and Visualisation	2-0-2-3	3
2	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
3	DC	EE-302	Control Systems	3-0-2-4	4
4	DC	ME-305	Design of Machine Elements	3-1-0-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 287: B. Tech General Engineering - Open Specialisation — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	HS-541	Technical Communication	1-0-0-1	1
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					5

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 288: B. Tech General Engineering - Open Specialisation — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 289: B. Tech General Engineering - Open Specialisation — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 26 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 290: General Engineering - Open Specialisation — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-502	Advanced Design Practicum	4
2	AR-505	Principles of Robot Autonomy	3
3	AR-519	Robot Manipulators	3
4	AR-521	Control of Robotic Systems	3
5	AR-522	Robot Vision	3
6	AR-526	Multimodal Learning	3
7	CS-212	Design and Analysis of Algorithms	4
8	CS-313	Computer Networks	4
9	CS-315	Artificial Intelligence (CS-362, CS-562)	4
10	CS-514	Advance Data Structure and Algorithm (CS-580)	4

General Engineering - Open Specialisation — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
11	DS-314	Probability Theory and Statistics	4
12	EE-203	Network Theory	3
13	EE-260	Signals and Systems	3
14	EE-301	Control Systems	3
15	EE-311	Device Electronics for Integrated Circuits	3
16	EE-503	Advance Communication Theory	3
17	EE-517	Wireless Communications and Networking	3
18	EE-529	Embedded Systems	4
19	EE-533	Acess Network	4
20	EE-536	IoT Systems	3
21	EE-607	Optical Communication Systems	3
22	EE-608	Digital Image Processing	4
23	EE-641	Advance Wireless Technologies	4
24	MA-546	Introduction to Quantitative Finance	4
25	MA-551	Numerical Analysis	3
26	ME-210	Fluid Mechanics	3
27	ME-452	Robot Kinematics, Dynamics and Control	4
28	ME-510	Advanced Manufacturing Processes	3
<i>Compulsory project components — counted as DE, not selectable</i>			
29	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
30	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Fintech

Table 291: General Engineering - Fintech — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	32
	Discipline Electives	34
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 292: B. Tech General Engineering - Fintech — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	DS-201	Data Handling and Visualisation	2-0-2-3	3
2	EE-261	Electrical Systems Around Us	3-0-0-3	3
3	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	CS-671	Deep Learning and Applications	3-1-0-4	4
5	DS-313	Statistical Foundations of Data Science	3-1-0-4	4
6	DS-412	Matrix Computations for Data Science	3-0-2-4	4
7	IC-253	Programming and Data Structures	2-0-2-3	3
8	MA-546	Introduction to Quantitative Finance		4
9	EE-302	Control Systems	3-0-2-4	4
Total				32

Table 293: B. Tech General Engineering - Fintech — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
7		HS-110	Japanese for Beginners	3	
8		HS-151	Introduction to English Literature	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 294: B. Tech General Engineering - Fintech — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 295: B. Tech General Engineering - Fintech — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	DS-201	Data Handling and Visualisation	2-0-2-3	3
4	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
5	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
6	DE	EE-203	Network Theory	3-0-0-3	3
7	DE	IC-241	Material Science for Engineers	3-0-0-3	3
8	DE	ME-212	Product Manufacturing Technology	2-0-3-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 296: B. Tech General Engineering - Fintech — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	CS-671	Deep Learning and Applications	3-1-0-4	4
3	DC	DS-313	Statistical Foundations of Data Science	3-1-0-4	4
4	DC	DS-412	Matrix Computations for Data Science	3-0-2-4	4
5	DC	IC-253	Programming and Data Structures	2-0-2-3	3
6	DC	MA-546	Introduction to Quantitative Finance		4
7	DE	MA-653P	Computational Financial Modelling Lab	0-0-2-1	1
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 297: B. Tech General Engineering - Fintech — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-302	Control Systems	3-0-2-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 298: B. Tech General Engineering - Fintech — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 299: B. Tech General Engineering - Fintech — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 300: B. Tech General Engineering - Fintech — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 34 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 301: General Engineering - Fintech — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-313	Computer Networks	4
2	DS-312	Computing Systems for Data Processing	4
3	EE-530	Optimization theory	3
4	MA-353	Modelling of infectious disease	3
5	MA-356	Modelling and Simulation	3
6	MA-460	Nonlinear Dynamics and Chaos	3
7	MA-548	Abstract Algebra	4
8	MA-550	Statistical Data Analysis	3
9	MA-552	Number Theory	3
10	MA-553	Mathematical Foundations of Financial Engineering	3
11	MA-560	Nonlinear Dynamics and Chaos	3

General Engineering - Fintech — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	MA-563	Statistical Computation and Simulation	4
13	MA-565	Numerical Methods in Quantitative Finance	3
14	MA-572	Applied Multivariate Statistical Analysis	4
15	MA-604	Introduction To Game Theory	3
16	MA-608	Computational Fluid Dynamics	3
17	MA-641	Operations Research	3
18	MA-644	Dynamical Systems	3
19	MA-650	Mathematical Models for Infectious Diseases	3
20	MA-651	Optimization Techniques	3
21	MA-652	Stability Theory of Differential Equations	3
22	MA-653	Computational Financial Modelling	3
23	MA-654	Financial Engineering	3
24	MA-655	Fixed Income Securities	3
25	MA-656	Stochastic Calculus for Financial Engineering	3
26	MA-665	Semigroup Of Bounded Linear Operators	3
<i>Compulsory project components — counted as DE, not selectable</i>			
27	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
28	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Materials Science and Engineering

Table 302: Materials Science and Engineering — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	45
	Discipline Electives	21
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 303: B. Tech Materials Science and Engineering — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
2	MT-201	Physics of Solids		3
3	MT-202	Quantum Mechanics and Applications		3
4	MT-203	Material Synthesis and Characterisation		4
5	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
6	ME-100	Reverse Engineering	0-0-2-1	1
7	MT-204	Thermodynamics and Kinetics and Materials	3	3
8	MT-205	Functional Properties of Materials	4	4
9	MT-206	Extraction and Materials Processing	4	4
10	ME-212	Product Realization (Manufacturing) Technology	2-0-3-3	3
11	MT-301	Phase Transformations		3
12	MT-302	Transport Phenomena		3
13	MT-303	Computational Materials Science		4
14	MT-304	Mechanical Behaviour of Materials		4
Total				45

Table 304: B. Tech Materials Science and Engineering — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
7		HS-110	Japanese for Beginners	3	
8		HS-151	Introduction to English Literature	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 305: B. Tech Materials Science and Engineering — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-241	Material Science for Engineers	3-0-0-3	3
8	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 306: B. Tech Materials Science and Engineering — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	DC	MT-201	Physics of Solids		3
5	DC	MT-202	Quantum Mechanics and Applications		3
6	DC	MT-203	Material Synthesis and Characterisation		4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 307: B. Tech Materials Science and Engineering — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
3	DC	ME-100	Reverse Engineering	0-0-2-1	1
4	DC	MT-204	Thermodynamics and Kinetics and Materials	3	3
5	DC	MT-205	Functional Properties of Materials	4	4
6	DC	MT-206	Extraction and Materials Processing	4	4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 308: B. Tech Materials Science and Engineering — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-212	Product Realization (Manufacturing) Technology	2-0-3-3	3
2	DC	MT-301	Phase Transformations		3
3	DC	MT-302	Transport Phenomena		3
4	DC	MT-303	Computational Materials Science		4
5	DC	MT-304	Mechanical Behaviour of Materials		4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 309: B. Tech Materials Science and Engineering — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 310: B. Tech Materials Science and Engineering — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	MS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 311: B. Tech Materials Science and Engineering — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	MS-010	Internship		2
2	MTP	MS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 21 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 312: Materials Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	ME-303	Heat Transfer	3
2	ME-308	Manufacturing Engineering	3
3	ME-353	Electronic Materials and Their Applications	3
4	ME-515	Carbon Materials and Technology	3
5	MT-501	Energy Conversion and Storage Technologies	3
6	MT-502	Recycling and Circular Economy	3
7	MT-503	Semiconductor Materials and Devices	3
8	MT-504	Powder metallurgical processing of Materials	3
9	MT-505	Thin Film Technology	3
10	MT-506	Biomaterials	3
11	MT-507	Modelling and Simulations in Material Science	3

Materials Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	MT-511	Sensor Materials and Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
13	MS-498P	Major Technical Project - I (compulsory; counted as DE)	4
14	MS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Science — Chemical Sciences

Table 313: Chemical Sciences — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	27
	IC Basket	9
	IKS + HSS	12
Discipline Courses	Discipline Core	65
	Discipline Electives	21
Electives	Free Electives	15
	Research Lit Presentation (1+1)	2
	RP (equivalent as DE)	12
Total		163

Table 314: B. S. Chemical Sciences — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CY-200	Foundations and Applications of Chemistry	3	3
2	CY-201P	Physical Chemistry Laboratory	0-0-4-2	2
3	CY-301	Principles and Theories of Physical Chemistry	3-0-0-3	3
4	CY-302	Principles of Organic Chemistry	3-0-0-3	3
5	CY-303	Fundamentals of Inorganic Chemistry	3-0-0-3	3
6	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
7	CY-202P	Organic Chemistry Laboratory	0-0-4-2	2
8	CY-203P	Inorganic Chemistry Laboratory	0-0-4-2	2
9	CY-304	Fundamental Analytical Chemistry	3-0-0-3	3
10	CY-401	Introduction to Quantum Chemistry and Molecular Spectroscopy	3-0-0-3	3
11	CY-512	Advanced Quantum Chemistry	3-0-0-3	3
12	CY-512P	Physical Chemistry Laboratory		3
13	CY-531	Organic Reactions and Mechanisms		3
14	CY-533	Chemistry of Main Group Elements		3
15	CY-533P	Inorganic Chemistry Laboratory		3
16	CY-511	Group Theory and Spectroscopy	3-0-0-3	3
17	CY-531P	Organic Chemistry Laboratory		3
18	CY-532	Photochemistry and Pericyclic Reactions		3
19	CY-534	Chemistry of Transition Elements		3
20	CY-514	Chemical and Statistical Thermodynamics	3-0-0-3	3
21	CY-535	Introduction to Organometallic Chemistry		3
22	CY-504	Heterocyclic Chemistry	2-0-0-2	2
23	CY-513	Chemical Kinetics and Reaction Dynamics	3-0-0-3	3
Total				65

Table 315: B. S. Chemical Sciences — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-110	Japanese for Beginners	3	
7		HS-151	Introduction to English Literature	3-0-0-3	
8		HS-256	Introduction to Spanish	3	
9	IKS + HSS	IC-181	Introduction to Consciousness and Holistic Wellbeing	2-0-2-3	3
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 316: B. S. Chemical Sciences — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
8	DC	CY-200	Foundations and Applications of Chemistry	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 317: B. S. Chemical Sciences — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-201P	Physical Chemistry Laboratory	0-0-4-2	2
2	DC	CY-301	Principles and Theories of Physical Chemistry	3-0-0-3	3
3	DC	CY-302	Principles of Organic Chemistry	3-0-0-3	3
4	DC	CY-303	Fundamentals of Inorganic Chemistry	3-0-0-3	3
5	DC	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 318: B. S. Chemical Sciences — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	CY-202P	Organic Chemistry Laboratory	0-0-4-2	2
3	DC	CY-203P	Inorganic Chemistry Laboratory	0-0-4-2	2
4	DC	CY-304	Fundamental Analytical Chemistry	3-0-0-3	3
5	DC	CY-401	Introduction to Quantum Chemistry and Molecular Spectroscopy	3-0-0-3	3
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 319: B. S. Chemical Sciences — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-512	Advanced Quantum Chemistry	3-0-0-3	3
2	DC	CY-512P	Physical Chemistry Laboratory		3
3	DC	CY-531	Organic Reactions and Mechanisms		3
4	DC	CY-533	Chemistry of Main Group Elements		3
5	DC	CY-533P	Inorganic Chemistry Laboratory		3
6	RP	DP-403P	Research Literature Presentation 1		1
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 320: B. S. Chemical Sciences — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-511	Group Theory and Spectroscopy	3-0-0-3	3
2	DC	CY-531P	Organic Chemistry Laboratory		3
3	DC	CY-532	Photochemistry and Pericyclic Reactions		3
4	DC	CY-534	Chemistry of Transition Elements		3
5	RP	DP-404P	Research Literature Presentation 2		1
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 321: B. S. Chemical Sciences — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-514	Chemical and Statistical Thermodynamics	3-0-0-3	3
2	DC	CY-535	Introduction to Organometallic Chemistry		3
3	RP	DP-551P	Undergraduate Research Project 1		6
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 322: B. S. Chemical Sciences — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-504	Heterocyclic Chemistry	2-0-0-2	2
2	DC	CY-513	Chemical Kinetics and Reaction Dynamics	3-0-0-3	3
3	RP	DP-552P	Undergraduate Research Project 2		6
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 21 credits of **Discipline Electives** and 15 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 323: Chemical Sciences — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BY-505	Nanobiotechnology	3
2	BY-514	Analytical Biotechniques	3
3	CE-519	Chemistry of Natural Waters	3
4	CH-101	General Chemistry	4
5	CY-001	Preparatory Chemistry - 1	4
6	CY-002	Preparatory Chemistry - 2	4
7	CY-100	Preparatory Chemistry 1	3
8	CY-101	Chemistry	5
9	CY-241	Nano-scale Science and Technology	3
10	CY-242	Introduction to Molecular Spectroscopy	3
11	CY-243	Engineering Chemistry	3
12	CY-247	Introduction to Molecular Thermodynamics	3
13	CY-248	Molecular Physical Chemistry for Engineers	3
14	CY-342	Understanding Small Systems	3
15	CY-344	Food Chemistry: Processing, Preservation and Storage	3
16	CY-402	Applied Materials Chemistry	3
17	CY-403	Numerical Methods and Data Analysis in Chemistry	3
18	CY-404	Fundamentals of Soft Matter Science and Applications	3
19	CY-405	Introduction to structural chemistry	3
20	CY-501	Organic Reaction and Mechanism	4
21	CY-501P	Organic Chemistry Laboratory	4
22	CY-502	Photochemistry and pericyclic reactions	4
23	CY-506	Chemistry of Transition Elements	4
24	CY-506P	Inorganic Chemistry Lab	4
25	CY-507	Chemistry of Main Group Elements	4
26	CY-508	Organometallic Chemistry	2
27	CY-511P	Physical Chemistry Laboratory	4
28	CY-515	Advanced Inorganic Spectroscopy	3
29	CY-521	Mathematics for Chemists	3
30	CY-522	Computational Chemistry	3
31	CY-523	Colloids and Interface Science and Technology	3
32	CY-524	Basic and Applied Electrochemistry	3
33	CY-541	Fundamentals of Organic Chemistry	3
34	CY-545	Reagents in organic synthesis	3
35	CY-547	Chemical Crystallography	3
36	CY-550	Bioinspired Materials	3
37	CY-552	Hydrogen Generation and Storage	3
38	CY-553	Organic Inorganic Spectroscopy	3
39	CY-554	Science and Technology of Nanomaterials	3
40	CY-555	Introduction to Polymer Science & Technology	3

Chemical Sciences — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
41	CY-556	Organic Spectroscopy	3
42	CY-560	Introduction to Supramolecular Chemistry	3
43	CY-561	Natural Products Synthesis	3
44	CY-562	Stereochemistry of organic compounds	3
45	CY-562P	Design Practicum - II	1
46	CY-598P	Post Graduate Project-1 (CH)	4
47	CY-599P	Post Graduate Project-2	9
48	CY-641	Polymer Synthesis	3
49	CY-642	Molecular- and Bio-electronics	3
50	CY-643	Advanced Analytical Techniques	3
51	CY-644	Bioinorganic chemistry	3
52	CY-645	Reagents in organic synthesis	3
53	CY-646	Advanced NMR Spectroscopy - A Problem Based Approach	3
54	CY-660	Photocatalysis: Concepts and Applications	3
55	CY-670	Fluorescence spectroscopy, microscopy and applications	3
56	CY-701	Advances Physical Methods in Chemistry Theory and	3
57	CY-702	Advanced Inorganic Chemistry: Theory and Applications	3
58	CY-703	Advanced Organic Chemistry	3
59	CY-704	Introduction to Theoretical Chemistry	3
60	CY-705	Modern Methods in Organic Synthesis	3
61	CY-746	Self Assembly of Surfactants and Polymers in Solution	3
62	EN-503	Energy Storage Technologies	3
63	ME-353	Electronic Materials and Their Applications	3
64	ME-509	Nanomanufacturing	3
65	ME-515	Carbon Materials and Technology	3
66	ME-516	Polymer Technology for Engineers	3
67	MT-201	Physics of Solids	3
68	MT-501	Energy Conversion and Storage Technologies	3
69	MT-502	Recycling and Circular Economy	3
70	MT-503	Semiconductor Materials and Devices	3
71	MT-505	Thin Film Technology	3
72	MT-506	Biomaterials	3
73	MT-509	Hydrogen Energy	3
74	MT-510	Colloids and Interfaces	3
75	PH-504	Organic Optoelectronics	3
<i>Compulsory project components — counted as DE, not selectable</i>			
76	DP-551P	Undergraduate Research Project 1 (compulsory; counted as DE)	6

Chemical Sciences — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
77	DP-552P	Undergraduate Research Project 2 (compulsory; counted as DE)	6

Bachelor of Technology — Microelectronics and VLSI

Table 324: Microelectronics and VLSI — B24 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	54
	Discipline Electives	12
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 325: B. Tech Microelectronics and VLSI — B24 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-212	Digital System Design	3-0-2-4	4
3	EE-260	Signals and Systems	2.5-0.5-0-3	3
4	EE-302	Control Systems	3-0-2-4	4
5	EE-311	Device Electronics	3-0-0-3	3
6	EE-202	Electromagnetic Theory and Transmission Lines	3-0-0-3	3
7	EE-211	Analog Circuit Design	2-0-2-3	4
8	VL-300	Reverse Engineering - E-waste management	0-0-2-1	1
9	VL-311	CMOS Processing and Practicum	3-0-2-4	4
10	VL-312	Electronic System Packaging	3-0-0-3	3
11	VL-326	Computer Organization	3-0-2-4	4
12	VL-401	RTL Design and Verification	2-0-2-3	3
13	VL-402	RF IC Design	3-0-0-3	3
14	VL-403	CMOS Digital IC Design	3-0-2-4	4
15	VL-404	CMOS Analog IC Design	3-0-2-4	4
16	VL-405	Design for Testability	3-0-2-4	4
Total				54

Table 326: B. Tech Microelectronics and VLSI — B24 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-110	Japanese for Beginners	3	
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-256	Introduction to Spanish	3	
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 327: B. Tech Microelectronics and VLSI — B24 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket (choose 1)	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
8		IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	
9		IC-241	Material Science for Engineers	3-0-0-3	
10		IC-253	Data Structures and Algorithms	2-0-2-3	
11	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 328: B. Tech Microelectronics and VLSI — B24 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-203	Network Theory	3-0-0-3	3
4	DC	EE-212	Digital System Design	3-0-2-4	4
5	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
6	DC	EE-302	Control Systems	3-0-2-4	4
7	DC	EE-311	Device Electronics	3-0-0-3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 329: B. Tech Microelectronics and VLSI — B24 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	EE-202	Electromagnetic Theory and Transmission Lines	3-0-0-3	3
3	DC	EE-211	Analog Circuit Design	2-0-2-3	4
4	DC	VL-300	Reverse Engineering - E-waste management	0-0-2-1	1
5	DC	VL-311	CMOS Processing and Practicum	3-0-2-4	4
6	DC	VL-312	Electronic System Packaging	3-0-0-3	3
7	DC	VL-326	Computer Organization	3-0-2-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 330: B. Tech Microelectronics and VLSI — B24 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	VL-401	RTL Design and Verification	2-0-2-3	3
2	DC	VL-402	RF IC Design	3-0-0-3	3
3	DC	VL-403	CMOS Digital IC Design	3-0-2-4	4
4	DC	VL-404	CMOS Analog IC Design	3-0-2-4	4
5	DC	VL-405	Design for Testability	3-0-2-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 331: B. Tech Microelectronics and VLSI — B24 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 332: B. Tech Microelectronics and VLSI — B24 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	VL-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 333: B. Tech Microelectronics and VLSI — B24 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	VL-010	Internship		2
2	MTP	VL-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 12 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 334: Microelectronics and VLSI — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	EE-210P	Digital Systems Design Lab	1
2	EE-507	Transmission Lines and Basic Microwave Engineering	4
3	EE-516	Biomedical Systems	4
4	EE-520	Microelectronic Devices and Modelling	3
5	EE-523	Digital VLSI Architecture Design	4
6	EE-529	Embedded Systems	4
7	EE-554	Low Power VLSI Design	3
8	EE-574	Biomedical Signal and Image Analysis	3
9	EE-585	Theory, Modelling and Characterization of Photo-voltaic Devices	3
10	EE-611	VLSI Technology	3

Microelectronics and VLSI — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	EE-615	Nanoelectronics and Micronanofabrication	3
12	EE-619	Mixed Signal VLSI Design	4
13	EE-621	Radiating systems	4
14	EE-622	Microwave Integrated Circuits	3
<i>Compulsory project components — counted as DE, not selectable</i>			
15	VL-498P	Major Technical Project - I (compulsory; counted as DE)	4
16	VL-499P	Major Technical Project - II (compulsory; counted as DE)	4

Batch 2025

Bachelor of Technology — Computer Science and Engineering

Table 335: Computer Science and Engineering — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	39
	Discipline Electives	27
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 336: B. Tech Computer Science and Engineering — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
2	CS-212	Design of Algorithms	3-0-2-4	4
3	CS-213	Reverse Engineering	0-0-2-1	1
4	CS-214	Computer Organization	3-0-2-4	4
5	CS-304	Formal Language and Automata Theory	3-0-0-3	3
6	CS-309	Information Systems and Databases	3-0-2-4	4
7	CS-315*	Artificial Intelligence	3-0-2-4	4
8	CS-312	Operating Systems	3-0-2-4	4
9	CS-313	Computer Networks	3-0-2-4	4
10	CS-302	Paradigms of Programming	3-0-2-4	4
11	CS-303	Software Engineering	2-0-2-3	3
Total				39

* CS-305 Artificial Intelligence has been recoded as CS-315 with L-T-P-C 3-0-2-4, w.e.f. batch 2025.

Table 337: B. Tech Computer Science and Engineering — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
5		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
6		IC-230	Environmental Science	3-0-0-3	
7	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
8		HS-112	Japanese Language Class I	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
10		HS-342	German I	3-0-0-3	
11	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
12		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 338: B. Tech Computer Science and Engineering — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 339: B. Tech Computer Science and Engineering — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-222P	Physics Practicum	0-0-3-2	2
3	IC	IC-272	Machine Learning	2-0-2-3	3
4	DC	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
5	DC	CS-212	Design of Algorithms	3-0-2-4	4
6	DC	CS-213	Reverse Engineering	0-0-2-1	1
7	DC	CS-214	Computer Organization	3-0-2-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 340: B. Tech Computer Science and Engineering — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-304	Formal Language and Automata Theory	3-0-0-3	3
2	DC	CS-309	Information Systems and Databases	3-0-2-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 341: B. Tech Computer Science and Engineering — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-315*	Artificial Intelligence	3-0-2-4	4
2	DC	CS-312	Operating Systems	3-0-2-4	4
3	DC	CS-313	Computer Networks	3-0-2-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 342: B. Tech Computer Science and Engineering — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-302	Paradigms of Programming	3-0-2-4	4
2	DC	CS-303	Software Engineering	2-0-2-3	3
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 343: B. Tech Computer Science and Engineering — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	CS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 344: B. Tech Computer Science and Engineering — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	CS-010	Internship		2
2	MTP	CS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 27 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 345: Computer Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-501	Access Networks	3
2	CS-502	Compiler Design	4
3	CS-506	Cognitive Modeling	3
4	CS-507	Computer Architecture	4
5	CS-508	Introduction to Network Security	3
6	CS-510	Randomized Algorithms	4
7	CS-511	Applied Probability	2
8	CS-512	Matrix Theory	2
9	CS-513	Discrete Mathematics	4

Computer Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
10	CS-514	Advance Data Structure and Algorithm (CS-580)	4
11	CS-515	Advanced Computer Science Practicum	3
12	CS-516	Exploratory Project	3
13	CS-520	Introduction to Quantum Computing	3
14	CS-521	Introduction to Post Quantum Security	3
15	CS-522	Distributed Algorithms	3
16	CS-524	Computational Complexity Theory	4
17	CS-525	Computational Foundations of Cryptography	3
18	CS-526	Mathematical Foundation of Cryptography	3
19	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3
20	CS-541	IoT systems and the Cloud	3
21	CS-542	Design Patterns for Scalable Systems	3
22	CS-544	Formal Concept Analysis: Theory and Practice	3
23	CS-545	Software Design Patterns	3
24	CS-546	Design of Concurrent Software	3
25	CS-547	Network Management Systems	3
26	CS-548	Cloud Networking	3
27	CS-549	Performance Analysis of Computer Networks	3
28	CS-550	Computer Graphics and Geometric Design	3
29	CS-551	Human Computer Interaction	3
30	CS-560	Text Retrieval and Mining	3
31	CS-561	Map Reduce and Big Data	3
32	CS-563	Scalable Data Science	4
33	CS-570	Algorithms for Curves and Surface	3
34	CS-571	Programming Practicum	3
35	CS-591	Special Topics in Knowledge Discovery and Data	1
36	CS-592	Selected Topics in Artificial Intelligence	2
37	CS-601	Probability and Random Processes	4
38	CS-603	Managerial Decision Making	3
39	CS-606	Cognitive Modeling	3
40	CS-609	Speech processing	3
41	CS-611	Program Analysis	4
42	CS-621P	Post Graduate Project-1	12
43	CS-630	Speech Technology	4
44	CS-660	Data Mining for Decision Making	4
45	CS-661	Knowledge Representation and Reasoning	3
46	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
47	CS-670	Kernal Methods for Pattern Recognition	4
48	CS-671	Deep Learning and Applications	4
49	CS-672	Advanced Topics in Deep Learning	4

Computer Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	CS-673	Advanced Computer Vision	4
51	CS-677	Soft Computing	3
52	CS-683	Generative AI	4
53	CS-685	Natural Language Processing	3
54	CS-686	Large Language Models: Core Concepts to Custom	4
55	CS-693	Compressed Sensing and its applications	3
56	DS-201	Data handling and visualization	3
57	DS-301	Mathematical Foundations of Data Science	4
58	DS-314	Probability Theory and Statistics	4
59	DS-401	Optimization for data science	3
60	DS-402	Matrix Computations for Data Science	3
61	DS-403	Introduction to Statistical Learning	3
62	DS-404	Information Security and Privacy	3
63	DS-412	Matrix Computations for Data Science	4
64	DS-415	Optimization for Machine Learning	4
65	DS-416	Fundamentals of Machine Learning (CS-669)	4
66	EE-511	Computer Vision	4
<i>Compulsory project components — counted as DE, not selectable</i>			
67	CS-498P	Major Technical Project - I (compulsory; counted as DE)	4
68	CS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Data Science and Artificial Intelligence

Table 346: Data Science and Artificial Intelligence — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	33
	Discipline Electives	33
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 347: B. Tech Data Science and Artificial Intelligence — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-213	Reverse Engineering	0-0-2-1	1
2	DS-314*	Probability Theory and Statistics	3-1-0-4	4
3	DS-412	Matrix Computations for Data Science	3-0-2-4	4
4	DS-312*	Computing Systems for Data Processing	3-0-2-4	4
5	DS-415*	Optimization for Machine Learning	3-0-2-4	4
6	DS-417 [†]	Deep Learning	3-0-2-4	4
7	CS-315*	Artificial Intelligence	3-0-2-4	4
8	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
9	DS-418 [†]	Introduction to Generative AI	3-0-2-4	4
Total				33

* Recoded w.e.f. batch 2025: CS-305 → CS-315 (Artificial Intelligence) and DS-302 → DS-312 (Computing Systems for Data Processing), both now 3-0-2-4; DS-313 → DS-314 (Probability Theory and Statistics, 3-1-0-4 unchanged); DS-411 → DS-415 (Optimization for Machine Learning) and DS-413 → DS-416 (Fundamentals of Machine Learning), both now 3-0-2-4.

[†]DS-417 (Deep Learning) and DS-418 (Introduction to Generative AI) are DSAI-only codes for the 600-level courses CS-671 and CS-683; DSAI students cannot also take CS-671 or CS-683 as electives. DS-417 is taken in Semester 4 (moved from Semester 6).

Table 348: B. Tech Data Science and Artificial Intelligence — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
6	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
7		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
8		IC-230	Environmental Science	3-0-0-3	
9	IKS + HSS	IC-182	History of Science and Technology	3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 349: B. Tech Data Science and Artificial Intelligence — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-252	Probability and Statistics	3-0-2-4	4
7	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
Total					20

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 350: B. Tech Data Science and Artificial Intelligence — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-222P	Physics Practicum	0-0-3-2	2
3	IC	IC-272	Machine Learning	2-0-2-3	3
4	DC	CS-213	Reverse Engineering	0-0-2-1	1
5	DC	DS-314*	Probability Theory and Statistics	3-1-0-4	4
6	DC	DS-412	Matrix Computations for Data Science	3-0-2-4	4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 351: B. Tech Data Science and Artificial Intelligence — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-312*	Computing Systems for Data Processing	3-0-2-4	4
2	DC	DS-415*	Optimization for Machine Learning	3-0-2-4	4
3	DC	DS-417 [†]	Deep Learning	3-0-2-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 352: B. Tech Data Science and Artificial Intelligence — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-315*	Artificial Intelligence	3-0-2-4	4
2	DC	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
Total					8

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 353: B. Tech Data Science and Artificial Intelligence — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-418 [†]	Introduction to Generative AI	3-0-2-4	4
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					8

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 354: B. Tech Data Science and Artificial Intelligence — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	DS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 355: B. Tech Data Science and Artificial Intelligence — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	DS-010	Internship		2
2	MTP	DS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 33 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 356: Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-507	Probabilistic Robotics	3
2	AR-516	Introduction to Blockchain and Web3	3
3	AR-517	Introduction to Cyber security	3
4	BE-303	Applied Biostatistics	4
5	BE-304	Bioinformatics	4
6	BE-502	Design and Analysis of Bioalgorithms	3
7	CS-212	Design and Analysis of Algorithms	4
8	CS-214	Computer Organization	4
9	CS-241	Introduction to Cryptography	3
10	CS-302	Paradigms of Programming	4

Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
11	CS-303	Software Engineering	3
12	CS-304	Formal Languages and Automata Theory	3
13	CS-306	Operating Systems	3
14	CS-307	Systems Practicum	2
15	CS-308	Large Application Practicum	2
16	CS-308P	Large Application Practicum	2
17	CS-309	Information and Database Systems	4
18	CS-310	Introduction to Communication and Distributed Processes	4
19	CS-313	Computer Networks	4
20	CS-347	Software Engineering	4
21	CS-350	Computer Graphics	3
22	CS-401	Architecture of High Performance Computers	3
23	CS-402	Compilers	2
24	CS-403	Algorithm Design and Analysis I	3
25	CS-405	Verification of Reactive Systems	3
26	CS-406	Computer Networks	4
27	CS-451	Computer Graphics and Game Design	3
28	CS-456	Distributed Databases	3
29	CS-501	Access Networks	3
30	CS-502	Compiler Design	4
31	CS-506	Cognitive Modeling	3
32	CS-507	Computer Architecture	4
33	CS-508	Introduction to Network Security	3
34	CS-510	Randomized Algorithms	4
35	CS-511	Applied Probability	2
36	CS-512	Matrix Theory	2
37	CS-513	Discrete Mathematics	4
38	CS-514	Advance Data Structure and Algorithm (CS-580)	4
39	CS-515	Advanced Computer Science Practicum	3
40	CS-516	Exploratory Project	3
41	CS-520	Introduction to Quantum Computing	3
42	CS-521	Introduction to Post Quantum Security	3
43	CS-522	Distributed Algorithms	3
44	CS-524	Computational Complexity Theory	4
45	CS-525	Computational Foundations of Cryptography	3
46	CS-526	Mathematical Foundation of Cryptography	3
47	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3
48	CS-541	IoT systems and the Cloud	3

Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
49	CS-542	Design Patterns for Scalable Systems	3
50	CS-544	Formal Concept Analysis: Theory and Practice	3
51	CS-545	Software Design Patterns	3
52	CS-546	Design of Concurrent Software	3
53	CS-547	Network Management Systems	3
54	CS-548	Cloud Networking	3
55	CS-549	Performance Analysis of Computer Networks	3
56	CS-550	Computer Graphics and Geometric Design	3
57	CS-551	Human Computer Interaction	3
58	CS-560	Text Retrieval and Mining	3
59	CS-561	Map Reduce and Big Data	3
60	CS-563	Scalable Data Science	4
61	CS-570	Algorithms for Curves and Surface	3
62	CS-571	Programming Practicum	3
63	CS-591	Special Topics in Knowledge Discovery and Data	1
64	CS-592	Selected Topics in Artificial Intelligence	2
65	CS-601	Probability and Random Processes	4
66	CS-603	Managerial Decision Making	3
67	CS-606	Cognitive Modeling	3
68	CS-609	Speech processing	3
69	CS-611	Program Analysis	4
70	CS-630	Speech Technology	4
71	CS-660	Data Mining for Decision Making	4
72	CS-661	Knowledge Representation and Reasoning	3
73	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
74	CS-670	Kernel Methods for Pattern Recognition	4
75	CS-672	Advanced Topics in Deep Learning	4
76	CS-673	Advanced Computer Vision	4
77	CS-677	Soft Computing	3
78	CS-685	Natural Language Processing	3
79	CS-686	Large Language Models: Core Concepts to Custom	4
80	CS-693	Compressed Sensing and its applications	3
81	EE-305	Digital Signal Processing	4
82	EE-511	Computer Vision	4
83	EE-530	Optimization theory	3
84	EE-531	Estimation and Detection Theory	3
85	EE-541	Tensors: Techniques, Algorithms, Applications for Signal	4
86	EE-543	Vision and Learning Based Control	3
87	EE-574	Biomedical Signal and Image Analysis	3

Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
88	EE-575	Applied Statistics for Data and Signals	4
89	EE-608	Digital Image Processing	4
90	EE-620	Advanced Digital Signal Processing	3
91	ME-522	High-Performance Scientific Computing	3
<i>Compulsory project components — counted as DE, not selectable</i>			
92	DS-498P	Major Technical Project - I (compulsory; counted as DE)	4
93	DS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Electrical Engineering

Table 357: Electrical Engineering — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	52
	Discipline Electives	20
Electives	Free Electives	17
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		161

Table 358: B. Tech Electrical Engineering — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-210	Digital Systems Design	3-0-0-3	3
3	EE-210P	Digital Systems Design Lab	0-0-2-1	1
4	EE-260	Signals and Systems	2.5-0.5-0-3	3
5	EE-261	Electrical Systems Around Us	3-0-0-3	3
6	EE-261P	Electrical Systems Around Us Lab	0-0-3-2	2
7	EE-311	Device Electronics	3-0-0-3	3
8	EE-201	Electromechanics	2.5-0.5-0-3	3
9	EE-201P	Electromechanics Lab	0-0-2-1	1
10	EE-205	Electromagnetics & Wave Propagation	3-0-0-3	3
11	EE-211	Analog Circuit Design	2-0-2-3	3
12	EE-223P	Reverse Engineering	1-0-0-1	1
13	EE-304	Communication Systems	3-0-1-4	4
14	EE-231	Measurement and Instrumentation	2-0-2-3	3
15	EE-302	Control Systems	3-0-2-4	4
16	EE-303	Power System	3-1-0-4	4
17	EE-314	Digital Signal Processing	3-0-2-4	4
18	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
Total				52

Table 359: B. Tech Electrical Engineering — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
5		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
6		IC-230	Environmental Science	3-0-0-3	
7	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
8		HS-112	Japanese Language Class I	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
10		HS-342	German I	3-0-0-3	
11	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
12		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 360: B. Tech Electrical Engineering — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket (choose 1)	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9		IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	
10		IC-241	Material Science for Engineers	3-0-0-3	
11		IC-253	Data Structures and Algorithms	2-0-2-3	
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 361: B. Tech Electrical Engineering — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-222P	Physics Practicum	0-0-3-2	2
3	IC	IC-272	Machine Learning	2-0-2-3	3
4	DC	EE-203	Network Theory	3-0-0-3	3
5	DC	EE-210	Digital Systems Design	3-0-0-3	3
6	DC	EE-210P	Digital Systems Design Lab	0-0-2-1	1
7	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
8	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
9	DC	EE-261P	Electrical Systems Around Us Lab	0-0-3-2	2
10	DC	EE-311	Device Electronics	3-0-0-3	3
Total					26

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 362: B. Tech Electrical Engineering — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
2	DC	EE-201P	Electromechanics Lab	0-0-2-1	1
3	DC	EE-205	Electromagnetics & Wave Propagation	3-0-0-3	3
4	DC	EE-211	Analog Circuit Design	2-0-2-3	3
5	DC	EE-223P	Reverse Engineering	1-0-0-1	1
6	DC	EE-304	Communication Systems	3-0-1-4	4
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 363: B. Tech Electrical Engineering — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	EE-302	Control Systems	3-0-2-4	4
3	DC	EE-303	Power System	3-1-0-4	4
4	DC	EE-314	Digital Signal Processing	3-0-2-4	4
5	DC	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 364: B. Tech Electrical Engineering — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 365: B. Tech Electrical Engineering — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	EE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 366: B. Tech Electrical Engineering — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	EE-010	Internship		2
2	MTP	EE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 20 credits of **Discipline Electives** and 17 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 367: Electrical Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BE-302	Bioelectric Systems Modeling	4
2	BE-304	Bioinformatics	4
3	CS-549	Performance Analysis of Computer Networks	3
4	CS-671	Deep Learning and Applications	4
5	EE-500	Network Control System	3
6	EE-501	Power System Operation and Control	3
7	EE-502	Analog System Design Laboratory	3
8	EE-503	Advance Communication Theory	3
9	EE-504	Switch Mode Power Conversion	3
10	EE-505	Materials for Semiconductor Industry	3
11	EE-506	Solar Photovoltaic Energy Systems	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	EE-507	Transmission Lines and Basic Microwave Engineering	4
13	EE-508	Fundamentals of Electric Drives	3
14	EE-508P	Practicum on fundamentals of Electrical Drives	2
15	EE-509	Linear Dynamical Systems	3
16	EE-511	Computer Vision	4
17	EE-512	CMOS Analog IC Design	4
18	EE-513	Special Electrical Machines	3
19	EE-514	Robust Control Systems	3
20	EE-515	Nonlinear Stability and Control	4
21	EE-516	Biomedical Systems	4
22	EE-517	Wireless Communications and Networking	3
23	EE-518	Information Theory	3
24	EE-519	CMOS Digital IC design Practicum	2
25	EE-519P	CMOS Digital IC Design Practicum	2
26	EE-520	Microelectronic Devices and Modelling	3
27	EE-522	Matrix Theory for Engineers	3
28	EE-523	Digital VLSI Architecture Design	4
29	EE-524	Digital MOS LSI Circuits	3
30	EE-526	Power Semiconductor Devices	3
31	EE-527	Analysis and Design of Power Electronic Converters	3
32	EE-527P	Practicum on Analysis and Design of Power Electronic Converters	2
33	EE-528	Modeling and Analysis of Electric Machines	3
34	EE-529	Embedded Systems	4
35	EE-530	Optimization theory	3
36	EE-531	Estimation and Detection Theory	3
37	EE-532	Supervised Research Exposure	3
38	EE-533	Access Network	4
39	EE-534	Probability and Random Processes	3
40	EE-535	Communication and Signal Processing Systems Design	3
41	EE-536	IoT Systems	3
42	EE-537	Power Quality Problems and Mitigation Techniques in Microgrids	3
43	EE-538	Data-driven Control	3
44	EE-540	Wide Band Gap Devices in Power Electronics Applications	3
45	EE-541	Sensors: Techniques, Algorithms, Applications for Signal	4
46	EE-542	Modelling, Simulation and Control of Hybrid Electric Vehicle	3
47	EE-543	Vision and Learning Based Control	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
48	EE-550	Optimal Control and Estimation	3
49	EE-551	Applied Photonics for Scientists and Engineers	3
50	EE-552	Power and Energy Systems	4
51	EE-553	Foundations of Intelligent Communication Systems-I	4
52	EE-554	Low Power VLSI Design	3
53	EE-555	Intelligent Control System	3
54	EE-556	Nuclear Reactor Control	1
55	EE-557	Adaptive Control	3
56	EE-560	Reconfigurable Computing	4
57	EE-570	Advanced Communication Theory	4
58	EE-574	Biomedical Signal and Image Analysis	3
59	EE-575	Applied Statistics for Data and Signals	4
60	EE-580	Network Systems: Modelling and Analysis	3
61	EE-581	Applied Statistics for Data and Signals	4
62	EE-582	Nonlinear Analysis and Control of Power Electronic Converters	3
63	EE-583	Smart Grids	3
64	EE-584	Power System Protection	3
65	EE-585	Theory, Modelling and Characterization of Photo-voltaic Devices	3
66	EE-593	Low Power VLSI Circuits	3
67	EE-597	Intellegent Control System	3
68	EE-601	Solid State Devices	3
69	EE-602	Control System Applications	3
70	EE-603	Renewable Energy and Smart Grid	3
71	EE-604	Practicum on Advanced Electric Drives	3
72	EE-604P	Practicum on Advanced Electric Drives	1
73	EE-605	Information Theory	3
74	EE-606	Introduction to High Voltage Engineering and Dielectric	3
75	EE-607	Optical Communication Systems	3
76	EE-608	Digital Image Processing	4
77	EE-609	Network Information Theory	3
78	EE-611	VLSI Technology	3
79	EE-611P	VLSI Fabrication Practicum	2
80	EE-612	OFDM For Optical Communications	3
81	EE-613	Wireless Communication	3
82	EE-614	Optical Communication Systems	3
83	EE-615	Nanoelectronics and Micronanofabrication	3
84	EE-616	Microwave engineering	2
85	EE-618	Industrial Process Control	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
86	EE-619	Mixed Signal VLSI Design	4
87	EE-620	Advanced Digital Signal Processing	3
88	EE-621	Radiating systems	4
89	EE-622	Microwave Integrated Circuits	3
90	EE-623P	Practicum on Digital Control of Power Electronics and Drives	3
91	EE-626P	Post Graduate Project - I	10
92	EE-630	HVDC Transmission and Flexible AC Transmission Systems	3
93	EE-641	Advance Wireless Technologies	4
94	EE-642	Research Study	3
95	EE-677	Analog Circuit Design	3
96	EE-678	Digital Circuit Design	3
97	EE-691	Advanced Topics in Dielectric Breakdown	1
98	EE-694	Photonics Computing	3
<i>Compulsory project components — counted as DE, not selectable</i>			
99	EE-498P	Major Technical Project - I (compulsory; counted as DE)	4
100	EE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Mechanical Engineering

Table 368: Mechanical Engineering — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	50
	Discipline Electives	16
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 369: B. Tech Mechanical Engineering — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-261	Electrical Systems Around Us	3-0-0-3	3
2	IC-241	Materials Science for Engineers	3-0-0-3	3
3	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	ME-210	Fluid Mechanics	2.5-0.5-0-3	3
5	ME-212	Product Manufacturing Technology	2-0-3-3	3
6	ME-213	Engineering Thermodynamics	3-1-0-4	4
7	ME-100	Reverse Engineering	0-0-2-1	1
8	ME-205	Machine Drawing	1-0-4-3	3
9	ME-210P	Fluid Mechanics Lab	0-0-2-1	1
10	ME-215	Manufacturing Engineering -1	3-0-0-3	3
11	ME-303	Heat Transfer	2.5-0.5-0-3	3
12	ME-307	Energy Conversion Devices	3-0-0-3	3
13	ME-303P	Heat Transfer Lab	0-0-3-1	1
14	ME-305	Design of Machine Elements	3-1-0-4	4
15	ME-309	Theory of Machines	3-1-0-4	4
16	ME-310	System Dynamics and Control	3-0-0-3	3
17	ME-311P	Mechanics of Solids Lab / Design Lab - 1	0-0-2-1	1
18	ME-315	Manufacturing Engineering 2	3-0-0-3	3
19	ME-312P	Theory of Machines Lab / Design Lab - 2	0-0-2-1	1
Total				50

Table 370: B. Tech Mechanical Engineering — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
5		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
6		IC-230	Environmental Science	3-0-0-3	
7	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
8		HS-112	Japanese Language Class I	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
10		HS-342	German I	3-0-0-3	
11	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
12		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 371: B. Tech Mechanical Engineering — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 372: B. Tech Mechanical Engineering — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-222P	Physics Practicum	0-0-3-2	2
3	IC	IC-272	Machine Learning	2-0-2-3	3
4	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
5	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
6	DC	ME-210	Fluid Mechanics	2.5-0.5-0-3	3
7	DC	ME-212	Product Manufacturing Technology	2-0-3-3	3
8	DC	ME-213	Engineering Thermodynamics	3-1-0-4	4
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 373: B. Tech Mechanical Engineering — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-100	Reverse Engineering	0-0-2-1	1
2	DC	ME-205	Machine Drawing	1-0-4-3	3
3	DC	ME-210P	Fluid Mechanics Lab	0-0-2-1	1
4	DC	ME-215	Manufacturing Engineering -1	3-0-0-3	3
5	DC	ME-303	Heat Transfer	2.5-0.5-0-3	3
6	DC	IC-241	Materials Science for Engineers	3-0-0-3	3
7	DC	ME-309	Theory of Machines	3-1-0-4	4
8	DC	ME-311P	Mechanics of Solids Lab / Design Lab - 1	0-0-2-1	1
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 374: B. Tech Mechanical Engineering — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-303P	Heat Transfer Lab	0-0-3-1	1
2	DC	ME-305	Design of Machine Elements	3-1-0-4	4
3	DC	ME-307	Energy Conversion Devices	3-0-0-3	3
4	DC	ME-310	System Dynamics and Control	3-0-0-3	3
5	DC	ME-312P	Theory of Machines Lab / Design Lab - 2	0-0-2-1	1
6	DC	ME-315	Manufacturing Engineering 2	3-0-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 375: B. Tech Mechanical Engineering — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 376: B. Tech Mechanical Engineering — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	ME-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 377: B. Tech Mechanical Engineering — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	ME-010	Internship		2
2	MTP	ME-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 16 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 378: Mechanical Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	ME-451	Refrigeration and Air Conditioning	3
2	ME-452	Robot Kinematics, Dynamics and Control	4
3	ME-501	Nanomanufacturing	3
4	ME-502	Functional Materials	3
5	ME-503	Micro/Nano Scale heat transfer	3
6	ME-504	Numerical Methods for Engineering Computation	3
7	ME-505	Applied Finite Element Method	3
8	ME-506	Fundamentals of Fracture Mechanics	3
9	ME-507	Micro and Nanoscale Fluid Mechanics	3
10	ME-508	Fundamentals of project management	3
11	ME-509	Nano-manufacturing	3
12	ME-510	Advanced Manufacturing Processes	3
13	ME-511	Manufacturing of composites	3
14	ME-513	Finite Element Methods in Engineering	4
15	ME-514	Fundamentals of Multiphase Flow	3
16	ME-516	Polymer Technology for Engineers	3
17	ME-517	Advanced Analytical Techniques for Engineers	4
18	ME-518	Conduction and Radiation	3
19	ME-519	Technical communication for Engineers	2
20	ME-520	Microwave based Manufacturing Processes	3
21	ME-521P	Laboratory course on Vehicle Design and Control	1
22	ME-522	High-Performance Scientific Computing	3
23	ME-523	Product Design	3
24	ME-524	Additive Manufacturing	3
25	ME-526	Programming Paradigm for Open-Source Software	1
26	ME-528	Thermal Management of Electronics	3
27	ME-530	Continuum Mechanics	3
28	ME-595	Experiential Learning in Composite Design, Manufacturing, Testing and Recycling	3
29	ME-600	Research Methodology	1
30	ME-601	Advanced Finite Element Methods	3
31	ME-602	Mechanical Vibration	3
32	ME-603	Advanced Fluid Mechanics	3
33	ME-604	Experimental Methods in Thermal Engineering	3
34	ME-605	Air Conditioning and Ventilation	3
35	ME-606	Advanced Solid Mechanics	3
36	ME-607	Materials Science For Failure Analysis	3
37	ME-608	Thin Films And Devices	3
38	ME-609	Functional Materials	3
39	ME-610	Advanced Thermodynamics	4

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
40	ME-611	Design and Optimization of Thermal Systems	3
41	ME-612	Introduction to Bio-materials	3
42	ME-613	Thermal Radiation	3
43	ME-614	Compressible Flow and Gas Dynamics	3
44	ME-615	Applied Computational Fluid Dynamics	3
45	ME-616	Dielectrics and Related Materials	3
46	ME-617	Mechanics of Composite Materials	3
47	ME-618	Stealth Technology: Infrared Signatures	3
48	ME-619	Experiments in Materials Science	3
49	ME-620	Modeling and Simulation	3
50	ME-621	Aircraft Propulsion	3
51	ME-622	Biomechanics of Musculoskeletal System	3
52	ME-625	Introduction to Turbulence and its Modeling	3
53	ME-626	Acoustics	3
54	ME-627	Mesh Independent Computational Techniques	3
55	ME-628	Impact Mechanics	3
56	ME-630	Machine learning for engineers	3
57	ME-631	Heat Transfer and Fluid Flow in Energy Systems	3
58	ME-632	Mechanics for Energy Systems	3
59	ME-633	Design of Energy Systems	3
60	ME-634	Thermodynamics for Energy Systems	3
61	ME-635	Manufacturing for Energy Systems	3
62	ME-636	Combustion Technology	3
63	ME-637	Wind Power Plant	3
64	ME-638	Solar Thermal Power Plant	3
65	ME-639	Thermal Power Plant Engineering	3
66	ME-640	Solar Power Utilization	3
67	ME-641	Finite Element Method	3
68	MT-201	Physics of Solids	3
69	MT-202	Applied Quantum Mechanics	3
70	MT-203	Materials Synthesis & Characterization	4
71	MT-204	Thermodynamics and Kinetics of Materials	3
72	MT-205	Functional Properties of Materials	4
73	MT-206	Extraction and Materials Processing	4
74	MT-301	Phase Transformations	3
75	MT-302	Transport Phenomena	3
76	MT-303	Computational Materials Science	4
77	MT-304	Mechanical Behavior of Materials	4
78	MT-501	Energy Conversion and Storage Technologies	3
79	MT-502	Recycling and Circular Economy	3

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
80	MT-503	Semiconductor Materials and Devices	3
81	MT-504	Powder metallurgical processing of Materials	3
82	MT-505	Thin Film Technology	3
83	MT-506	Biomaterials	3
84	MT-507	Modelling and Simulations in Material Science	3
85	MT-508	Iron and Steel Making	3
86	MT-509	Hydrogen Energy	3
87	MT-510	Colloids and Interfaces	3
88	MT-511	Sensor Materials and Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
89	ME-498P	Major Technical Project - I (compulsory; counted as DE)	4
90	ME-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Civil Engineering

Table 379: Civil Engineering — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	36
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	49
	Discipline Electives	17
Electives	Free Electives	25
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 380: B. Tech Civil Engineering — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CE-202	Introduction to Civil Engineering Profession	1-0-0-1	1
2	CE-203	Construction Materials	3-0-0-3	3
3	CE-203P	Building (Constructions) Materials Lab	0-0-2-1	1
4	CE-252	Geology and Geomorphology	2-0-2-3	3
5	CE-310	Strength of Materials and Structures	3-0-0-3	3
6	CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
7	CE-201	Surveying Traditional and Digital	2-0-4-4	4
8	CE-251	Hydraulics Engineering	3-0-0-3	3
9	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
10	CE-311	Geotechnical Engineering I	3-0-0-3	3
11	CE-311P	Geotechnical Engineering Lab	0-0-2-1	1
12	CE-404	Analysis of Structures	3-0-0-3	3
13	CE-303	Water Resources Engineering	3-0-0-3	3
14	CE-351	Design of Reinforced Concrete Structures	2-1-0-3	3
15	CE-352	Transportation Engineering	3-0-0-3	3
16	CE-352P	Transportation Engineering Lab	0-0-2-1	1
17	CE-353P	Civil Engineering Drawing	0-0-2-1	1
18	CE-402	Geotechnical Engineering II	2-1-0-3	3
19	CE-305P	Environmental Engineering Lab	0-0-2-1	1
20	CE-356	Reverse Engineering	1-0-0-1	1
21	CE-401	Design of Steel Structures	2-1-0-3	3
22	CE-403	Water and Wastewater Engineering	3-0-0-3	3
Total				49

Table 381: B. Tech Civil Engineering — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS	IC-181	Indian Knowledge Systems	2-0-2-3	3
10	(choose 1)	IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 382: B. Tech Civil Engineering — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	DC	CE-202	Introduction to Civil Engineering Profession	1-0-0-1	1
Total					25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 383: B. Tech Civil Engineering — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CE-203	Construction Materials	3-0-0-3	3
4	DC	CE-203P	Building (Constructions) Materials Lab	0-0-2-1	1
5	DC	CE-252	Geology and Geomorphology	2-0-2-3	3
6	DC	CE-310	Strength of Materials and Structures	3-0-0-3	3
7	DC	CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 384: B. Tech Civil Engineering — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-201	Surveying Traditional and Digital	2-0-4-4	4
2	DC	CE-251	Hydraulics Engineering	3-0-0-3	3
3	DC	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
4	DC	CE-311	Geotechnical Engineering I	3-0-0-3	3
5	DC	CE-311P	Geotechnical Engineering Lab	0-0-2-1	1
6	DC	CE-404	Analysis of Structures	3-0-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 385: B. Tech Civil Engineering — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-303	Water Resources Engineering	3-0-0-3	3
2	DC	CE-351	Design of Reinforced Concrete Structures	2-1-0-3	3
3	DC	CE-352	Transportation Engineering	3-0-0-3	3
4	DC	CE-352P	Transportation Engineering Lab	0-0-2-1	1
5	DC	CE-353P	Civil Engineering Drawing	0-0-2-1	1
6	DC	CE-402	Geotechnical Engineering II	2-1-0-3	3
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 386: B. Tech Civil Engineering — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-305P	Environmental Engineering Lab	0-0-2-1	1
2	DC	CE-356	Reverse Engineering	1-0-0-1	1
3	DC	CE-401	Design of Steel Structures	2-1-0-3	3
4	DC	CE-403	Water and Wastewater Engineering	3-0-0-3	3
5	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 387: B. Tech Civil Engineering — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	CE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 388: B. Tech Civil Engineering — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	CE-010	Internship		2
2	MTP	CE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 17 credits of **Discipline Electives** and 25 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 389: Civil Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CD-591	Special Topics in Disaster Resilient Hill Roads	1
2	CE-304	Hydraulics Engineering Lab	3
3	CE-305	Environment and Earth Science Lab	3
4	CE-306	Environmental Engineering Lab	3
5	CE-353	Civil Engineering Drawing	3
6	CE-354	Building and Pavement Materials Laboratory	3

Civil Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
7	CE-355	Hydrology	3
8	CE-405	Water and Wastewater Engineering	3
9	CE-451	Irrigation Engineering and Hydraulic Structures	3
10	CE-452	Open Channel Flow	3
11	CE-501	Remote Sensing	3
12	CE-502	Groundwater Flow and Contaminant Transport	3
13	CE-503	Fundamentals of Project Management	3
14	CE-504	Slope Stability and Retaining Structures	3
15	CE-505	Engineering of Ground Modification	3
16	CE-506	Analysis of Indeterminate Structures	3
17	CE-507	Advanced Concrete Science	3
18	CE-508	Photogeology and Photogrammetry	3
19	CE-509	Bridge Engineering	3
20	CE-510	Modelling and Simulation in Water Resources Engineering	3
21	CE-511	Structural Dynamics with Application to Earthquake Engineering	3
22	CE-512	Advanced Soil Mechanics	3
23	CE-513	Advanced Foundation Engineering	3
24	CE-514	Rock Mechanics	3
25	CE-515	Environmental Impact Assessment	3
26	CE-516	Uncertainty Analysis in Civil Engineering	3
27	CE-517	Hydroinformatics	4
28	CE-518	Structural Reliability and Risk Assessment	3
29	CE-519	Chemistry of Natural Waters	3
30	CE-520	Environmental Reaction Modeling	3
31	CE-521	Ecology and Environment Microbiology	4
32	CE-522	Matrix Methods for Structural Analysis	3
33	CE-523	Building Science	3
34	CE-524	Applied Hydroclimatology	3
35	CE-524P	Computational Hydroclimatology Lab	1
36	CE-526	Critical Zone Science	3
37	CE-551	Geosynthetics and their applications	3
38	CE-552	Concrete Technology	3
39	CE-554	Prestressed Concrete Structures	3
40	CE-555	Advanced Design of Structures	3
41	CE-556	Structural Engineering Laboratory	3
42	CE-556P	Structural Engineering laboratory	2
43	CE-557	Solid Mechanics in Structural Engineering	3
44	CE-558	Air pollution and its mitigation	3
45	CE-559	Biological Wastewater Treatment	3

Civil Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
46	CE-560	Soil Dynamics	3
47	CE-561	The science of climate change	3
48	CE-562	Numerical Methods for Engineering Computation	3
49	CE-563	Finite Element Methods in Engineering	4
50	CE-564	Fundamentals of Fracture Mechanics	3
51	CE-567	Highway Safety and Modeling	3
52	CE-568P	Experimental Structural Dynamics	1
53	CE-586	Mini Project	3
54	CE-586P	Mini Project	3
55	CE-587	Industrial / Academic Internship	3
56	CE-591	Special Topics in Civil and Mechanical Engineering	3
57	CE-600	Research Methodology for Civil Engineering	1
58	CE-601	Geo-Informatics	3
59	CE-602	Blast Engineering	3
60	CE-604	Theory of Plasticity	3
61	CE-605	Engineering Seismology and Seismic Hazard Assessment	3
62	CE-606	Constitutive Modeling of Frictional Material	3
63	CE-610	Analysis and Design for Earthquake Resistant Structures	3
64	CE-611	Structural Health Monitoring	3
65	CE-612	Theory of Plates and Shells	3
66	CE-613	Mechanics of Unsaturated Soils	3
67	CE-617	Mechanics of Composite Materials	3
68	CE-688	Post Graduate Project - 1	3
69	CE-689	Post Graduate Project - II	3
<i>Compulsory project components — counted as DE, not selectable</i>			
70	CE-498P	Major Technical Project - I (compulsory; counted as DE)	4
71	CE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Engineering Physics

Table 390: Engineering Physics — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	36
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	37
	Discipline Electives	29
Electives	Free Electives	25
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 391: B. Tech Engineering Physics — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EP-301	Engineering Mathematics 2	3-1-0-4	4
2	EP-321	Foundations of Electrodynamics	3-0-0-3	3
3	PH-301	Quantum Mechanics and Applications	2.5-0.5-0-3	3
4	EE-223P	Reverse Engineering	1-0-0-1	1
5	EP-403	Physics of Atoms and Molecules	3-0-0-3	3
6	PH-302	Introduction to Statistical Mechanics	2.5-0.5-0-3	3
7	PH-501	Solid State Physics	3	3
8	EE-311	Device Electronics for Integrated Circuits	3-0-0-3	3
9	EP-302	Computational Methods for Engineering	2-0-2-3	3
10	EP-402P	Engineering Physics Practicum	1-0-5-4	4
11	EP-401P	Engineering of Instrumentation	1-0-5-4	4
12	PH-502	Photonics	3-0-0-3	3
Total				37

Table 392: B. Tech Engineering Physics — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS	IC-181	Indian Knowledge Systems	2-0-2-3	3
10	(choose 1)	IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 393: B. Tech Engineering Physics — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 394: B. Tech Engineering Physics — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EP-301	Engineering Mathematics 2	3-1-0-4	4
4	DC	EP-321	Foundations of Electrodynamics	3-0-0-3	3
5	DC	PH-301	Quantum Mechanics and Applications	2.5-0.5-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 395: B. Tech Engineering Physics — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-223P	Reverse Engineering	1-0-0-1	1
2	DC	EP-403	Physics of Atoms and Molecules	3-0-0-3	3
3	DC	PH-302	Introduction to Statistical Mechanics	2.5-0.5-0-3	3
4	DC	PH-501	Solid State Physics	3	3
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 396: B. Tech Engineering Physics — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-311	Device Electronics for Integrated Circuits	3-0-0-3	3
2	DC	EP-302	Computational Methods for Engineering	2-0-2-3	3
3	DC	EP-402P	Engineering Physics Practicum	1-0-5-4	4
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 397: B. Tech Engineering Physics — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EP-401P	Engineering of Instrumentation	1-0-5-4	4
2	DC	PH-502	Photonics	3-0-0-3	3
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 398: B. Tech Engineering Physics — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	EP-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 399: B. Tech Engineering Physics — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	EP-010	Internship		2
2	MTP	EP-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 29 credits of **Discipline Electives** and 25 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 400: Engineering Physics — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-208	Mathematical Foundations of Computer Science	4
2	CS-241	Introduction to Cryptography	3
3	CS-304	Formal Languages and Automata Theory	3
4	CS-309	Information and Database Systems	4
5	CS-511	Applied Probability	2
6	CS-671	Deep Learning and Applications	4
7	CS-672	Advanced Topics in Deep Learning	4
8	DS-201	Data handling and visualization	3
9	DS-301	Mathematical Foundations of Data Science	4

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
10	DS-401	Optimization for data science	3
11	DS-404	Information Security and Privacy	3
12	DS-416	Fundamentals of Machine Learning (CS-669)	4
13	EE-203	Network Theory	3
14	EE-211	Analog Circuit Design	3
15	EE-307	Theory of Measurement	3
16	EE-511	Computer Vision	4
17	EE-512	CMOS Analog IC Design	4
18	EE-519P	CMOS Digital IC Design Practicum	2
19	EE-520	Microelectronic Devices and Modelling	3
20	EE-524	Digital MOS LSI Circuits	3
21	EE-534	Probability and Random Processes	3
22	EE-551	Applied Photonics for Scientists and Engineers	3
23	EE-593	Low Power VLSI Circuits	3
24	EE-611	VLSI Technology	3
25	EE-614	Optical Communication Systems	3
26	EE-621	Radiating systems	4
27	EN-502	Emerging Energy Sources	3
28	EP-201	Reverse Engineering	1
29	EP-401	Engineering of Instrumentation	3
30	EP-402	Engineering Physics Practicum	3
31	EP-502	Informatics for Materials Design	3
32	IC-253	Programming and Data Structures	3
33	MA-210	Real and Complex Analysis	3
34	MA-511	Real Analysis	4
35	MA-512	Linear Algebra	4
36	MA-513	Ordinary Differential Equation	4
37	MA-516	Topology	4
38	MA-521	Topology and Functional Analysis	4
39	MA-522	Partial Differential Equations	4
40	MA-560	Nonlinear Dynamics and Chaos	3
41	ME-210	Fluid Mechanics	3
42	ME-303	Heat Transfer	3
43	ME-307	Energy Conversion Devices	3
44	ME-308	Manufacturing Engineering	3
45	ME-503	Micro/Nano Scale heat transfer	3
46	ME-509	Nano-manufacturing	3
47	ME-510	Advanced Manufacturing Processes	3
48	ME-603	Advanced Fluid Mechanics	3
49	ME-615	Applied Computational Fluid Dynamics	3

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	PH-503	Laser and Applications	3
51	PH-504	Organic Optoelectronics	3
52	PH-507	X-rays as a Probe to Study Material Properties	3
53	PH-508	Magnetism and Magnetic Materials	3
54	PH-511	Mathematical Physics	4
55	PH-512	Classical Mechanics	4
56	PH-513	Quantum Mechanics	3
57	PH-514	Electronics	3
58	PH-515P	Physics Laboratory	3
59	PH-516	Research Project I	2
60	PH-521	Electromagnetic Theory	4
61	PH-525P	Electronics Lab	3
62	PH-528	Introduction to General Relativity	3
63	PH-530	Cosmology-I	3
64	PH-550	Introduction to Quantum Optics	3
65	PH-579	Quantum Computation and Information	3
66	PH-600	Research Methodology	1
67	PH-601	Mesoscopic Physics and Quantum Transport	3
68	PH-603	Advanced Condensed Matter Physics	3
69	PH-604	Optical Properties of Solids	3
70	PH-605	Superconductivity	3
71	PH-606	Quantum Field Theory	3
72	PH-607	Physics of Ultracold Quantum Gases	3
73	PH-608	Computer assisted quantum mechanics	3
74	PH-609	Theory of quantum collision and spectroscopy	3
75	PH-612	Nuclear and Particle Physics	3
76	PH-613	Special Topics in Quantum Mechanics	3
77	PH-615P	Mini Thesis I	3
78	PH-621	Computational Methods for Physicists	4
79	PH-625	Data Analysis in Particle Physics	4
80	PH-626	Elementary Theoretical Particle Physics	3
81	PH-627	Topological Quantum Matter	3
82	PH-701	Introduction to Molecular Simulations	4
83	PH-706	Introduction to Stochastic Problems in Physics	3
84	QS-501P	Experiments in Quantum Optics	3
85	QT-301	Survey of Quantum Technologies and Applications	3
86	QT-406	Introduction to Quantum communication	3
87	QT-407	Introduction to Quantum Sensing	3
88	QT-509	Engineering Foundations of Quantum Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
89	EP-498P	Major Technical Project - I (compulsory; counted as DE)	4
90	EP-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Bioengineering

Table 401: Bioengineering — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	36
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	42
	Discipline Electives	24
Electives	Free Electives	25
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 402: B. Tech Bioengineering — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	BE-201	Cell Biology	3-0-2-4	4
2	BE-202	Biochemistry and Molecular Biology	3-0-2-4	4
3	BE-308	Introduction to Biomanufacturing	3-0-2-4	4
4	BE-309	Biosensing & Bioinstrumentation	3-0-2-4	4
5	BE-203	Enzymology and Bioprocessing	3-0-2-4	4
6	BE-301	Biomechanics	3-0-2-4	4
7	BE-304	Bioinformatics	3-0-2-4	4
8	BE-101P	Reverse Engineering	0-0-2-1	1
9	BE-303	Applied Biostatistics	3-0-2-4	4
10	BE-305	Bioethics and Regulatory Affairs	1-0-0-1	1
11	BE-306	Fundamentals of Genetic Engineering	3-1-0-4	4
12	BE-310	Biomaterials	3-0-2-4	4
Total				42

Table 403: B. Tech Bioengineering — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS	IC-181	Indian Knowledge Systems	2-0-2-3	3
10	(choose 1)	IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 404: B. Tech Bioengineering — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 405: B. Tech Bioengineering — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	BE-201	Cell Biology	3-0-2-4	4
4	DC	BE-202	Biochemistry and Molecular Biology	3-0-2-4	4
5	DC	BE-308	Introduction to Biomanufacturing	3-0-2-4	4
6	DC	BE-309	Biosensing & Bioinstrumentation	3-0-2-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 406: B. Tech Bioengineering — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	BE-203	Enzymology and Bioprocessing	3-0-2-4	4
2	DC	BE-301	Biomechanics	3-0-2-4	4
3	DC	BE-304	Bioinformatics	3-0-2-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 407: B. Tech Bioengineering — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	BE-101P	Reverse Engineering	0-0-2-1	1
2	DC	BE-303	Applied Biostatistics	3-0-2-4	4
3	DC	BE-305	Bioethics and Regulatory Affairs	1-0-0-1	1
4	DC	BE-306	Fundamentals of Genetic Engineering	3-1-0-4	4
5	DC	BE-310	Biomaterials	3-0-2-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 408: B. Tech Bioengineering — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 409: B. Tech Bioengineering — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	BE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 410: B. Tech Bioengineering — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	BE-010	Internship		2
2	MTP	BE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 24 credits of **Discipline Electives** and 25 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 411: Bioengineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-506	Cognitive Robotics	3
2	BE-101	Reverse Engineering for Bioengineers	3
3	BE-302	Bioelectric Systems Modeling	4
4	BE-307	Reverse Engineering for Bioengineers	3
5	BE-401	Bioengineering Mini Project, Term Paper and Seminar	4
6	BE-501	Anatomy and Physiology	3
7	BE-502	Design and Analysis of Bioalgorithms	3
8	BE-503	Biosensing & Bioinstrumentation	4
9	BE-504	Biomaterials	4
10	BE-505	Computational Biology	3
11	BE-506	Biological Modelling and Simulation	3

Bioengineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	BE-507	Environmental Impact Assessment	3
13	BE-598	Thesis Project Part I	3
14	BE-599	Thesis Project Part II	3
15	BE-698	Thesis Project Part I	3
16	BE-699	Thesis Project Part II	17
17	BY-501	DNA Nanotechnology	3
18	BY-502	Biophysics and Protein Engineering	3
19	BY-503	Cellular Fuel and Cellular Communication	3
20	BY-504	Metabolic Systems Biology	4
21	BY-506	Advanced Immunology	3
22	BY-507	Genetic Engineering	4
23	BY-508	Practical Metabolomics	3
24	BY-509	Practical OMICs	3
25	BY-510	Advanced Cell Biology	3
26	BY-511	Cell Physiology in health and disease	3
27	BY-512	Quantitative and Computational Biology	3
28	BY-514	Analytical Biotechniques	3
29	BY-515	Molecular Biotechnology	3
30	BY-516	Introduction to OMICS and Systems Analysis	3
31	BY-517	Proteomics	3
32	BY-518	Disease Biology	3
33	BY-519	Protein Sciences in Therapeutics	3
34	BY-520	Cell Biology and Physiology Lab	3
35	BY-520P	Cell Biology and Physiology	1
36	BY-521	Computational Biology and Cellular Bioprocess Technology Lab	3
37	BY-521P	Computational Biology and Cellular Bioprocess Technology Lab	1
38	BY-522	Analytical and Molecular Biotechnology Lab	3
39	BY-523	Systems Biology Lab	3
40	BY-524	Medical and NanoBiotechnology Lab	3
41	BY-527	Gene silencing and genome editing: principles and applications	3
42	BY-528	Sensory Biology	3
43	BY-529	Mechanobiology of the Cell (MBoC)	3
44	BY-531	Quantitative Biology and Data Analytics	3
45	BY-532	Immunotechnology	3
46	BY-535P	Analytical Biotechniques Lab	1
47	BY-536P	Immunotechnology Lab	1
48	BY-537	Computational Biology-1	4
49	BY-606	Bioinformatics Applications for Systems	3

Bioengineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	BY-613	Metagenomics and Next Generation Sequencing Technologies	3
51	CE-521	Ecology and Environment Microbiology	4
52	CE-559	Biological Wastewater Treatment	3
53	CS-309	Information and Database Systems	4
54	CS-606	Cognitive Modeling	3
55	CS-671	Deep Learning and Applications	4
56	CY-344	Food Chemistry: Processing, Preservation and Storage	3
57	CY-550	Bioinspired Materials	3
58	CY-642	Molecular- and Bio-electronics	3
59	CY-643	Advanced Analytical Techniques	3
60	CY-644	Bioinorganic chemistry	3
61	CY-670	Fluorescence spectroscopy, microscopy and applications	3
62	DS-303	Statistical Foundations of Data Science	3
63	DS-314	Probability Theory and Statistics	4
64	EE-305	Digital Signal Processing	4
65	EE-314	Digital Signal Processing	4
66	EE-516	Biomedical Systems	4
67	EE-574	Biomedical Signal and Image Analysis	3
68	EE-608	Digital Image Processing	4
69	MA-621	Modeling Population Dynamics	3
70	MA-650	Mathematical Models for Infectious Diseases	3
71	ME-303	Heat Transfer	3
72	ME-308	Manufacturing Engineering	3
73	ME-527	Biofluid Dynamics	3
74	ME-612	Introduction to Bio-materials	3
75	ME-622	Biomechanics of Musculoskeletal System	3
76	MT-506	Biomaterials	3
<i>Compulsory project components — counted as DE, not selectable</i>			
77	BE-498P	Major Technical Project - I (compulsory; counted as DE)	4
78	BE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Mathematics and Computing

Table 412: Mathematics and Computing — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	51
	Discipline Electives	15
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 413: B. Tech Mathematics and Computing — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
2	MA-210	Real and Complex Analysis	2.5-0.5-0-3	3
3	MA-211	Ordinary Differential Equations	3-1-0-4	4
4	MA-312 / CS-212	Design of Algorithms	3-0-2-4	4
5	CS-201	Computer Organization	3-0-0-3	3
5b	CS-201P	Computer Organization Laboratory	0-0-2-1	1
6	MA-220	Partial Differential Equations	3-1-0-4	4
7	MA-221	Numerical Analysis	3-1-0-4	4
8	MA-222	Applied Linear Programming	3-1-0-4	4
9	MA-310	Matrix Computation and Lab	3-0-2-4	4
10	MA-311	Mathematical Modelling	3-0-0-3	3
11	MA-313 / CS-304	Formal Language and Automata Theory	3-0-0-3	3
12	MA-322	Applied Graph Theory		4
13	MA-321	Numerics of Differential Equation		4
14	MA-323P	Applied Databases Practicum	0-0-3-2	2
Total				51

Table 414: B. Tech Mathematics and Computing — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS	IC-181	Indian Knowledge Systems	2-0-2-3	3
10	(choose 1)	IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 415: B. Tech Mathematics and Computing — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 416: B. Tech Mathematics and Computing — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-222P	Physics Practicum	0-0-3-2	2
3	IC	IC-272	Machine Learning	2-0-2-3	3
4	DC	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
5	DC	MA-210	Real and Complex Analysis	2.5-0.5-0-3	3
6	DC	MA-211	Ordinary Differential Equations	3-1-0-4	4
7	DC	MA-312 / CS-212	Design of Algorithms	3-0-2-4	4
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 417: B. Tech Mathematics and Computing — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-201	Computer Organization	3-0-0-3	3
2	DC	CS-201P	Computer Organization Laboratory	0-0-2-1	1
3	DC	MA-220	Partial Differential Equations	3-1-0-4	4
4	DC	MA-221	Numerical Analysis	3-1-0-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 418: B. Tech Mathematics and Computing — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	MA-310	Matrix Computation and Lab	3-0-2-4	4
2	DC	MA-311	Mathematical Modelling	3-0-0-3	3
3	DC	MA-313 / CS-304	Formal Language and Automata Theory	3-0-0-3	3
4	DC	MA-222	Applied Linear Programming	3-1-0-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 419: B. Tech Mathematics and Computing — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	MA-321	Numerics of Differential Equation	3-0-2-4	4
2	DC	MA-323P	Applied Databases Practicum	0-0-3-2	2
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
4	DC	MA-322	Applied Graph Theory	3-0-2-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 420: B. Tech Mathematics and Computing — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	MC-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 421: B. Tech Mathematics and Computing — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	MC-010	Internship		2
2	MTP	MC-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 15 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

This programme groups its Discipline Electives into the named baskets below; the remaining approved electives follow as a general list.

Table 422: Mathematics and Computing — Discipline Elective Basket I — Foundation Module (choose from these)

Sl. No	Course Code	Course Name	Credits
1	MA-251	Abstract Algebra	3
2	MA-252	Functional Analysis	3
3	MA-253	Measure Theory	3
4	MA-254	Topology	3
5	MA-255	Number Theory	3

Table 423: Mathematics and Computing — Discipline Elective Basket II — Advanced Modelling Module (choose from these)

Sl. No	Course Code	Course Name	Credits
1	MA-351	Climate Modelling	3
2	MA-352	Computational Financial Modelling & Lab	3
3	MA-353	Modelling of Infectious Disease	3
4	MA-354	Mathematical Image Processing	3
5	MA-355	Mathematical Control Theory	3
6	MA-356	Modelling and Simulation	3
7	MA-357	Modelling Population Dynamics	3
8	MA-546	Introduction to Quantitative Finance	4
9	MA-570	Data Driven Dynamical Systems	3

Table 424: Mathematics and Computing — Other Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BE-303	Applied Biostatistics	4
2	BE-304	Bioinformatics	4
3	CE-251	Hydraulics Engineering	3
4	CS-241	Introduction to Cryptography	3
5	CS-302	Paradigms of Programming	4
6	CS-306	Operating Systems	3
7	CS-309	Information and Database Systems	4
8	CS-313	Computer Networks	4
9	CS-315	Artificial Intelligence (CS-362, CS-562)	4
10	CS-502	Compiler Design	4
11	CS-514	Advance Data Structure and Algorithm (CS-580)	4
12	CS-563	Scalable Data Science	4
13	CS-606	Cognitive Modeling	3
14	CS-609	Speech processing	3
15	CS-677	Soft Computing	3
16	DS-201	Data handling and visualization	3
17	DS-312	Computing Systems for Data Processing	4
18	DS-416	Fundamentals of Machine Learning (CS-669)	4
19	EE-203	Network Theory	3
20	EE-260	Signals and Systems	3
21	EE-301	Control Systems	3
22	EE-511	Computer Vision	4
23	EE-530	Optimization theory	3
24	EE-608	Digital Image Processing	4
25	MA-001	Preparatory Mathematics - 1	4
26	MA-002	Preparatory Mathematics - 2	4

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
27	MA-101	Calculus I	4
28	MA-102	Linear Algebra	4
29	MA-120	Introduction to Computing Systems and Databases	4
30	MA-201	Probability and Statistics	4
31	MA-460	Nonlinear Dynamics and Chaos	3
32	MA-465	Ordinary Differential Equations	3
33	MA-510	Climate Change Analysis	3
34	MA-516	Topology	4
35	MA-521	Topology and Functional Analysis	4
36	MA-522	Partial Differential Equations	4
37	MA-523	Numerical Analysis	4
38	MA-524	Probability and Statistics	4
39	MA-525	Heuristic Optimization	3
40	MA-526	An Introduction to Wavelets	3
41	MA-527	Field and Galois Theory	3
42	MA-528	Measure Theory and Integration	4
43	MA-529	Statistical Inference	4
44	MA-530	Graph Theory	4
45	MA-546	Introduction to Quantitative Finance	4
46	MA-548	Abstract Algebra	4
47	MA-549	Abstract Algebra	4
48	MA-550	Statistical Data Analysis	3
49	MA-551	Numerical Analysis	3
50	MA-552	Number Theory	3
51	MA-553	Mathematical Foundations of Financial Engineering	3
52	MA-555	Introduction to Partial Differential Equations for Engineers	3
53	MA-560	Nonlinear Dynamics and Chaos	3
54	MA-563	Statistical Computation and Simulation	4
55	MA-565	Numerical Methods in Quantitative Finance	3
56	MA-568	Real Analysis	3
57	MA-570	Data-driven Dynamical Systems	3
58	MA-571	Introduction to Approximation algorithms	4
59	MA-572	Applied Multivariate Statistical Analysis	4
60	MA-581P	Post Graduate Project I	6
61	MA-582P	Post Graduate Project-2	8
62	MA-588	Mathematical Control Theory	4
63	MA-595	Calculus as Ganita	3
64	MA-601	Real and Functional Analysis	3
65	MA-603	Advanced Partial Differential Equations	3
66	MA-604	Introduction To Game Theory	3

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
67	MA-605	Statistical Data Analysis	3
68	MA-607	Numerical Analysis	3
69	MA-608	Computational Fluid Dynamics	3
70	MA-609	Numerics of Partial Differential Equations	3
71	MA-610	Mathematical Modeling	3
72	MA-611	Statistical tools and Computing	4
73	MA-612	Operator Theory	4
74	MA-621	Modeling Population Dynamics	3
75	MA-641	Operations Research	3
76	MA-644	Dynamical Systems	3
77	MA-650	Mathematical Models for Infectious Diseases	3
78	MA-651	Optimization Techniques	3
79	MA-652	Stability Theory of Differential Equations	3
80	MA-653	Computational Financial Modelling	3
81	MA-653P	Computational Financial Modelling Lab	1
82	MA-654	Financial Engineering	3
83	MA-655	Fixed Income Securities	3
84	MA-656	Stochastic Calculus for Financial Engineering	3
85	MA-665	Semigroup Of Bounded Linear Operators	3
86	MA-704	Dynamical System	3
87	MA-709	Numerical Linear Algebra	3
88	MA-765	Fractional Differential Equations	4
89	MA-780	Topics on Semigroup Theory	3
90	ME-210	Fluid Mechanics	3
91	ME-213	Engineering Thermodynamics	4
92	ME-303	Heat Transfer	3
93	ME-308	Manufacturing Engineering	3
94		Time Series Analysis	3
95		Mathematical Method for Signal Processing	4
96	CS-669	Pattern Recognition	4
97	BE-301	Biomechanics	4
98	BY-507	Genetic Engineering	4
99	CS-352	Transportation Engineering	3
100	CS-352P	Transporting Engg. Lab	1
101		Information Systems and Databases	4
102		Times Series Analysis and Applications / Bayesian Data Analysis and Applications	3
103		Big Data: Management and Analytics	4
104	EP-302	Computational Methods for Engineering	3
105	ME-310	System Dynamics and Control	3
106	ME-210P	Fluid Mechanics Lab	1

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
107	ME-303P	Heat Transfer Lab	1
108		Cellular Automata	3
109	CS-606	Computational Modeling of Social Systems	3
110	MA-562	An Introduction to Bayesian Analysis	4
111	MA-561	Fluidics and Microfluidics	4
112	MA-573	Advanced Mathematical Foundations of Machine Learning	4
113	MA-574	An Introduction to Probabilistic Machine Learning	4
114	MA-576	Nonlinear Optimization	4
<i>Compulsory project components — counted as DE, not selectable</i>			
115	MC-498P	Major Technical Project - I (compulsory; counted as DE)	4
116	MC-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - AI and Robotics

Table 425: General Engineering - AI and Robotics — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	34
	Discipline Electives	32
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 426: B. Tech General Engineering - AI and Robotics — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-261	Electrical System Around Us	3-0-0-3	3
2	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
3	EE-201	Electromechanics	2.5-0.5-0-3	3
4	IC-253	Programming and Data Structures	2-0-2-3	3
5	ME-100	Reverse Engineering	0-0-2-1	1
6	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
7	AR-521	Control of Robotic Systems	3-0-0-3	3
8	AR-523	Robot Manipulators	2-1-0-3	3
9	EE-302	Control Systems	3-0-2-4	4
10	ME-305	Design of Machine Elements	3-1-0-4	4
11	ME-309	Theory of Machines	3-1-0-4	4
Total				34

Table 427: B. Tech General Engineering - AI and Robotics — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Computing and Data Science	3-0-2-4	4
4	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
5		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
6		IC-230	Environmental Science	3-0-0-3	
7	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
8		HS-112	Japanese Language Class I	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
10		HS-342	German I	3-0-0-3	
11	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
12		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 428: B. Tech General Engineering - AI and Robotics — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 429: B. Tech General Engineering - AI and Robotics — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-261	Electrical System Around Us	3-0-0-3	3
4	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 430: B. Tech General Engineering - AI and Robotics — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
2	DC	IC-253	Programming and Data Structures	2-0-2-3	3
3	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 431: B. Tech General Engineering - AI and Robotics — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
2	DC	AR-521	Control of Robotic Systems	3-0-0-3	3
3	DC	AR-523	Robot Manipulators	2-1-0-3	3
4	DC	EE-302	Control Systems	3-0-2-4	4
5	DC	ME-305	Design of Machine Elements	3-1-0-4	4
6	DC	ME-309	Theory of Machines	3-1-0-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 432: B. Tech General Engineering - AI and Robotics — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 433: B. Tech General Engineering - AI and Robotics — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 434: B. Tech General Engineering - AI and Robotics — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 32 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Odd-semester electives are offered in Semesters 3, 5 and 7; even-semester electives in Semesters 4, 6 and 8.

Table 435: General Engineering - AI and Robotics — Discipline Electives, Odd Semesters (3, 5, 7)

Sl. No	Course Code	Course Name	Credits
1	AR-511	Autonomous Mobile Robots	3
2	AR-522	Robot Vision	3
3	CS-212	Design of Algorithms	4
4	CS-315	Artificial Intelligence	4
5	EE-203	Network Theory	3
6	EE-260	Signals and Systems	3
7	EE-511	Computer Vision	4
8	GE-303	Modern Powertrains: Electric, Hybrid & IC/Aeroengines	3
9	IC-241	Material Science for Engineers	3
10	ME-210	Fluid Mechanics	3
11	ME-212	Product Manufacturing Technology	3
12	ME-213	Thermodynamics	4

Table 436: General Engineering - AI and Robotics — Discipline Electives, Even Semesters (4, 6, 8)

Sl. No	Course Code	Course Name	Credits
1	AR-508	Marine Robotics	3

General Engineering - AI and Robotics — Discipline Electives, Even Semesters (4, 6, 8) (continued)

Sl. No	Course Code	Course Name	Credits
2	AR-513	Unmanned Aerial System	3
3	AR-524	Robot Simulators	3
4	AR-525	Reinforcement Learning for Robotics	3
5	AR-526	Multimodal Learning	3
6	CE-251	Hydraulics Engineering	3
7	CS-671	Deep Learning and Applications	4
8	CS-673	Advanced Computer Vision	4
9	CS-683	Generative AI	4
10	DS-314	Probability Theory and Statistics	4
11	EE-211	Analog Circuit Design	4
12	EE-608	Digital Image Processing	4
13	IC-241	Material Science for Engineers	3
<i>Compulsory project components — counted as DE, not selectable</i>			
14	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
15	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Mechatronics

Table 437: General Engineering - Mechatronics — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	38
	Discipline Electives	28
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 438: B. Tech General Engineering - Mechatronics — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-260	Signals and Systems	2.5-0.5-0-3	3
3	EE-261	Electrical System Around Us	3-0-0-3	3
4	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
5	EE-201	Electromechanics	2.5-0.5-0-3	3
6	EE-211	Analog Circuit Design	2-0-2-3	3
7	ME-100	Reverse Engineering	0-0-2-1	1
8	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
9	CS-315 [†]	Artificial Intelligence	3-0-2-4	4
10	EE-302	Control Systems	3-0-2-4	4
11	ME-305	Design of Machine Elements	3-1-0-4	4
12	ME-309	Theory of Machines	3-1-0-4	4
Total				38

[†]CS-305 Artificial Intelligence (3-0-0-3) has been recoded as CS-315 (3-0-2-4) w.e.f. batch 2025; the extra credit is adjusted against Discipline Electives, keeping the total at 160.

Table 439: B. Tech General Engineering - Mechatronics — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Computing and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
10		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 440: B. Tech General Engineering - Mechatronics — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 441: B. Tech General Engineering - Mechatronics — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-203	Network Theory	3-0-0-3	3
4	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
5	DC	EE-261	Electrical System Around Us	3-0-0-3	3
6	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 442: B. Tech General Engineering - Mechatronics — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
2	DC	EE-211	Analog Circuit Design	2-0-2-3	3
3	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 443: B. Tech General Engineering - Mechatronics — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
2	DC	CS-315 [†]	Artificial Intelligence	3-0-2-4	4
3	DC	EE-302	Control Systems	3-0-2-4	4
4	DC	ME-305	Design of Machine Elements	3-1-0-4	4
5	DC	ME-309	Theory of Machines	3-1-0-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 444: B. Tech General Engineering - Mechatronics — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 445: B. Tech General Engineering - Mechatronics — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 446: B. Tech General Engineering - Mechatronics — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 28 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Odd-semester electives are offered in Semesters 3, 5 and 7; even-semester electives in Semesters 4, 6 and 8.

Table 447: General Engineering - Mechatronics — Discipline Electives, Odd Semesters (3, 5, 7)

Sl. No	Course Code	Course Name	Credits
1	AR-521	Control of Robotic Systems	3
2	CS-212	Design of Algorithms	4
3	CS-312	Operating System	4
4	CS-313	Computer Networks	4
5	DS-314	Probability Theory and Statistics	4
6	EE-210	Digital Systems Design	4
7	EE-231	Measurement and Instrumentation	3
8	EE-311	Device Electronics for Integrated Circuits	3

General Engineering - Mechatronics — Discipline Electives, Odd Semesters (3, 5, 7) (continued)

Sl. No	Course Code	Course Name	Credits
9	EE-326	Computer Organisation and Processor Architecture Design	4
10	ME-210	Fluid Mechanics	3
11	ME-212	Product Manufacturing Technology	3
12	ME-213	Thermodynamics	3
13	ME-311P	Design Lab – I	1
14	ME-315	Manufacturing Engineering – II	3
15	VL-401	RTL Design and Verification	3
16	VL-403	CMOS Digital IC Design	4
17	VL-404	CMOS Analog IC Design	4

Table 448: General Engineering - Mechatronics — Discipline Electives, Even Semesters (4, 6, 8)

Sl. No	Course Code	Course Name	Credits
1	AR-509	Deep Learning for Computer Vision	4
2	AR-523	Robot Manipulators	3
3	AR-526	Multimodal Learning	3
4	EE-529	Embedded Systems	4
5	GE-303	Modern Powertrains: Electric, Hybrid & IC/Aeroengines	3
6	IC-121	Mechanics of Particles and Waves	3
7	IC-253	Programming and Data Structures	3
8	ME-215	Manufacturing Engineering – I	3
9	ME-312P	Design Lab – II	1
<i>Compulsory project components — counted as DE, not selectable</i>			
10	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
11	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Communication Technology

Table 449: General Engineering - Communication Technology — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	34
	Discipline Electives	32
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 450: B. Tech General Engineering - Communication Technology — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	ME-100	Reverse Engineering	0-0-2-1	1
2	EE-261	Electrical System Around Us	3-0-0-3	3
3	EE-231	Measurement and Instrumentation	2-0-2-3	3
4	EE-316	Communication Systems	3-0-2-4	4
5	EE-201	Electromechanics	2.5-0.5-0-3	3
6	EE-203	Network Theory	3-0-0-3	3
7	IC-253	Programming and Data Structures	2-0-2-3	3
8	EE-260	Signals and Systems	2.5-0.5-0-3	3
9	CS-313	Computer Networks	3-0-2-4	4
10	EE-314	Digital Signal Processing	3-0-2-4	4
11	EE-205	Electromagnetics and Wave Propagation	3-0-0-3	3
Total				34

Table 451: B. Tech General Engineering - Communication Technology — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Computing and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
10		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 452: B. Tech General Engineering - Communication Technology — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 453: B. Tech General Engineering - Communication Technology — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-313	Computer Networks	3-0-2-4	4
4	DC	EE-203	Network Theory	3-0-0-3	3
5	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
6	DC	EE-261	Electrical System Around Us	3-0-0-3	3
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 454: B. Tech General Engineering - Communication Technology — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
2	DC	EE-205	Electromagnetics and Wave Propagation	3-0-0-3	3
3	DC	EE-316	Communication Systems	3-0-2-4	4
4	DC	IC-253	Programming and Data Structures	2-0-2-3	3
5	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 455: B. Tech General Engineering - Communication Technology — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	EE-314	Digital Signal Processing	3-0-2-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 456: B. Tech General Engineering - Communication Technology — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 457: B. Tech General Engineering - Communication Technology — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 458: B. Tech General Engineering - Communication Technology — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 32 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Odd-semester electives are offered in Semesters 3, 5 and 7; even-semester electives in Semesters 4, 6 and 8.

Table 459: General Engineering - Communication Technology — Discipline Electives, Odd Semesters (3, 5, 7)

Sl. No	Course Code	Course Name	Credits
1	EE-503	Advanced Communication Theory	3
2	EE-517	Wireless Communications & Networks	3
3	EE-531	Estimation and Detection Theory	3
4	EE-553	Fundamentals of Intelligent Communication System-I	4
5	EE-641	Advanced Wireless Technologies	3
6	GE-303	Modern Powertrains: Electric, Hybrid & IC/Aeroengines	3
7	VL-402	RFIC Design	3

Table 460: General Engineering - Communication Technology — Discipline Electives, Even Semesters (4, 6, 8)

Sl. No	Course Code	Course Name	Credits
1	CS-549	Performance Analysis of Computer Networks	3
2	DS-314	Probability Theory and Statistics	4
3	EE-211	Analog Circuit Design	4
4	EE-518	Information Theory	3
5	EE-530	Applied Optimization	3
6	EE-533	Access Network	4
7	EE-536	IoT Systems	3
8	EE-541	Tensors: Applications for Signal Processing & ML	3
9	EE-574	Biomedical Signal & Image Processing	3
10	EE-581	Fundamentals of Intelligent Communication System-II	3
11	EE-607	Optical Communication System	3
12	EE-608	Digital Image Processing	4
13	IC-241	Material Science for Engineers	3
<i>Compulsory project components — counted as DE, not selectable</i>			
14	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
15	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Open Specialisation

Table 461: General Engineering - Open Specialisation — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	35
	Discipline Electives (merged into Free Electives)	—
Electives	Free Electives (including merged DE pool)	53
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 462: B. Tech General Engineering - Open Specialisation — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	ME-100	Reverse Engineering	0-0-2-1	1
2	EE-261	Electrical System Around Us	3-0-0-3	3
3	EE-231	Measurement and Instrumentation	2-0-2-3	3
4	ME-213	Thermodynamics	3-1-0-4	4
5	EE-201	Electromechanics	2.5-0.5-0-3	3
6	EE-211	Analog Circuit Design	2-0-2-3	3
7	ME-212	Product Manufacturing Technology	2-0-3-3	3
8	IC-241	Material Science for Engineers	3-0-0-3	3
9	IC-253	Programming and Data Structures	2-0-2-3	3
10	HS-307	Macroeconomics	3-0-0-3	3
11	HS-541	Technical Communication	1-0-0-1	1
12	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total				33

Table 463: B. Tech General Engineering - Open Specialisation — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Computing and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
10		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 464: B. Tech General Engineering - Open Specialisation — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 465: B. Tech General Engineering - Open Specialisation — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-261	Electrical System Around Us	3-0-0-3	3
4	DC	IC-241	Material Science for Engineers	3-0-0-3	3
5	DC	ME-212	Product Manufacturing Technology	2-0-3-3	3
6	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 466: B. Tech General Engineering - Open Specialisation — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
2	DC	EE-211	Analog Circuit Design	2-0-2-3	3
3	DC	HS-307	Macroeconomics	3-0-0-3	3
4	DC	IC-253	Programming and Data Structures	2-0-2-3	3
5	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 467: B. Tech General Engineering - Open Specialisation — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	ME-213	Thermodynamics	3-1-0-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 468: B. Tech General Engineering - Open Specialisation — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	HS-541	Technical Communication	1-0-0-1	1
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					5

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 469: B. Tech General Engineering - Open Specialisation — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 470: B. Tech General Engineering - Open Specialisation — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 53 credits of **Free Electives** — for this major the Discipline Elective and Free Elective pools are merged into a single pool — from across the institute, subject to the elective policy below.*

Elective Policy. There is no prescribed Discipline Elective list for this major. Students may choose electives from any department, provided that no more than 40% of their elective credits come from the Discipline Core list of any single named major. The Discipline Elective and Free Elective pools are merged into one pool of 53 credits of Free Electives, giving students maximum flexibility in course selection.

The following compulsory project components count toward the DE credit requirement; they are not selectable DE options.

Table 471: General Engineering - Open Specialisation — Compulsory project components credited as DE

Sl. No	Course Code	Course Name	Credits
1	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
2	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Data Analytics for Management and Finance

Table 472: General Engineering - Data Analytics for Management and Finance — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	41
	Discipline Electives	25
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 473: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	DS-314*	Probability Theory and Statistics	3-1-0-4	4
2	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
3	HS-202	Principles of Economics	3-0-0-3	3
4	HS-307	Macroeconomics – I	3-0-0-3	3
5	IC-253	Programming and Data Structures	2-0-2-3	3
6	MA-222	Applied Mathematical Programming	3-1-0-4	4
7	MA-546	Introduction to Quantitative Finance		4
8	MA-653P	Computational Financial Modelling Lab	0-0-2-1	1
9	MA-654	Financial Engineering		3
10	MA-545	Technical Analysis of Financial Market		3
11	MA-579	Computational Methods in Financial Time Series Analysis		3
12	MB-311	Management Fundamentals for Engineers	3-0-0-3	3
13	MB-312	Economic Decision-Making for Engineers	3-0-0-3	3
Total				41

* Recoded w.e.f. batch 2025: DS-313 → DS-314 (Probability Theory and Statistics, 3-1-0-4 unchanged) and DS-413 → DS-416 (Fundamentals of Machine Learning, now 3-0-2-4).

Table 474: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Computing and Data Science	3-0-2-4	4
4	IC Basket	IC-230	Environmental Science	3-0-0-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
10		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 475: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 476: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	DS-314*	Probability Theory and Statistics	3-1-0-4	4
4	DC	HS-202	Principles of Economics	3-0-0-3	3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 477: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	HS-307	Macroeconomics – I	3-0-0-3	3
2	DC	IC-253	Programming and Data Structures	2-0-2-3	3
3	DC	MA-222	Applied Mathematical Programming	3-1-0-4	4
4	DC	MA-546	Introduction to Quantitative Finance		4
5	DC	MA-653P	Computational Financial Modelling Lab	0-0-2-1	1
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 478: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
2	DC	MA-654	Financial Engineering		3
3	DC	MB-311	Management Fundamentals for Engineers	3-0-0-3	3
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 479: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	MB-312	Economic Decision-Making for Engineers	3-0-0-3	3
2	DC	MA-545	Technical Analysis of Financial Market		3
3	DC	MA-579	Computational Methods in Financial Time Series Analysis		3
4	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 480: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 481: B. Tech General Engineering - Data Analytics for Management and Finance — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 25 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Odd-semester electives are offered in Semesters 3, 5 and 7; even-semester electives in Semesters 4, 6 and 8.

Table 482: General Engineering - Data Analytics for Management and Finance — Discipline Electives, Odd Semesters (3, 5, 7)

Sl. No	Course Code	Course Name	Credits
1	DS-412	Matrix Computations for Data Science	4
2	HS-510	Essentials of Entrepreneurship	3
3	MA-211	Ordinary Differential Equations	4
4	MA-311	Mathematical Modelling	3

Table 483: General Engineering - Data Analytics for Management and Finance — Discipline Electives, Even Semesters (4, 6, 8)

Sl. No	Course Code	Course Name	Credits
1	DS-312	Computing Systems for Data Processing	4
2	DS-415	Optimization for Machine Learning	4
3	HS-481	International Economics	3
4	HS-504	Personal Finance and Portfolio Management	3
5	HS-550	Statistical Methods	4
6	MA-220	Partial Differential Equations	4
7	MA-221	Numerical Analysis	4
8	MA-524	Probability and Statistics	4
9	MA-552	Number Theory	3
10	MA-604	Introduction to Game Theory	3
11	ME-100	Reverse Engineering	1

Table 484: General Engineering - Data Analytics for Management and Finance — Discipline Electives, Any Semester (Business)

Sl. No	Course Code	Course Name	Credits
1	MB-410	Marketing for Engineers	3
2	MB-411	Human Resource Management and Organizational Behaviour for Engineers	3
3	MB-412	Consumer Behaviour in Technology Markets	3
4	MB-413	Technology Entrepreneurship and Venture Creation	3
5	MB-414	Accounting Practices for Technology Businesses	3
6	MB-415	Financial Management for Technology Businesses	3
7	MB-416	Industrial Operations and Supply Chain Systems	3
<i>Compulsory project components — counted as DE, not selectable</i>			
8	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
9	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Materials Science and Engineering

Table 485: Materials Science and Engineering — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	45
	Discipline Electives	21
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 486: B. Tech Materials Science and Engineering — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
2	MT-201	Physics of Solids		3
3	MT-202	Applied Quantum Mechanics		3
4	MT-203	Materials Synthesis & Characterization		4
5	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
6	ME-100	Reverse Engineering	0-0-2-1	1
7	MT-204	Thermodynamics and Kinetics of Materials	3	3
8	MT-205	Functional Properties of Materials	4	4
9	MT-206	Extraction and Materials Processing	4	4
10	ME-212	Product Manufacturing Technology	2-0-3-3	3
11	MT-301	Phase Transformations		3
12	MT-302	Transport Phenomena		3
13	MT-303	Computational Materials Science		4
14	MT-304	Mechanical Behaviour of Materials		4
Total				45

Table 487: B. Tech Materials Science and Engineering — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Computing and Data Science	3-0-2-4	4
4	IC Basket	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
10		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 488: B. Tech Materials Science and Engineering — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-241	Material Science for Engineers	3-0-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 489: B. Tech Materials Science and Engineering — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-222P	Physics Practicum	0-0-3-2	2
3	IC	IC-272	Machine Learning	2-0-2-3	3
4	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
5	DC	MT-201	Physics of Solids		3
6	DC	MT-202	Applied Quantum Mechanics		3
7	DC	MT-203	Materials Synthesis & Characterization		4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 490: B. Tech Materials Science and Engineering — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
2	DC	ME-100	Reverse Engineering	0-0-2-1	1
3	DC	MT-204	Thermodynamics and Kinetics of Materials	3	3
4	DC	MT-205	Functional Properties of Materials	4	4
5	DC	MT-206	Extraction and Materials Processing	4	4
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 491: B. Tech Materials Science and Engineering — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-212	Product Manufacturing Technology	2-0-3-3	3
2	DC	MT-301	Phase Transformations		3
3	DC	MT-302	Transport Phenomena		3
4	DC	MT-303	Computational Materials Science		4
5	DC	MT-304	Mechanical Behaviour of Materials		4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 492: B. Tech Materials Science and Engineering — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 493: B. Tech Materials Science and Engineering — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	MS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 494: B. Tech Materials Science and Engineering — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	MS-010	Internship		2
2	MTP	MS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 21 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 495: Materials Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	ME-303	Heat Transfer	3
2	ME-308	Manufacturing Engineering	3
3	ME-353	Electronic Materials and Their Applications	3
4	ME-515	Carbon Materials and Technology	3
5	MT-501	Energy Conversion and Storage Technologies	3
6	MT-502	Recycling and Circular Economy	3
7	MT-503	Semiconductor Materials and Devices	3
8	MT-504	Powder metallurgical processing of Materials	3
9	MT-505	Thin Film Technology	3
10	MT-506	Biomaterials	3
11	MT-507	Modelling and Simulations in Material Science	3

Materials Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	MT-511	Sensor Materials and Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
13	MS-498P	Major Technical Project - I (compulsory; counted as DE)	4
14	MS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Science — Chemical Sciences

Table 496: Chemical Sciences — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	31
	IC Basket	9
	IKS + HSS	12
Discipline Courses	Discipline Core	59
	Discipline Electives	23
Electives	Free Electives	15
	Research Lit Presentation (1+1)	2
	RP (equivalent as DE)	12
Total		163

Table 497: B. S. Chemical Sciences — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CY-201P	Physical Chemistry Laboratory	0-0-4-2	2
2	CY-301	Principles and Theories of Physical Chemistry	3-0-0-3	3
3	CY-302	Principles of Organic Chemistry	3-0-0-3	3
4	CY-303	Fundamentals of Inorganic Chemistry	3-0-0-3	3
5	CY-202P	Organic Chemistry Laboratory	0-0-4-2	2
6	CY-203P	Inorganic Chemistry Laboratory	0-0-4-2	2
7	CY-304	Fundamental Analytical Chemistry	3-0-0-3	3
8	CY-401	Introduction to Quantum Chemistry and Molecular Spectroscopy	3-0-0-3	3
9	CY-512	Advanced Quantum Chemistry	3-0-0-3	3
10	CY-512P	Physical Chemistry Laboratory		3
11	CY-531	Organic Reactions and Mechanisms		3
12	CY-533	Chemistry of Main Group Elements		3
13	CY-533P	Inorganic Chemistry Laboratory		3
14	CY-511	Group Theory and Spectroscopy	3-0-0-3	3
15	CY-531P	Organic Chemistry Laboratory		3
16	CY-532	Photochemistry and Pericyclic Reactions		3
17	CY-534	Chemistry of Transition Elements		3
18	CY-514	Chemical and Statistical Thermodynamics	3-0-0-3	3
19	CY-535	Introduction to Organometallic Chemistry		3
20	CY-504	Heterocyclic Chemistry	2-0-0-2	2
21	CY-513	Chemical Kinetics and Reaction Dynamics	3-0-0-3	3
Total				59

Table 498: B. S. Chemical Sciences — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
5	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
6		HS-112	Japanese Language Class I	3-0-0-3	
7		HS-256	Introduction to Spanish	3	
8		HS-342	German I	3-0-0-3	
9	IKS + HSS	IC-181	Indian Knowledge Systems	2-0-2-3	3
10	(choose 1)	IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 499: B. S. Chemical Sciences — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 500: B. S. Chemical Sciences — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
3	DC	CY-201P	Physical Chemistry Laboratory	0-0-4-2	2
4	DC	CY-301	Principles and Theories of Physical Chemistry	3-0-0-3	3
5	DC	CY-302	Principles of Organic Chemistry	3-0-0-3	3
6	DC	CY-303	Fundamentals of Inorganic Chemistry	3-0-0-3	3
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 501: B. S. Chemical Sciences — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-202P	Organic Chemistry Laboratory	0-0-4-2	2
2	DC	CY-203P	Inorganic Chemistry Laboratory	0-0-4-2	2
3	DC	CY-304	Fundamental Analytical Chemistry	3-0-0-3	3
4	DC	CY-401	Introduction to Quantum Chemistry and Molecular Spectroscopy	3-0-0-3	3
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 502: B. S. Chemical Sciences — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-512	Advanced Quantum Chemistry	3-0-0-3	3
2	DC	CY-512P	Physical Chemistry Laboratory		3
3	DC	CY-531	Organic Reactions and Mechanisms		3
4	DC	CY-533	Chemistry of Main Group Elements		3
5	DC	CY-533P	Inorganic Chemistry Laboratory		3
6	RP	DP-403P	Research Literature Presentation 1		1
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 503: B. S. Chemical Sciences — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-511	Group Theory and Spectroscopy	3-0-0-3	3
2	DC	CY-531P	Organic Chemistry Laboratory		3
3	DC	CY-532	Photochemistry and Pericyclic Reactions		3
4	DC	CY-534	Chemistry of Transition Elements		3
5	RP	DP-404P	Research Literature Presentation 2		1
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 504: B. S. Chemical Sciences — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-514	Chemical and Statistical Thermodynamics	3-0-0-3	3
2	DC	CY-535	Introduction to Organometallic Chemistry		3
3	RP	DP-551P	Undergraduate Research Project 1		6
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 505: B. S. Chemical Sciences — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-504	Heterocyclic Chemistry	2-0-0-2	2
2	DC	CY-513	Chemical Kinetics and Reaction Dynamics	3-0-0-3	3
3	RP	DP-552P	Undergraduate Research Project 2		6
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 23 credits of **Discipline Electives** and 15 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 506: Chemical Sciences — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BY-505	Nanobiotechnology	3
2	BY-514	Analytical Biotechniques	3
3	CE-519	Chemistry of Natural Waters	3
4	CH-101	General Chemistry	4
5	CY-001	Preparatory Chemistry - 1	4
6	CY-002	Preparatory Chemistry - 2	4
7	CY-100	Preparatory Chemistry 1	3
8	CY-101	Chemistry	5
9	CY-200	Foundations and applications of Chemistry	3
10	CY-241	Nano-scale Science and Technology	3
11	CY-242	Introduction to Molecular Spectroscopy	3
12	CY-243	Engineering Chemistry	3
13	CY-247	Introduction to Molecular Thermodynamics	3
14	CY-248	Molecular Physical Chemistry for Engineers	3
15	CY-342	Understanding Small Systems	3
16	CY-344	Food Chemistry: Processing, Preservation and Storage	3
17	CY-402	Applied Materials Chemistry	3
18	CY-403	Numerical Methods and Data Analysis in Chemistry	3
19	CY-404	Fundamentals of Soft Matter Science and Applications	3
20	CY-405	Introduction to structural chemistry	3
21	CY-501	Organic Reaction and Mechanism	4
22	CY-501P	Organic Chemistry Laboratory	4
23	CY-502	Photochemistry and pericyclic reactions	4
24	CY-506	Chemistry of Transition Elements	4
25	CY-506P	Inorganic Chemistry Lab	4
26	CY-507	Chemistry of Main Group Elements	4
27	CY-508	Organometallic Chemistry	2
28	CY-511P	Physical Chemistry Laboratory	4
29	CY-515	Advanced Inorganic Spectroscopy	3
30	CY-521	Mathematics for Chemists	3
31	CY-522	Computational Chemistry	3
32	CY-523	Colloids and Interface Science and Technology	3
33	CY-524	Basic and Applied Electrochemistry	3
34	CY-541	Fundamentals of Organic Chemistry	3
35	CY-545	Reagents in organic synthesis	3
36	CY-547	Chemical Crystallography	3
37	CY-550	Bioinspired Materials	3
38	CY-552	Hydrogen Generation and Storage	3
39	CY-553	Organic Inorganic Spectroscopy	3
40	CY-554	Science and Technology of Nanomaterials	3

Chemical Sciences — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
41	CY-555	Introduction to Polymer Science & Technology	3
42	CY-556	Organic Spectroscopy	3
43	CY-560	Introduction to Supramolecular Chemistry	3
44	CY-561	Natural Products Synthesis	3
45	CY-562	Stereochemistry of organic compounds	3
46	CY-562P	Design Practicum - II	1
47	CY-598P	Post Graduate Project-1 (CH)	4
48	CY-599P	Post Graduate Project-2	9
49	CY-641	Polymer Synthesis	3
50	CY-642	Molecular- and Bio-electronics	3
51	CY-643	Advanced Analytical Techniques	3
52	CY-644	Bioinorganic chemistry	3
53	CY-645	Reagents in organic synthesis	3
54	CY-646	Advanced NMR Spectroscopy - A Problem Based Approach	3
55	CY-660	Photocatalysis: Concepts and Applications	3
56	CY-670	Fluorescence spectroscopy, microscopy and applications	3
57	CY-701	Advances Physical Methods in Chemistry Theory and	3
58	CY-702	Advanced Inorganic Chemistry: Theory and Applications	3
59	CY-703	Advanced Organic Chemistry	3
60	CY-704	Introduction to Theoretical Chemistry	3
61	CY-705	Modern Methods in Organic Synthesis	3
62	CY-746	Self Assembly of Surfactants and Polymers in Solution	3
63	EN-503	Energy Storage Technologies	3
64	ME-353	Electronic Materials and Their Applications	3
65	ME-509	Nanomanufacturing	3
66	ME-515	Carbon Materials and Technology	3
67	ME-516	Polymer Technology for Engineers	3
68	MT-201	Physics of Solids	3
69	MT-501	Energy Conversion and Storage Technologies	3
70	MT-502	Recycling and Circular Economy	3
71	MT-503	Semiconductor Materials and Devices	3
72	MT-505	Thin Film Technology	3
73	MT-506	Biomaterials	3
74	MT-509	Hydrogen Energy	3
75	MT-510	Colloids and Interfaces	3
76	PH-504	Organic Optoelectronics	3
<i>Compulsory project components — counted as DE, not selectable</i>			
77	DP-551P	Undergraduate Research Project 1 (compulsory; counted as DE)	6

Chemical Sciences — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
78	DP-552P	Undergraduate Research Project 2 (compulsory; counted as DE)	6

Bachelor of Technology — Microelectronics and VLSI

Table 507: Microelectronics and VLSI — B25 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	54
	Discipline Electives	12
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 508: B. Tech Microelectronics and VLSI — B25 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-212	Digital System Design	3-0-2-4	4
3	EE-260	Signals and Systems	2.5-0.5-0-3	3
4	EE-302	Control Systems	3-0-2-4	4
5	VL-201	Semiconductor Device for ICs	3-0-0-3	3
6	EE-202	Electromagnetic Theory and Transmission Lines	3-0-0-3	3
7	EE-211	Analog Circuit Design	2-0-2-3	4
8	VL-300	Reverse Engineering - E-waste management	0-0-2-1	1
9	VL-311	CMOS Processing and Practicum	3-0-2-4	4
10	VL-312	Electronic System Packaging	3-0-0-3	3
11	VL-326	Computer Organization	3-0-2-4	4
12	VL-401	RTL Design and Verification	2-0-2-3	3
13	VL-402	RF IC Design	3-0-0-3	3
14	VL-403	CMOS Digital IC Design	3-0-2-4	4
15	VL-404	CMOS Analog IC Design	3-0-2-4	4
16	VL-405	Design for Testability	3-0-2-4	4
Total				54

Table 509: B. Tech Microelectronics and VLSI — B25 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
4	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
5		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
6		IC-230	Environmental Science	3-0-0-3	
7	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
8		HS-112	Japanese Language Class I	3-0-0-3	
9		HS-256	Introduction to Spanish	3	
10		HS-342	German I	3-0-0-3	
11	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
12		IC-182	History of Science and Technology	3	
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 510: B. Tech Microelectronics and VLSI — B25 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
5	IC	IC-161	Applied Electronics	3	3
6	IC	IC-161P	Applied Electronics Lab	2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket (choose 1)	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9		IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	
10		IC-241	Material Science for Engineers	3-0-0-3	
11		IC-253	Data Structures and Algorithms	2-0-2-3	
Total					24

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 511: B. Tech Microelectronics and VLSI — B25 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-222P	Physics Practicum	0-0-3-2	2
3	IC	IC-272	Machine Learning	2-0-2-3	3
4	DC	EE-203	Network Theory	3-0-0-3	3
5	DC	EE-212	Digital System Design	3-0-2-4	4
6	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
7	DC	EE-302	Control Systems	3-0-2-4	4
8	DC	VL-201	Semiconductor Device for ICs	3-0-0-3	3
Total					25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 512: B. Tech Microelectronics and VLSI — B25 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-202	Electromagnetic Theory and Transmission Lines	3-0-0-3	3
2	DC	EE-211	Analog Circuit Design	2-0-2-3	4
3	DC	VL-300	Reverse Engineering - E-waste management	0-0-2-1	1
4	DC	VL-311	CMOS Processing and Practicum	3-0-2-4	4
5	DC	VL-312	Electronic System Packaging	3-0-0-3	3
6	DC	VL-326	Computer Organization	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 513: B. Tech Microelectronics and VLSI — B25 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	VL-401	RTL Design and Verification	2-0-2-3	3
2	DC	VL-402	RF IC Design	3-0-0-3	3
3	DC	VL-403	CMOS Digital IC Design	3-0-2-4	4
4	DC	VL-404	CMOS Analog IC Design	3-0-2-4	4
5	DC	VL-405	Design for Testability	3-0-2-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 514: B. Tech Microelectronics and VLSI — B25 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 515: B. Tech Microelectronics and VLSI — B25 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	VL-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 516: B. Tech Microelectronics and VLSI — B25 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	VL-010	Internship		2
2	MTP	VL-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 12 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 517: Microelectronics and VLSI — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	EE-210P	Digital Systems Design Lab	1
2	EE-507	Transmission Lines and Basic Microwave Engineering	4
3	EE-516	Biomedical Systems	4
4	EE-520	Microelectronic Devices and Modelling	3
5	EE-523	Digital VLSI Architecture Design	4
6	EE-529	Embedded Systems	4
7	EE-554	Low Power VLSI Design	3
8	EE-574	Biomedical Signal and Image Analysis	3
9	EE-585	Theory, Modelling and Characterization of Photo-voltaic Devices	3
10	EE-611	VLSI Technology	3

Microelectronics and VLSI — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	EE-615	Nanoelectronics and Micronanofabrication	3
12	EE-619	Mixed Signal VLSI Design	4
13	EE-621	Radiating systems	4
14	EE-622	Microwave Integrated Circuits	3
<i>Compulsory project components — counted as DE, not selectable</i>			
15	VL-498P	Major Technical Project - I (compulsory; counted as DE)	4
16	VL-499P	Major Technical Project - II (compulsory; counted as DE)	4

Batch 2026

Bachelor of Technology — Computer Science and Engineering

Table 518: Computer Science and Engineering — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	39
	Discipline Electives	27
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 519: B. Tech Computer Science and Engineering — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
2	CS-212	Design of Algorithms	3-0-2-4	4
3	CS-213	Reverse Engineering	0-0-2-1	1
4	CS-214	Computer Organization	3-0-2-4	4
5	CS-304	Formal Language and Automata Theory	3-0-0-3	3
6	CS-309	Information Systems and Databases	3-0-2-4	4
7	CS-315*	Artificial Intelligence	3-0-2-4	4
8	CS-312	Operating Systems	3-0-2-4	4
9	CS-313	Computer Networks	3-0-2-4	4
10	CS-302	Paradigms of Programming	3-0-2-4	4
11	CS-303	Software Engineering	2-0-2-3	3
Total				39

* CS-305 Artificial Intelligence has been recoded as CS-315 with L-T-P-C 3-0-2-4, w.e.f. batch 2025.

Table 520: B. Tech Computer Science and Engineering — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		IC-182	History of Science and Technology	3	
10	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
11		HS-112	Japanese Language I	3-0-0-3	
12		HS-342	German I	3-0-0-3	
13		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 521: B. Tech Computer Science and Engineering — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 522: B. Tech Computer Science and Engineering — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
4	DC	CS-212	Design of Algorithms	3-0-2-4	4
5	DC	CS-213	Reverse Engineering	0-0-2-1	1
6	DC	CS-214	Computer Organization	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 523: B. Tech Computer Science and Engineering — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-304	Formal Language and Automata Theory	3-0-0-3	3
2	DC	CS-309	Information Systems and Databases	3-0-2-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 524: B. Tech Computer Science and Engineering — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-315*	Artificial Intelligence	3-0-2-4	4
2	DC	CS-312	Operating Systems	3-0-2-4	4
3	DC	CS-313	Computer Networks	3-0-2-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 525: B. Tech Computer Science and Engineering — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-302	Paradigms of Programming	3-0-2-4	4
2	DC	CS-303	Software Engineering	2-0-2-3	3
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 526: B. Tech Computer Science and Engineering — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	CS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 527: B. Tech Computer Science and Engineering — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	CS-010	Internship		2
2	MTP	CS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 27 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 528: Computer Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-501	Access Networks	3
2	CS-502	Compiler Design	4
3	CS-506	Cognitive Modeling	3
4	CS-507	Computer Architecture	4
5	CS-508	Introduction to Network Security	3
6	CS-510	Randomized Algorithms	4
7	CS-511	Applied Probability	2
8	CS-512	Matrix Theory	2
9	CS-513	Discrete Mathematics	4

Computer Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
10	CS-514	Advance Data Structure and Algorithm (CS-580)	4
11	CS-515	Advanced Computer Science Practicum	3
12	CS-516	Exploratory Project	3
13	CS-520	Introduction to Quantum Computing	3
14	CS-521	Introduction to Post Quantum Security	3
15	CS-522	Distributed Algorithms	3
16	CS-524	Computational Complexity Theory	4
17	CS-525	Computational Foundations of Cryptography	3
18	CS-526	Mathematical Foundation of Cryptography	3
19	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3
20	CS-541	IoT systems and the Cloud	3
21	CS-542	Design Patterns for Scalable Systems	3
22	CS-544	Formal Concept Analysis: Theory and Practice	3
23	CS-545	Software Design Patterns	3
24	CS-546	Design of Concurrent Software	3
25	CS-547	Network Management Systems	3
26	CS-548	Cloud Networking	3
27	CS-549	Performance Analysis of Computer Networks	3
28	CS-550	Computer Graphics and Geometric Design	3
29	CS-551	Human Computer Interaction	3
30	CS-560	Text Retrieval and Mining	3
31	CS-561	Map Reduce and Big Data	3
32	CS-563	Scalable Data Science	4
33	CS-570	Algorithms for Curves and Surface	3
34	CS-571	Programming Practicum	3
35	CS-591	Special Topics in Knowledge Discovery and Data	1
36	CS-592	Selected Topics in Artificial Intelligence	2
37	CS-601	Probability and Random Processes	4
38	CS-603	Managerial Decision Making	3
39	CS-606	Cognitive Modeling	3
40	CS-609	Speech processing	3
41	CS-611	Program Analysis	4
42	CS-621P	Post Graduate Project-1	12
43	CS-630	Speech Technology	4
44	CS-660	Data Mining for Decision Making	4
45	CS-661	Knowledge Representation and Reasoning	3
46	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
47	CS-670	Kernal Methods for Pattern Recognition	4
48	CS-671	Deep Learning and Applications	4
49	CS-672	Advanced Topics in Deep Learning	4

Computer Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	CS-673	Advanced Computer Vision	4
51	CS-677	Soft Computing	3
52	CS-683	Generative AI	4
53	CS-685	Natural Language Processing	3
54	CS-686	Large Language Models: Core Concepts to Custom	4
55	CS-693	Compressed Sensing and its applications	3
56	DS-201	Data handling and visualization	3
57	DS-301	Mathematical Foundations of Data Science	4
58	DS-314	Probability Theory and Statistics	4
59	DS-401	Optimization for data science	3
60	DS-402	Matrix Computations for Data Science	3
61	DS-403	Introduction to Statistical Learning	3
62	DS-412	Matrix Computations for Data Science	4
63	DS-415	Optimization for Machine Learning	4
64	DS-416	Fundamentals of Machine Learning (CS-669)	4
65	EE-511	Computer Vision	4
<i>Compulsory project components — counted as DE, not selectable</i>			
66	CS-498P	Major Technical Project - I (compulsory; counted as DE)	4
67	CS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Data Science and Artificial Intelligence

Table 529: Data Science and Artificial Intelligence — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	33
	Discipline Electives	33
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 530: B. Tech Data Science and Artificial Intelligence — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-213	Reverse Engineering	0-0-2-1	1
2	DS-314*	Probability Theory and Statistics	3-1-0-4	4
3	DS-412	Matrix Computations for Data Science	3-0-2-4	4
4	DS-312*	Computing Systems for Data Processing	3-0-2-4	4
5	DS-415*	Optimization for Machine Learning	3-0-2-4	4
6	DS-417 [†]	Deep Learning	3-0-2-4	4
7	CS-315*	Artificial Intelligence	3-0-2-4	4
8	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
9	DS-418 [†]	Introduction to Generative AI	3-0-2-4	4
Total				33

* Recoded w.e.f. batch 2025: CS-305 → CS-315 (Artificial Intelligence) and DS-302 → DS-312 (Computing Systems for Data Processing), both now 3-0-2-4; DS-313 → DS-314 (Probability Theory and Statistics, 3-1-0-4 unchanged); DS-411 → DS-415 (Optimization for Machine Learning) and DS-413 → DS-416 (Fundamentals of Machine Learning), both now 3-0-2-4.

[†]DS-417 (Deep Learning) and DS-418 (Introduction to Generative AI) are DSAI-only codes for the 600-level courses CS-671 and CS-683; DSAI students cannot also take CS-671 or CS-683 as electives. DS-417 is taken in Semester 4 (moved from Semester 6).

Table 531: B. Tech Data Science and Artificial Intelligence — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		IC-182	History of Science and Technology	3	
10	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
11		HS-112	Japanese Language I	3-0-0-3	
12		HS-342	German I	3-0-0-3	
13		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 532: B. Tech Data Science and Artificial Intelligence — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 533: B. Tech Data Science and Artificial Intelligence — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	CS-213	Reverse Engineering	0-0-2-1	1
3	DC	DS-314*	Probability Theory and Statistics	3-1-0-4	4
4	DC	DS-412	Matrix Computations for Data Science	3-0-2-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 534: B. Tech Data Science and Artificial Intelligence — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	DS-312*	Computing Systems for Data Processing	3-0-2-4	4
3	DC	DS-415*	Optimization for Machine Learning	3-0-2-4	4
4	DC	DS-417 [†]	Deep Learning	3-0-2-4	4
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 535: B. Tech Data Science and Artificial Intelligence — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-315*	Artificial Intelligence	3-0-2-4	4
2	DC	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
Total					8

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 536: B. Tech Data Science and Artificial Intelligence — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-418 [†]	Introduction to Generative AI	3-0-2-4	4
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					8

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 537: B. Tech Data Science and Artificial Intelligence — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	DS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 538: B. Tech Data Science and Artificial Intelligence — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	DS-010	Internship		2
2	MTP	DS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 33 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 539: Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-507	Probabilistic Robotics	3
2	AR-516	Introduction to Blockchain and Web3	3
3	AR-517	Introduction to Cyber security	3
4	BE-303	Applied Biostatistics	4
5	BE-304	Bioinformatics	4
6	BE-502	Design and Analysis of Bioalgorithms	3
7	CS-212	Design and Analysis of Algorithms	4
8	CS-214	Computer Organization	4
9	CS-241	Introduction to Cryptography	3
10	CS-302	Paradigms of Programming	4
11	CS-303	Software Engineering	3
12	CS-304	Formal Languages and Automata Theory	3
13	CS-306	Operating Systems	3
14	CS-307	Systems Practicum	2
15	CS-308	Large Application Practicum	2
16	CS-308P	Large Application Practicum	2
17	CS-309	Information and Database Systems	4
18	CS-310	Introduction to Communication and Distributed Processes	4
19	CS-313	Computer Networks	4

Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
20	CS-347	Software Engineering	4
21	CS-350	Computer Graphics	3
22	CS-401	Architecture of High Performance Computers	3
23	CS-402	Compilers	2
24	CS-403	Algorithm Design and Analysis I	3
25	CS-405	Verification of Reactive Systems	3
26	CS-406	Computer Networks	4
27	CS-451	Computer Graphics and Game Design	3
28	CS-456	Distributed Databases	3
29	CS-501	Access Networks	3
30	CS-502	Compiler Design	4
31	CS-506	Cognitive Modeling	3
32	CS-507	Computer Architecture	4
33	CS-508	Introduction to Network Security	3
34	CS-510	Randomized Algorithms	4
35	CS-511	Applied Probability	2
36	CS-512	Matrix Theory	2
37	CS-513	Discrete Mathematics	4
38	CS-514	Advance Data Structure and Algorithm (CS-580)	4
39	CS-515	Advanced Computer Science Practicum	3
40	CS-516	Exploratory Project	3
41	CS-520	Introduction to Quantum Computing	3
42	CS-521	Introduction to Post Quantum Security	3
43	CS-522	Distributed Algorithms	3
44	CS-524	Computational Complexity Theory	4
45	CS-525	Computational Foundations of Cryptography	3
46	CS-526	Mathematical Foundation of Cryptography	3
47	CS-528	Unconditional Lower Bounds for Algorithms and Data Structures	3
48	CS-541	IoT systems and the Cloud	3
49	CS-542	Design Patterns for Scalable Systems	3
50	CS-544	Formal Concept Analysis: Theory and Practice	3
51	CS-545	Software Design Patterns	3
52	CS-546	Design of Concurrent Software	3
53	CS-547	Network Management Systems	3
54	CS-548	Cloud Networking	3
55	CS-549	Performance Analysis of Computer Networks	3
56	CS-550	Computer Graphics and Geometric Design	3
57	CS-551	Human Computer Interaction	3
58	CS-560	Text Retrieval and Mining	3

Data Science and Artificial Intelligence — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
59	CS-561	Map Reduce and Big Data	3
60	CS-563	Scalable Data Science	4
61	CS-570	Algorithms for Curves and Surface	3
62	CS-571	Programming Practicum	3
63	CS-591	Special Topics in Knowledge Discovery and Data	1
64	CS-592	Selected Topics in Artificial Intelligence	2
65	CS-601	Probability and Random Processes	4
66	CS-603	Managerial Decision Making	3
67	CS-606	Cognitive Modeling	3
68	CS-609	Speech processing	3
69	CS-611	Program Analysis	4
70	CS-630	Speech Technology	4
71	CS-660	Data Mining for Decision Making	4
72	CS-661	Knowledge Representation and Reasoning	3
73	CS-662	Mobile Virtual Reality and Artificial Intelligence	3
74	CS-670	Kernel Methods for Pattern Recognition	4
75	CS-672	Advanced Topics in Deep Learning	4
76	CS-673	Advanced Computer Vision	4
77	CS-677	Soft Computing	3
78	CS-685	Natural Language Processing	3
79	CS-686	Large Language Models: Core Concepts to Custom	4
80	CS-693	Compressed Sensing and its applications	3
81	EE-305	Digital Signal Processing	4
82	EE-511	Computer Vision	4
83	EE-530	Optimization theory	3
84	EE-531	Estimation and Detection Theory	3
85	EE-541	Tensors: Techniques, Algorithms, Applications for Signal	4
86	EE-543	Vision and Learning Based Control	3
87	EE-574	Biomedical Signal and Image Analysis	3
88	EE-575	Applied Statistics for Data and Signals	4
89	EE-608	Digital Image Processing	4
90	EE-620	Advanced Digital Signal Processing	3
91	ME-522	High-Performance Scientific Computing	3
<i>Compulsory project components — counted as DE, not selectable</i>			
92	DS-498P	Major Technical Project - I (compulsory; counted as DE)	4
93	DS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Electrical Engineering

Table 540: Electrical Engineering — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	52
	Discipline Electives	20
Electives	Free Electives	17
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		161

Table 541: B. Tech Electrical Engineering — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-210	Digital Systems Design	3-0-0-3	3
3	EE-210P	Digital Systems Design Lab	0-0-2-1	1
4	EE-260	Signals and Systems	2.5-0.5-0-3	3
5	EE-261	Electrical Systems Around Us	3-0-0-3	3
6	EE-261P	Electrical Systems Around Us Lab	0-0-3-2	2
7	EE-311	Device Electronics	3-0-0-3	3
8	EE-201	Electromechanics	2.5-0.5-0-3	3
9	EE-201P	Electromechanics Lab	0-0-2-1	1
10	EE-205	Electromagnetics & Wave Propagation	3-0-0-3	3
11	EE-211	Analog Circuit Design	2-0-2-3	3
12	EE-223P	Reverse Engineering	1-0-0-1	1
13	EE-304	Communication Systems	3-0-1-4	4
14	EE-231	Measurement and Instrumentation	2-0-2-3	3
15	EE-302	Control Systems	3-0-2-4	4
16	EE-303	Power System	3-1-0-4	4
17	EE-314	Digital Signal Processing	3-0-2-4	4
18	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
Total				52

Table 542: B. Tech Electrical Engineering — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		IC-182	History of Science and Technology	3	
10	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
11		HS-112	Japanese Language I	3-0-0-3	
12		HS-342	German I	3-0-0-3	
13		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 543: B. Tech Electrical Engineering — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket (choose 1)	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9		IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	
10		IC-241	Material Science for Engineers	3-0-0-3	
11		IC-253	Data Structures and Algorithms	2-0-2-3	
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 544: B. Tech Electrical Engineering — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	EE-203	Network Theory	3-0-0-3	3
3	DC	EE-210	Digital Systems Design	3-0-0-3	3
4	DC	EE-210P	Digital Systems Design Lab	0-0-2-1	1
5	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
6	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
7	DC	EE-261P	Electrical Systems Around Us Lab	0-0-3-2	2
8	DC	EE-311	Device Electronics	3-0-0-3	3
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 545: B. Tech Electrical Engineering — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
3	DC	EE-201P	Electromechanics Lab	0-0-2-1	1
4	DC	EE-205	Electromagnetics & Wave Propagation	3-0-0-3	3
5	DC	EE-211	Analog Circuit Design	2-0-2-3	3
6	DC	EE-223P	Reverse Engineering	1-0-0-1	1
7	DC	EE-304	Communication Systems	3-0-1-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 546: B. Tech Electrical Engineering — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	EE-302	Control Systems	3-0-2-4	4
3	DC	EE-303	Power System	3-1-0-4	4
4	DC	EE-314	Digital Signal Processing	3-0-2-4	4
5	DC	EE-326	Computer Organization and Processor Architecture Design	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 547: B. Tech Electrical Engineering — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 548: B. Tech Electrical Engineering — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	EE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 549: B. Tech Electrical Engineering — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	EE-010	Internship		2
2	MTP	EE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 20 credits of **Discipline Electives** and 17 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 550: Electrical Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BE-302	Bioelectric Systems Modeling	4
2	BE-304	Bioinformatics	4
3	CS-549	Performance Analysis of Computer Networks	3
4	CS-671	Deep Learning and Applications	4
5	EE-500	Network Control System	3
6	EE-501	Power System Operation and Control	3
7	EE-502	Analog System Design Laboratory	3
8	EE-503	Advance Communication Theory	3
9	EE-504	Switch Mode Power Conversion	3
10	EE-505	Materials for Semiconductor Industry	3
11	EE-506	Solar Photovoltaic Energy Systems	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	EE-507	Transmission Lines and Basic Microwave Engineering	4
13	EE-508	Fundamentals of Electric Drives	3
14	EE-508P	Practicum on fundamentals of Electrical Drives	2
15	EE-509	Linear Dynamical Systems	3
16	EE-511	Computer Vision	4
17	EE-512	CMOS Analog IC Design	4
18	EE-513	Special Electrical Machines	3
19	EE-514	Robust Control Systems	3
20	EE-515	Nonlinear Stability and Control	4
21	EE-516	Biomedical Systems	4
22	EE-517	Wireless Communications and Networking	3
23	EE-518	Information Theory	3
24	EE-519	CMOS Digital IC design Practicum	2
25	EE-519P	CMOS Digital IC Design Practicum	2
26	EE-520	Microelectronic Devices and Modelling	3
27	EE-522	Matrix Theory for Engineers	3
28	EE-523	Digital VLSI Architecture Design	4
29	EE-524	Digital MOS LSI Circuits	3
30	EE-526	Power Semiconductor Devices	3
31	EE-527	Analysis and Design of Power Electronic Converters	3
32	EE-527P	Practicum on Analysis and Design of Power Electronic Converters	2
33	EE-528	Modeling and Analysis of Electric Machines	3
34	EE-529	Embedded Systems	4
35	EE-530	Optimization theory	3
36	EE-531	Estimation and Detection Theory	3
37	EE-532	Supervised Research Exposure	3
38	EE-533	Access Network	4
39	EE-534	Probability and Random Processes	3
40	EE-535	Communication and Signal Processing Systems Design	3
41	EE-536	IoT Systems	3
42	EE-537	Power Quality Problems and Mitigation Techniques in Microgrids	3
43	EE-538	Data-driven Control	3
44	EE-540	Wide Band Gap Devices in Power Electronics Applications	3
45	EE-541	Sensors: Techniques, Algorithms, Applications for Signal	4
46	EE-542	Modelling, Simulation and Control of Hybrid Electric Vehicle	3
47	EE-543	Vision and Learning Based Control	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
48	EE-550	Optimal Control and Estimation	3
49	EE-551	Applied Photonics for Scientists and Engineers	3
50	EE-552	Power and Energy Systems	4
51	EE-553	Foundations of Intelligent Communication Systems-I	4
52	EE-554	Low Power VLSI Design	3
53	EE-555	Intelligent Control System	3
54	EE-556	Nuclear Reactor Control	1
55	EE-557	Adaptive Control	3
56	EE-560	Reconfigurable Computing	4
57	EE-570	Advanced Communication Theory	4
58	EE-574	Biomedical Signal and Image Analysis	3
59	EE-575	Applied Statistics for Data and Signals	4
60	EE-580	Network Systems: Modelling and Analysis	3
61	EE-581	Applied Statistics for Data and Signals	4
62	EE-582	Nonlinear Analysis and Control of Power Electronic Converters	3
63	EE-583	Smart Grids	3
64	EE-584	Power System Protection	3
65	EE-585	Theory, Modelling and Characterization of Photo-voltaic Devices	3
66	EE-593	Low Power VLSI Circuits	3
67	EE-597	Intellegent Control System	3
68	EE-601	Solid State Devices	3
69	EE-602	Control System Applications	3
70	EE-603	Renewable Energy and Smart Grid	3
71	EE-604	Practicum on Advanced Electric Drives	3
72	EE-604P	Practicum on Advanced Electric Drives	1
73	EE-605	Information Theory	3
74	EE-606	Introduction to High Voltage Engineering and Dielectric	3
75	EE-607	Optical Communication Systems	3
76	EE-608	Digital Image Processing	4
77	EE-609	Network Information Theory	3
78	EE-611	VLSI Technology	3
79	EE-611P	VLSI Fabrication Practicum	2
80	EE-612	OFDM For Optical Communications	3
81	EE-613	Wireless Communication	3
82	EE-614	Optical Communication Systems	3
83	EE-615	Nanoelectronics and Micronanofabrication	3
84	EE-616	Microwave engineering	2
85	EE-618	Industrial Process Control	3

Electrical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
86	EE-619	Mixed Signal VLSI Design	4
87	EE-620	Advanced Digital Signal Processing	3
88	EE-621	Radiating systems	4
89	EE-622	Microwave Integrated Circuits	3
90	EE-623P	Practicum on Digital Control of Power Electronics and Drives	3
91	EE-626P	Post Graduate Project - I	10
92	EE-630	HVDC Transmission and Flexible AC Transmission Systems	3
93	EE-641	Advance Wireless Technologies	4
94	EE-642	Research Study	3
95	EE-677	Analog Circuit Design	3
96	EE-678	Digital Circuit Design	3
97	EE-691	Advanced Topics in Dielectric Breakdown	1
98	EE-694	Photonics Computing	3
<i>Compulsory project components — counted as DE, not selectable</i>			
99	EE-498P	Major Technical Project - I (compulsory; counted as DE)	4
100	EE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Mechanical Engineering

Table 551: Mechanical Engineering — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	50
	Discipline Electives	16
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 552: B. Tech Mechanical Engineering — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-261	Electrical Systems Around Us	3-0-0-3	3
2	IC-241	Materials Science for Engineers	3-0-0-3	3
3	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	ME-210	Fluid Mechanics	2.5-0.5-0-3	3
5	ME-212	Product Manufacturing Technology	2-0-3-3	3
6	ME-213	Engineering Thermodynamics	3-1-0-4	4
7	ME-100	Reverse Engineering	0-0-2-1	1
8	ME-205	Machine Drawing	1-0-4-3	3
9	ME-210P	Fluid Mechanics Lab	0-0-2-1	1
10	ME-215	Manufacturing Engineering -1	3-0-0-3	3
11	ME-303	Heat Transfer	2.5-0.5-0-3	3
12	ME-307	Energy Conversion Devices	3-0-0-3	3
13	ME-303P	Heat Transfer Lab	0-0-3-1	1
14	ME-305	Design of Machine Elements	3-1-0-4	4
15	ME-309	Theory of Machines	3-1-0-4	4
16	ME-310	System Dynamics and Control	3-0-0-3	3
17	ME-311P	Mechanics of Solids Lab / Design Lab - 1	0-0-2-1	1
18	ME-315	Manufacturing Engineering 2	3-0-0-3	3
19	ME-312P	Theory of Machines Lab / Design Lab - 2	0-0-2-1	1
Total				50

Table 553: B. Tech Mechanical Engineering — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		IC-182	History of Science and Technology	3	
10	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
11		HS-112	Japanese Language I	3-0-0-3	
12		HS-342	German I	3-0-0-3	
13		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 554: B. Tech Mechanical Engineering — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 555: B. Tech Mechanical Engineering — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-261	Electrical Systems Around Us	3-0-0-3	3
4	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
5	DC	ME-210	Fluid Mechanics	2.5-0.5-0-3	3
6	DC	ME-212	Product Manufacturing Technology	2-0-3-3	3
7	DC	ME-213	Engineering Thermodynamics	3-1-0-4	4
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 556: B. Tech Mechanical Engineering — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-100	Reverse Engineering	0-0-2-1	1
2	DC	ME-205	Machine Drawing	1-0-4-3	3
3	DC	ME-210P	Fluid Mechanics Lab	0-0-2-1	1
4	DC	ME-215	Manufacturing Engineering -1	3-0-0-3	3
5	DC	ME-303	Heat Transfer	2.5-0.5-0-3	3
6	DC	IC-241	Materials Science for Engineers	3-0-0-3	3
7	DC	ME-309	Theory of Machines	3-1-0-4	4
8	DC	ME-311P	Mechanics of Solids Lab / Design Lab - 1	0-0-2-1	1
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 557: B. Tech Mechanical Engineering — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-303P	Heat Transfer Lab	0-0-3-1	1
2	DC	ME-305	Design of Machine Elements	3-1-0-4	4
3	DC	ME-307	Energy Conversion Devices	3-0-0-3	3
4	DC	ME-310	System Dynamics and Control	3-0-0-3	3
5	DC	ME-312P	Theory of Machines Lab / Design Lab - 2	0-0-2-1	1
6	DC	ME-315	Manufacturing Engineering 2	3-0-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 558: B. Tech Mechanical Engineering — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 559: B. Tech Mechanical Engineering — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	ME-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 560: B. Tech Mechanical Engineering — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	ME-010	Internship		2
2	MTP	ME-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 16 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 561: Mechanical Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	ME-451	Refrigeration and Air Conditioning	3
2	ME-452	Robot Kinematics, Dynamics and Control	4
3	ME-501	Nanomanufacturing	3
4	ME-502	Functional Materials	3
5	ME-503	Micro/Nano Scale heat transfer	3
6	ME-504	Numerical Methods for Engineering Computation	3
7	ME-505	Applied Finite Element Method	3
8	ME-506	Fundamentals of Fracture Mechanics	3
9	ME-507	Micro and Nanoscale Fluid Mechanics	3
10	ME-508	Fundamentals of project management	3
11	ME-509	Nano-manufacturing	3

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	ME-510	Advanced Manufacturing Processes	3
13	ME-511	Manufacturing of composites	3
14	ME-513	Finite Element Methods in Engineering	4
15	ME-514	Fundamentals of Multiphase Flow	3
16	ME-516	Polymer Technology for Engineers	3
17	ME-517	Advanced Analytical Techniques for Engineers	4
18	ME-518	Conduction and Radiation	3
19	ME-519	Technical communication for Engineers	2
20	ME-520	Microwave based Manufacturing Processes	3
21	ME-521P	Laboratory course on Vehicle Design and Control	1
22	ME-522	High-Performance Scientific Computing	3
23	ME-523	Product Design	3
24	ME-524	Additive Manufacturing	3
25	ME-526	Programming Paradigm for Open-Source Software	1
26	ME-528	Thermal Management of Electronics	3
27	ME-530	Continuum Mechanics	3
28	ME-595	Experiential Learning in Composite Design, Manufacturing, Testing and Recycling	3
29	ME-600	Research Methodology	1
30	ME-601	Advanced Finite Element Methods	3
31	ME-602	Mechanical Vibration	3
32	ME-603	Advanced Fluid Mechanics	3
33	ME-604	Experimental Methods in Thermal Engineering	3
34	ME-605	Air Conditioning and Ventilation	3
35	ME-606	Advanced Solid Mechanics	3
36	ME-607	Materials Science For Failure Analysis	3
37	ME-608	Thin Films And Devices	3
38	ME-609	Functional Materials	3
39	ME-610	Advanced Thermodynamics	4
40	ME-611	Design and Optimization of Thermal Systems	3
41	ME-612	Introduction to Bio-materials	3
42	ME-613	Thermal Radiation	3
43	ME-614	Compressible Flow and Gas Dynamics	3
44	ME-615	Applied Computational Fluid Dynamics	3
45	ME-616	Dielectrics and Related Materials	3
46	ME-617	Mechanics of Composite Materials	3
47	ME-618	Stealth Technology: Infrared Signatures	3
48	ME-619	Experiments in Materials Science	3
49	ME-620	Modeling and Simulation	3
50	ME-621	Aircraft Propulsion	3
51	ME-622	Biomechanics of Musculoskeletal System	3

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
52	ME-625	Introduction to Turbulence and its Modeling	3
53	ME-626	Acoustics	3
54	ME-627	Mesh Independent Computational Techniques	3
55	ME-628	Impact Mechanics	3
56	ME-630	Machine learning for engineers	3
57	ME-631	Heat Transfer and Fluid Flow in Energy Systems	3
58	ME-632	Mechanics for Energy Systems	3
59	ME-633	Design of Energy Systems	3
60	ME-634	Thermodynamics for Energy Systems	3
61	ME-635	Manufacturing for Energy Systems	3
62	ME-636	Combustion Technology	3
63	ME-637	Wind Power Plant	3
64	ME-638	Solar Thermal Power Plant	3
65	ME-639	Thermal Power Plant Engineering	3
66	ME-640	Solar Power Utilization	3
67	ME-641	Finite Element Method	3
68	MT-201	Physics of Solids	3
69	MT-202	Applied Quantum Mechanics	3
70	MT-203	Materials Synthesis & Characterization	4
71	MT-204	Thermodynamics and Kinetics of Materials	3
72	MT-205	Functional Properties of Materials	4
73	MT-206	Extraction and Materials Processing	4
74	MT-301	Phase Transformations	3
75	MT-302	Transport Phenomena	3
76	MT-303	Computational Materials Science	4
77	MT-304	Mechanical Behavior of Materials	4
78	MT-501	Energy Conversion and Storage Technologies	3
79	MT-502	Recycling and Circular Economy	3
80	MT-503	Semiconductor Materials and Devices	3
81	MT-504	Powder metallurgical processing of Materials	3
82	MT-505	Thin Film Technology	3
83	MT-506	Biomaterials	3
84	MT-507	Modelling and Simulations in Material Science	3
85	MT-508	Iron and Steel Making	3
86	MT-509	Hydrogen Energy	3
87	MT-510	Colloids and Interfaces	3
88	MT-511	Sensor Materials and Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
89	ME-498P	Major Technical Project - I (compulsory; counted as DE)	4

Mechanical Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
90	ME-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Civil Engineering

Table 562: Civil Engineering — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	49
	Discipline Electives	17
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 563: B. Tech Civil Engineering — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CE-202	Introduction to Civil Engineering Profession	1-0-0-1	1
2	CE-203	Construction Materials	3-0-0-3	3
3	CE-203P	Building (Constructions) Materials Lab	0-0-2-1	1
4	CE-252	Geology and Geomorphology	2-0-2-3	3
5	CE-310	Strength of Materials and Structures	3-0-0-3	3
6	CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
7	CE-201	Surveying Traditional and Digital	2-0-4-4	4
8	CE-251	Hydraulics Engineering	3-0-0-3	3
9	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
10	CE-311	Geotechnical Engineering I	3-0-0-3	3
11	CE-311P	Geotechnical Engineering Lab	0-0-2-1	1
12	CE-404	Analysis of Structures	3-0-0-3	3
13	CE-303	Water Resources Engineering	3-0-0-3	3
14	CE-351	Design of Reinforced Concrete Structures	2-1-0-3	3
15	CE-352	Transportation Engineering	3-0-0-3	3
16	CE-352P	Transportation Engineering Lab	0-0-2-1	1
17	CE-353P	Civil Engineering Drawing	0-0-2-1	1
18	CE-402	Geotechnical Engineering II	2-1-0-3	3
19	CE-305P	Environmental Engineering Lab	0-0-2-1	1
20	CE-356	Reverse Engineering	1-0-0-1	1
21	CE-401	Design of Steel Structures	2-1-0-3	3
22	CE-403	Water and Wastewater Engineering	3-0-0-3	3
Total				49

Table 564: B. Tech Civil Engineering — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 565: B. Tech Civil Engineering — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	DC	CE-202	Introduction to Civil Engineering Profession	1-0-0-1	1
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 566: B. Tech Civil Engineering — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CE-203	Construction Materials	3-0-0-3	3
4	DC	CE-203P	Building (Constructions) Materials Lab	0-0-2-1	1
5	DC	CE-252	Geology and Geomorphology	2-0-2-3	3
6	DC	CE-310	Strength of Materials and Structures	3-0-0-3	3
7	DC	CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 567: B. Tech Civil Engineering — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-201	Surveying Traditional and Digital	2-0-4-4	4
2	DC	CE-251	Hydraulics Engineering	3-0-0-3	3
3	DC	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
4	DC	CE-311	Geotechnical Engineering I	3-0-0-3	3
5	DC	CE-311P	Geotechnical Engineering Lab	0-0-2-1	1
6	DC	CE-404	Analysis of Structures	3-0-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 568: B. Tech Civil Engineering — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-303	Water Resources Engineering	3-0-0-3	3
2	DC	CE-351	Design of Reinforced Concrete Structures	2-1-0-3	3
3	DC	CE-352	Transportation Engineering	3-0-0-3	3
4	DC	CE-352P	Transportation Engineering Lab	0-0-2-1	1
5	DC	CE-353P	Civil Engineering Drawing	0-0-2-1	1
6	DC	CE-402	Geotechnical Engineering II	2-1-0-3	3
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 569: B. Tech Civil Engineering — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-305P	Environmental Engineering Lab	0-0-2-1	1
2	DC	CE-356	Reverse Engineering	1-0-0-1	1
3	DC	CE-401	Design of Steel Structures	2-1-0-3	3
4	DC	CE-403	Water and Wastewater Engineering	3-0-0-3	3
5	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 570: B. Tech Civil Engineering — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	CE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 571: B. Tech Civil Engineering — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	CE-010	Internship		2
2	MTP	CE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 17 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 572: Civil Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CD-591	Special Topics in Disaster Resilient Hill Roads	1
2	CE-304	Hydraulics Engineering Lab	3
3	CE-305	Environment and Earth Science Lab	3
4	CE-306	Environmental Engineering Lab	3
5	CE-353	Civil Engineering Drawing	3
6	CE-354	Building and Pavement Materials Laboratory	3

Civil Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
7	CE-355	Hydrology	3
8	CE-405	Water and Wastewater Engineering	3
9	CE-451	Irrigation Engineering and Hydraulic Structures	3
10	CE-452	Open Channel Flow	3
11	CE-501	Remote Sensing	3
12	CE-502	Groundwater Flow and Contaminant Transport	3
13	CE-503	Fundamentals of Project Management	3
14	CE-504	Slope Stability and Retaining Structures	3
15	CE-505	Engineering of Ground Modification	3
16	CE-506	Analysis of Indeterminate Structures	3
17	CE-507	Advanced Concrete Science	3
18	CE-508	Photogeology and Photogrammetry	3
19	CE-509	Bridge Engineering	3
20	CE-510	Modelling and Simulation in Water Resources Engineering	3
21	CE-511	Structural Dynamics with Application to Earthquake Engineering	3
22	CE-512	Advanced Soil Mechanics	3
23	CE-513	Advanced Foundation Engineering	3
24	CE-514	Rock Mechanics	3
25	CE-515	Environmental Impact Assessment	3
26	CE-516	Uncertainty Analysis in Civil Engineering	3
27	CE-517	Hydroinformatics	4
28	CE-518	Structural Reliability and Risk Assessment	3
29	CE-519	Chemistry of Natural Waters	3
30	CE-520	Environmental Reaction Modeling	3
31	CE-521	Ecology and Environment Microbiology	4
32	CE-522	Matrix Methods for Structural Analysis	3
33	CE-523	Building Science	3
34	CE-524	Applied Hydroclimatology	3
35	CE-524P	Computational Hydroclimatology Lab	1
36	CE-526	Critical Zone Science	3
37	CE-551	Geosynthetics and their applications	3
38	CE-552	Concrete Technology	3
39	CE-554	Prestressed Concrete Structures	3
40	CE-555	Advanced Design of Structures	3
41	CE-556	Structural Engineering Laboratory	3
42	CE-556P	Structural Engineering laboratory	2
43	CE-557	Solid Mechanics in Structural Engineering	3
44	CE-558	Air pollution and its mitigation	3
45	CE-559	Biological Wastewater Treatment	3

Civil Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
46	CE-560	Soil Dynamics	3
47	CE-561	The science of climate change	3
48	CE-562	Numerical Methods for Engineering Computation	3
49	CE-563	Finite Element Methods in Engineering	4
50	CE-564	Fundamentals of Fracture Mechanics	3
51	CE-567	Highway Safety and Modeling	3
52	CE-568P	Experimental Structural Dynamics	1
53	CE-586	Mini Project	3
54	CE-586P	Mini Project	3
55	CE-587	Industrial / Academic Internship	3
56	CE-591	Special Topics in Civil and Mechanical Engineering	3
57	CE-600	Research Methodology for Civil Engineering	1
58	CE-601	Geo-Informatics	3
59	CE-602	Blast Engineering	3
60	CE-604	Theory of Plasticity	3
61	CE-605	Engineering Seismology and Seismic Hazard Assessment	3
62	CE-606	Constitutive Modeling of Frictional Material	3
63	CE-610	Analysis and Design for Earthquake Resistant Structures	3
64	CE-611	Structural Health Monitoring	3
65	CE-612	Theory of Plates and Shells	3
66	CE-613	Mechanics of Unsaturated Soils	3
67	CE-617	Mechanics of Composite Materials	3
68	CE-688	Post Graduate Project - 1	3
69	CE-689	Post Graduate Project - II	3
<i>Compulsory project components — counted as DE, not selectable</i>			
70	CE-498P	Major Technical Project - I (compulsory; counted as DE)	4
71	CE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Engineering Physics

Table 573: Engineering Physics — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	37
	Discipline Electives	29
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 574: B. Tech Engineering Physics — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EP-301	Engineering Mathematics 2	3-1-0-4	4
2	EP-321	Foundations of Electrodynamics	3-0-0-3	3
3	PH-301	Quantum Mechanics and Applications	2.5-0.5-0-3	3
4	EE-223P	Reverse Engineering	1-0-0-1	1
5	EP-403	Physics of Atoms and Molecules	3-0-0-3	3
6	PH-302	Introduction to Statistical Mechanics	2.5-0.5-0-3	3
7	PH-501	Solid State Physics	3	3
8	EE-311	Device Electronics for Integrated Circuits	3-0-0-3	3
9	EP-302	Computational Methods for Engineering	2-0-2-3	3
10	EP-402P	Engineering Physics Practicum	1-0-5-4	4
11	EP-401P	Engineering of Instrumentation	1-0-5-4	4
12	PH-502	Photonics	3-0-0-3	3
Total				37

Table 575: B. Tech Engineering Physics — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 576: B. Tech Engineering Physics — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 577: B. Tech Engineering Physics — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	EP-301	Engineering Mathematics 2	3-1-0-4	4
3	DC	EP-321	Foundations of Electrodynamics	3-0-0-3	3
4	DC	PH-301	Quantum Mechanics and Applications	2.5-0.5-0-3	3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 578: B. Tech Engineering Physics — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	DC	EE-223P	Reverse Engineering	1-0-0-1	1
3	DC	EP-403	Physics of Atoms and Molecules	3-0-0-3	3
4	DC	PH-302	Introduction to Statistical Mechanics	2.5-0.5-0-3	3
5	DC	PH-501	Solid State Physics	3	3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 579: B. Tech Engineering Physics — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-311	Device Electronics for Integrated Circuits	3-0-0-3	3
2	DC	EP-302	Computational Methods for Engineering	2-0-2-3	3
3	DC	EP-402P	Engineering Physics Practicum	1-0-5-4	4
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 580: B. Tech Engineering Physics — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EP-401P	Engineering of Instrumentation	1-0-5-4	4
2	DC	PH-502	Photonics	3-0-0-3	3
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 581: B. Tech Engineering Physics — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	EP-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 582: B. Tech Engineering Physics — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	EP-010	Internship		2
2	MTP	EP-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 29 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 583: Engineering Physics — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-208	Mathematical Foundations of Computer Science	4
2	CS-241	Introduction to Cryptography	3
3	CS-304	Formal Languages and Automata Theory	3
4	CS-309	Information and Database Systems	4
5	CS-511	Applied Probability	2
6	CS-671	Deep Learning and Applications	4
7	CS-672	Advanced Topics in Deep Learning	4
8	DS-201	Data handling and visualization	3
9	DS-301	Mathematical Foundations of Data Science	4

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
10	DS-401	Optimization for data science	3
11	DS-416	Fundamentals of Machine Learning (CS-669)	4
12	EE-203	Network Theory	3
13	EE-211	Analog Circuit Design	3
14	EE-307	Theory of Measurement	3
15	EE-511	Computer Vision	4
16	EE-512	CMOS Analog IC Design	4
17	EE-519P	CMOS Digital IC Design Practicum	2
18	EE-520	Microelectronic Devices and Modelling	3
19	EE-524	Digital MOS LSI Circuits	3
20	EE-534	Probability and Random Processes	3
21	EE-551	Applied Photonics for Scientists and Engineers	3
22	EE-593	Low Power VLSI Circuits	3
23	EE-611	VLSI Technology	3
24	EE-614	Optical Communication Systems	3
25	EE-621	Radiating systems	4
26	EN-502	Emerging Energy Sources	3
27	EP-201	Reverse Engineering	1
28	EP-401	Engineering of Instrumentation	3
29	EP-402	Engineering Physics Practicum	3
30	EP-502	Informatics for Materials Design	3
31	IC-253	Programming and Data Structures	3
32	MA-210	Real and Complex Analysis	3
33	MA-511	Real Analysis	4
34	MA-512	Linear Algebra	4
35	MA-513	Ordinary Differential Equation	4
36	MA-516	Topology	4
37	MA-521	Topology and Functional Analysis	4
38	MA-522	Partial Differential Equations	4
39	MA-560	Nonlinear Dynamics and Chaos	3
40	ME-210	Fluid Mechanics	3
41	ME-303	Heat Transfer	3
42	ME-307	Energy Conversion Devices	3
43	ME-308	Manufacturing Engineering	3
44	ME-503	Micro/Nano Scale heat transfer	3
45	ME-509	Nano-manufacturing	3
46	ME-510	Advanced Manufacturing Processes	3
47	ME-603	Advanced Fluid Mechanics	3
48	ME-615	Applied Computational Fluid Dynamics	3
49	PH-503	Laser and Applications	3

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
50	PH-504	Organic Optoelectronics	3
51	PH-507	X-rays as a Probe to Study Material Properties	3
52	PH-508	Magnetism and Magnetic Materials	3
53	PH-511	Mathematical Physics	4
54	PH-512	Classical Mechanics	4
55	PH-513	Quantum Mechanics	3
56	PH-514	Electronics	3
57	PH-515P	Physics Laboratory	3
58	PH-516	Research Project I	2
59	PH-521	Electromagnetic Theory	4
60	PH-525P	Electronics Lab	3
61	PH-528	Introduction to General Relativity	3
62	PH-530	Cosmology-I	3
63	PH-550	Introduction to Quantum Optics	3
64	PH-579	Quantum Computation and Information	3
65	PH-600	Research Methodology	1
66	PH-601	Mesoscopic Physics and Quantum Transport	3
67	PH-603	Advanced Condensed Matter Physics	3
68	PH-604	Optical Properties of Solids	3
69	PH-605	Superconductivity	3
70	PH-606	Quantum Field Theory	3
71	PH-607	Physics of Ultracold Quantum Gases	3
72	PH-608	Computer assisted quantum mechanics	3
73	PH-609	Theory of quantum collision and spectroscopy	3
74	PH-612	Nuclear and Particle Physics	3
75	PH-613	Special Topics in Quantum Mechanics	3
76	PH-615P	Mini Thesis I	3
77	PH-621	Computational Methods for Physicists	4
78	PH-625	Data Analysis in Particle Physics	4
79	PH-626	Elementary Theoretical Particle Physics	3
80	PH-627	Topological Quantum Matter	3
81	PH-701	Introduction to Molecular Simulations	4
82	PH-706	Introduction to Stochastic Problems in Physics	3
83	QS-501P	Experiments in Quantum Optics	3
84	QT-301	Survey of Quantum Technologies and Applications	3
85	QT-406	Introduction to Quantum communication	3
86	QT-407	Introduction to Quantum Sensing	3
87	QT-509	Engineering Foundations of Quantum Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
88	EP-498P	Major Technical Project - I (compulsory; counted as DE)	4

Engineering Physics — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
89	EP-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Bioengineering

Table 584: Bioengineering — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	42
	Discipline Electives	24
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 585: B. Tech Bioengineering — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	BE-201	Cell Biology	3-0-2-4	4
2	BE-202	Biochemistry and Molecular Biology	3-0-2-4	4
3	BE-308	Introduction to Biomanufacturing	3-0-2-4	4
4	BE-309	Biosensing & Bioinstrumentation	3-0-2-4	4
5	BE-203	Enzymology and Bioprocessing	3-0-2-4	4
6	BE-301	Biomechanics	3-0-2-4	4
7	BE-304	Bioinformatics	3-0-2-4	4
8	BE-101P	Reverse Engineering	0-0-2-1	1
9	BE-303	Applied Biostatistics	3-0-2-4	4
10	BE-305	Bioethics and Regulatory Affairs	1-0-0-1	1
11	BE-306	Fundamentals of Genetic Engineering	3-1-0-4	4
12	BE-310	Biomaterials	3-0-2-4	4
Total				42

Table 586: B. Tech Bioengineering — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 587: B. Tech Bioengineering — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 588: B. Tech Bioengineering — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	BE-201	Cell Biology	3-0-2-4	4
3	DC	BE-202	Biochemistry and Molecular Biology	3-0-2-4	4
4	DC	BE-308	Introduction to Biomanufacturing	3-0-2-4	4
5	DC	BE-309	Biosensing & Bioinstrumentation	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 589: B. Tech Bioengineering — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	BE-203	Enzymology and Bioprocessing	3-0-2-4	4
2	DC	BE-301	Biomechanics	3-0-2-4	4
3	DC	BE-304	Bioinformatics	3-0-2-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 590: B. Tech Bioengineering — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	BE-101P	Reverse Engineering	0-0-2-1	1
2	DC	BE-303	Applied Biostatistics	3-0-2-4	4
3	DC	BE-305	Bioethics and Regulatory Affairs	1-0-0-1	1
4	DC	BE-306	Fundamentals of Genetic Engineering	3-1-0-4	4
5	DC	BE-310	Biomaterials	3-0-2-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 591: B. Tech Bioengineering — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 592: B. Tech Bioengineering — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	BE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 593: B. Tech Bioengineering — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	BE-010	Internship		2
2	MTP	BE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 24 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 594: Bioengineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-506	Cognitive Robotics	3
2	BE-101	Reverse Engineering for Bioengineers	3
3	BE-302	Bioelectric Systems Modeling	4
4	BE-307	Reverse Engineering for Bioengineers	3
5	BE-401	Bioengineering Mini Project, Term Paper and Seminar	4
6	BE-501	Anatomy and Physiology	3
7	BE-502	Design and Analysis of Bioalgorithms	3
8	BE-503	Biosensing & Bioinstrumentation	4
9	BE-504	Biomaterials	4
10	BE-505	Computational Biology	3
11	BE-506	Biological Modelling and Simulation	3
12	BE-507	Environmental Impact Assessment	3
13	BE-598	Thesis Project Part I	3
14	BE-599	Thesis Project Part II	3
15	BE-698	Thesis Project Part I	3
16	BE-699	Thesis Project Part II	17
17	BY-501	DNA Nanotechnology	3
18	BY-502	Biophysics and Protein Engineering	3
19	BY-503	Cellular Fuel and Cellular Communication	3

Bioengineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
20	BY-504	Metabolic Systems Biology	4
21	BY-506	Advanced Immunology	3
22	BY-507	Genetic Engineering	4
23	BY-508	Practical Metabolomics	3
24	BY-509	Practical OMICS	3
25	BY-510	Advanced Cell Biology	3
26	BY-511	Cell Physiology in health and disease	3
27	BY-512	Quantitative and Computational Biology	3
28	BY-514	Analytical Biotechniques	3
29	BY-515	Molecular Biotechnology	3
30	BY-516	Introduction to OMICS and Systems Analysis	3
31	BY-517	Proteomics	3
32	BY-518	Disease Biology	3
33	BY-519	Protein Sciences in Therapeutics	3
34	BY-520	Cell Biology and Physiology Lab	3
35	BY-520P	Cell Biology and Physiology	1
36	BY-521	Computational Biology and Cellular Bioprocess Technology Lab	3
37	BY-521P	Computational Biology and Cellular Bioprocess Technology Lab	1
38	BY-522	Analytical and Molecular Biotechnology Lab	3
39	BY-523	Systems Biology Lab	3
40	BY-524	Medical and NanoBiotechnology Lab	3
41	BY-527	Gene silencing and genome editing: principles and applications	3
42	BY-528	Sensory Biology	3
43	BY-529	Mechanobiology of the Cell (MBoC)	3
44	BY-531	Quantitative Biology and Data Analytics	3
45	BY-532	Immunotechnology	3
46	BY-535P	Analytical Biotechniques Lab	1
47	BY-536P	Immunotechnology Lab	1
48	BY-537	Computational Biology-1	4
49	BY-606	Bioinformatics Applications for Systems	3
50	BY-613	Metagenomics and Next Generation Sequencing Technologies	3
51	CE-521	Ecology and Environment Microbiology	4
52	CE-559	Biological Wastewater Treatment	3
53	CS-309	Information and Database Systems	4
54	CS-606	Cognitive Modeling	3
55	CS-671	Deep Learning and Applications	4
56	CY-344	Food Chemistry: Processing, Preservation and Storage	3

Bioengineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
57	CY-550	Bioinspired Materials	3
58	CY-642	Molecular- and Bio-electronics	3
59	CY-643	Advanced Analytical Techniques	3
60	CY-644	Bioinorganic chemistry	3
61	CY-670	Fluorescence spectroscopy, microscopy and applications	3
62	DS-303	Statistical Foundations of Data Science	3
63	DS-314	Probability Theory and Statistics	4
64	EE-305	Digital Signal Processing	4
65	EE-314	Digital Signal Processing	4
66	EE-516	Biomedical Systems	4
67	EE-574	Biomedical Signal and Image Analysis	3
68	EE-608	Digital Image Processing	4
69	MA-621	Modeling Population Dynamics	3
70	MA-650	Mathematical Models for Infectious Diseases	3
71	ME-303	Heat Transfer	3
72	ME-308	Manufacturing Engineering	3
73	ME-527	Biofluid Dynamics	3
74	ME-612	Introduction to Bio-materials	3
75	ME-622	Biomechanics of Musculoskeletal System	3
76	MT-506	Biomaterials	3
<i>Compulsory project components — counted as DE, not selectable</i>			
77	BE-498P	Major Technical Project - I (compulsory; counted as DE)	4
78	BE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Mathematics and Computing

Table 595: Mathematics and Computing — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	51
	Discipline Electives	15
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 596: B. Tech Mathematics and Computing — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
2	MA-210	Real and Complex Analysis	2.5-0.5-0-3	3
3	MA-211	Ordinary Differential Equations	3-1-0-4	4
4	MA-312 / CS-212	Design of Algorithms	3-0-2-4	4
5	CS-201	Computer Organization	3-0-0-3	3
5b	CS-201P	Computer Organization Laboratory	0-0-2-1	1
6	MA-220	Partial Differential Equations	3-1-0-4	4
7	MA-221	Numerical Analysis	3-1-0-4	4
8	MA-222	Applied Linear Programming	3-1-0-4	4
9	MA-310	Matrix Computation and Lab	3-0-2-4	4
10	MA-311	Mathematical Modelling	3-0-0-3	3
11	MA-313 / CS-304	Formal Language and Automata Theory	3-0-0-3	3
12	MA-322	Applied Graph Theory		4
13	MA-321	Numerics of Differential Equation		4
14	MA-323P	Applied Databases Practicum	0-0-3-2	2
Total				51

Table 597: B. Tech Mathematics and Computing — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 598: B. Tech Mathematics and Computing — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-253	Data Structures and Algorithms	2-0-2-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 599: B. Tech Mathematics and Computing — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-208	Mathematical Foundations of Computer Science	3-1-0-4	4
4	DC	MA-210	Real and Complex Analysis	2.5-0.5-0-3	3
5	DC	MA-211	Ordinary Differential Equations	3-1-0-4	4
6	DC	MA-312 / CS-212	Design of Algorithms	3-0-2-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 600: B. Tech Mathematics and Computing — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-201	Computer Organization	3-0-0-3	3
2	DC	CS-201P	Computer Organization Laboratory	0-0-2-1	1
3	DC	MA-220	Partial Differential Equations	3-1-0-4	4
4	DC	MA-221	Numerical Analysis	3-1-0-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 601: B. Tech Mathematics and Computing — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	MA-310	Matrix Computation and Lab	3-0-2-4	4
2	DC	MA-311	Mathematical Modelling	3-0-0-3	3
3	DC	MA-313 / CS-304	Formal Language and Automata Theory	3-0-0-3	3
4	DC	MA-222	Applied Linear Programming	3-1-0-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 602: B. Tech Mathematics and Computing — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	MA-321	Numerics of Differential Equation	3-0-2-4	4
2	DC	MA-323P	Applied Databases Practicum	0-0-3-2	2
3	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
4	DC	MA-322	Applied Graph Theory	3-0-2-4	4
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 603: B. Tech Mathematics and Computing — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	MC-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 604: B. Tech Mathematics and Computing — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	MC-010	Internship		2
2	MTP	MC-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 15 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

This programme groups its Discipline Electives into the named baskets below; the remaining approved electives follow as a general list.

Table 605: Mathematics and Computing — Discipline Elective Basket I — Foundation Module (choose from these)

Sl. No	Course Code	Course Name	Credits
1	MA-251	Abstract Algebra	3
2	MA-252	Functional Analysis	3
3	MA-253	Measure Theory	3
4	MA-254	Topology	3
5	MA-255	Number Theory	3

Table 606: Mathematics and Computing — Discipline Elective Basket II — Advanced Modelling Module (choose from these)

Sl. No	Course Code	Course Name	Credits
1	MA-351	Climate Modelling	3
2	MA-352	Computational Financial Modelling & Lab	3
3	MA-353	Modelling of Infectious Disease	3
4	MA-354	Mathematical Image Processing	3
5	MA-355	Mathematical Control Theory	3
6	MA-356	Modelling and Simulation	3
7	MA-357	Modelling Population Dynamics	3
8	MA-546	Introduction to Quantitative Finance	4
9	MA-570	Data Driven Dynamical Systems	3

Table 607: Mathematics and Computing — Other Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BE-303	Applied Biostatistics	4
2	BE-304	Bioinformatics	4
3	CE-251	Hydraulics Engineering	3
4	CS-241	Introduction to Cryptography	3
5	CS-302	Paradigms of Programming	4
6	CS-306	Operating Systems	3
7	CS-309	Information and Database Systems	4
8	CS-313	Computer Networks	4
9	CS-315	Artificial Intelligence (CS-362, CS-562)	4
10	CS-502	Compiler Design	4
11	CS-514	Advance Data Structure and Algorithm (CS-580)	4
12	CS-563	Scalable Data Science	4
13	CS-606	Cognitive Modeling	3
14	CS-609	Speech processing	3
15	CS-677	Soft Computing	3
16	DS-201	Data handling and visualization	3
17	DS-312	Computing Systems for Data Processing	4
18	DS-416	Fundamentals of Machine Learning (CS-669)	4
19	EE-203	Network Theory	3
20	EE-260	Signals and Systems	3
21	EE-301	Control Systems	3
22	EE-511	Computer Vision	4
23	EE-530	Optimization theory	3
24	EE-608	Digital Image Processing	4
25	MA-001	Preparatory Mathematics - 1	4
26	MA-002	Preparatory Mathematics - 2	4

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
27	MA-101	Calculus I	4
28	MA-102	Linear Algebra	4
29	MA-120	Introduction to Computing Systems and Databases	4
30	MA-201	Probability and Statistics	4
31	MA-460	Nonlinear Dynamics and Chaos	3
32	MA-465	Ordinary Differential Equations	3
33	MA-510	Climate Change Analysis	3
34	MA-516	Topology	4
35	MA-521	Topology and Functional Analysis	4
36	MA-522	Partial Differential Equations	4
37	MA-523	Numerical Analysis	4
38	MA-524	Probability and Statistics	4
39	MA-525	Heuristic Optimization	3
40	MA-526	An Introduction to Wavelets	3
41	MA-527	Field and Galois Theory	3
42	MA-528	Measure Theory and Integration	4
43	MA-529	Statistical Inference	4
44	MA-530	Graph Theory	4
45	MA-546	Introduction to Quantitative Finance	4
46	MA-548	Abstract Algebra	4
47	MA-549	Abstract Algebra	4
48	MA-550	Statistical Data Analysis	3
49	MA-551	Numerical Analysis	3
50	MA-552	Number Theory	3
51	MA-553	Mathematical Foundations of Financial Engineering	3
52	MA-555	Introduction to Partial Differential Equations for Engineers	3
53	MA-560	Nonlinear Dynamics and Chaos	3
54	MA-563	Statistical Computation and Simulation	4
55	MA-565	Numerical Methods in Quantitative Finance	3
56	MA-568	Real Analysis	3
57	MA-570	Data-driven Dynamical Systems	3
58	MA-571	Introduction to Approximation algorithms	4
59	MA-572	Applied Multivariate Statistical Analysis	4
60	MA-581P	Post Graduate Project I	6
61	MA-582P	Post Graduate Project-2	8
62	MA-588	Mathematical Control Theory	4
63	MA-595	Calculus as Ganita	3
64	MA-601	Real and Functional Analysis	3
65	MA-603	Advanced Partial Differential Equations	3
66	MA-604	Introduction To Game Theory	3

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
67	MA-605	Statistical Data Analysis	3
68	MA-607	Numerical Analysis	3
69	MA-608	Computational Fluid Dynamics	3
70	MA-609	Numerics of Partial Differential Equations	3
71	MA-610	Mathematical Modeling	3
72	MA-611	Statistical tools and Computing	4
73	MA-612	Operator Theory	4
74	MA-621	Modeling Population Dynamics	3
75	MA-641	Operations Research	3
76	MA-644	Dynamical Systems	3
77	MA-650	Mathematical Models for Infectious Diseases	3
78	MA-651	Optimization Techniques	3
79	MA-652	Stability Theory of Differential Equations	3
80	MA-653	Computational Financial Modelling	3
81	MA-653P	Computational Financial Modelling Lab	1
82	MA-654	Financial Engineering	3
83	MA-655	Fixed Income Securities	3
84	MA-656	Stochastic Calculus for Financial Engineering	3
85	MA-665	Semigroup Of Bounded Linear Operators	3
86	MA-704	Dynamical System	3
87	MA-709	Numerical Linear Algebra	3
88	MA-765	Fractional Differential Equations	4
89	MA-780	Topics on Semigroup Theory	3
90	ME-210	Fluid Mechanics	3
91	ME-213	Engineering Thermodynamics	4
92	ME-303	Heat Transfer	3
93	ME-308	Manufacturing Engineering	3
94		Time Series Analysis	3
95		Mathematical Method for Signal Processing	4
96	CS-669	Pattern Recognition	4
97	BE-301	Biomechanics	4
98	BY-507	Genetic Engineering	4
99	CS-352	Transportation Engineering	3
100	CS-352P	Transporting Engg. Lab	1
101		Information Systems and Databases	4
102		Times Series Analysis and Applications / Bayesian Data Analysis and Applications	3
103		Big Data: Management and Analytics	4
104	EP-302	Computational Methods for Engineering	3
105	ME-310	System Dynamics and Control	3
106	ME-210P	Fluid Mechanics Lab	1

Mathematics and Computing — Other Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
107	ME-303P	Heat Transfer Lab	1
108		Cellular Automata	3
109	CS-606	Computational Modeling of Social Systems	3
110	MA-562	An Introduction to Bayesian Analysis	4
111	MA-561	Fluidics and Microfluidics	4
112	MA-573	Advanced Mathematical Foundations of Machine Learning	4
113	MA-574	An Introduction to Probabilistic Machine Learning	4
114	MA-576	Nonlinear Optimization	4
<i>Compulsory project components — counted as DE, not selectable</i>			
115	MC-498P	Major Technical Project - I (compulsory; counted as DE)	4
116	MC-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - AI and Robotics

Table 608: General Engineering - AI and Robotics — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	34*
	Discipline Electives	32*
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

*EE-261 (Electrical System Around Us) and EE-201 (Electromechanics) are shown here at 3 credits each, as per the course catalogue; the programme brochure lists them at 5 and 4 credits respectively. If these are offered at the higher credits, the Discipline Core total rises accordingly and the Discipline Elective requirement falls by the same amount, so that $DC + DE = 66$ credits either way.

Table 609: B. Tech General Engineering - AI and Robotics — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-261	Electrical System Around Us	3-0-0-3	3*
2	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
3	EE-201	Electromechanics	2.5-0.5-0-3	3*
4	IC-253	Programming and Data Structures	2-0-2-3	3
5	ME-100	Reverse Engineering	0-0-2-1	1
6	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
7	AR-521	Control of Robotic Systems	3-0-0-3	3
8	AR-523	Robot Manipulators	2-1-0-3	3
9	EE-302	Control Systems	3-0-2-4	4
10	ME-305	Design of Machine Elements	3-1-0-4	4
11	ME-309	Theory of Machines	3-1-0-4	4
Total				34*

Table 610: B. Tech General Engineering - AI and Robotics — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Computing and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 611: B. Tech General Engineering - AI and Robotics — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 612: B. Tech General Engineering - AI and Robotics — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-261	Electrical System Around Us	3-0-0-3	3*
4	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					12*

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 613: B. Tech General Engineering - AI and Robotics — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3*
2	DC	IC-253	Programming and Data Structures	2-0-2-3	3
3	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					7*

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 614: B. Tech General Engineering - AI and Robotics — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
2	DC	AR-521	Control of Robotic Systems	3-0-0-3	3
3	DC	AR-523	Robot Manipulators	2-1-0-3	3
4	DC	EE-302	Control Systems	3-0-2-4	4
5	DC	ME-305	Design of Machine Elements	3-1-0-4	4
6	DC	ME-309	Theory of Machines	3-1-0-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 615: B. Tech General Engineering - AI and Robotics — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 616: B. Tech General Engineering - AI and Robotics — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 617: B. Tech General Engineering - AI and Robotics — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Beyond the core courses above, students choose 32 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Odd-semester electives are offered in Semesters 3, 5 and 7; even-semester electives in Semesters 4, 6 and 8.

Table 618: General Engineering - AI and Robotics — Discipline Electives, Odd Semesters (3, 5, 7)

Sl. No	Course Code	Course Name	Credits
1	AR-511	Autonomous Mobile Robots	3
2	AR-522	Robot Vision	3
3	CS-212	Design of Algorithms	4
4	CS-315	Artificial Intelligence	4
5	EE-203	Network Theory	3
6	EE-260	Signals and Systems	3
7	EE-511	Computer Vision	4
8	GE-303	Modern Powertrains: Electric, Hybrid & IC/Aeroengines	3
9	IC-241	Material Science for Engineers	3
10	ME-210	Fluid Mechanics	3
11	ME-212	Product Manufacturing Technology	3
12	ME-213	Thermodynamics	4

Table 619: General Engineering - AI and Robotics — Discipline Electives, Even Semesters (4, 6, 8)

Sl. No	Course Code	Course Name	Credits
1	AR-508	Marine Robotics	3

General Engineering - AI and Robotics — Discipline Electives, Even Semesters (4, 6, 8) (continued)

Sl. No	Course Code	Course Name	Credits
2	AR-513	Unmanned Aerial System	3
3	AR-524	Robot Simulators	3
4	AR-525	Reinforcement Learning for Robotics	3
5	AR-526	Multimodal Learning	3
6	CE-251	Hydraulics Engineering	3
7	CS-671	Deep Learning and Applications	4
8	CS-673	Advanced Computer Vision	4
9	CS-683	Generative AI	4
10	DS-314	Probability Theory and Statistics	4
11	EE-211	Analog Circuit Design	4
12	EE-608	Digital Image Processing	4
13	IC-241	Material Science for Engineers	3
<i>Compulsory project components — counted as DE, not selectable</i>			
14	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
15	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Mechatronics

Table 620: General Engineering - Mechatronics — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	38*
	Discipline Electives	28*
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

*EE-261 (Electrical System Around Us) and EE-201 (Electromechanics) are shown here at 3 credits each, as per the course catalogue; the programme brochure lists them at 5 and 4 credits respectively. If these are offered at the higher credits, the Discipline Core total rises accordingly and the Discipline Elective requirement falls by the same amount, so that $DC + DE = 66$ credits either way.

Table 621: B. Tech General Engineering - Mechatronics — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-260	Signals and Systems	2.5-0.5-0-3	3
3	EE-261	Electrical System Around Us	3-0-0-3	3*
4	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
5	EE-201	Electromechanics	2.5-0.5-0-3	3*
6	EE-211	Analog Circuit Design	2-0-2-3	3
7	ME-100	Reverse Engineering	0-0-2-1	1
8	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
9	CS-315 [†]	Artificial Intelligence	3-0-2-4	4
10	EE-302	Control Systems	3-0-2-4	4
11	ME-305	Design of Machine Elements	3-1-0-4	4
12	ME-309	Theory of Machines	3-1-0-4	4
Total				38*

[†]CS-305 Artificial Intelligence (3-0-0-3) has been recoded as CS-315 (3-0-2-4) w.e.f. batch 2025; the extra credit is adjusted against Discipline Electives, keeping the total at 160.

Table 622: B. Tech General Engineering - Mechatronics — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Computing and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 623: B. Tech General Engineering - Mechatronics — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 624: B. Tech General Engineering - Mechatronics — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-203	Network Theory	3-0-0-3	3
4	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
5	DC	EE-261	Electrical System Around Us	3-0-0-3	3*
6	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					18*

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 625: B. Tech General Engineering - Mechatronics — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3*
2	DC	EE-211	Analog Circuit Design	2-0-2-3	3
3	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					7*

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 626: B. Tech General Engineering - Mechatronics — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
2	DC	CS-315 [†]	Artificial Intelligence	3-0-2-4	4
3	DC	EE-302	Control Systems	3-0-2-4	4
4	DC	ME-305	Design of Machine Elements	3-1-0-4	4
5	DC	ME-309	Theory of Machines	3-1-0-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 627: B. Tech General Engineering - Mechatronics — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 628: B. Tech General Engineering - Mechatronics — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 629: B. Tech General Engineering - Mechatronics — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Beyond the core courses above, students choose 28 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Odd-semester electives are offered in Semesters 3, 5 and 7; even-semester electives in Semesters 4, 6 and 8.

Table 630: General Engineering - Mechatronics — Discipline Electives, Odd Semesters (3, 5, 7)

Sl. No	Course Code	Course Name	Credits
1	AR-521	Control of Robotic Systems	3
2	CS-212	Design of Algorithms	4
3	CS-312	Operating System	4
4	CS-313	Computer Networks	4
5	DS-314	Probability Theory and Statistics	4
6	EE-210	Digital Systems Design	4
7	EE-231	Measurement and Instrumentation	3
8	EE-311	Device Electronics for Integrated Circuits	3

General Engineering - Mechatronics — Discipline Electives, Odd Semesters (3, 5, 7) (continued)

Sl. No	Course Code	Course Name	Credits
9	EE-326	Computer Organisation and Processor Architecture Design	4
10	ME-210	Fluid Mechanics	3
11	ME-212	Product Manufacturing Technology	3
12	ME-213	Thermodynamics	3
13	ME-311P	Design Lab – I	1
14	ME-315	Manufacturing Engineering – II	3
15	VL-401	RTL Design and Verification	3
16	VL-403	CMOS Digital IC Design	4
17	VL-404	CMOS Analog IC Design	4

Table 631: General Engineering - Mechatronics — Discipline Electives, Even Semesters (4, 6, 8)

Sl. No	Course Code	Course Name	Credits
1	AR-509	Deep Learning for Computer Vision	4
2	AR-523	Robot Manipulators	3
3	AR-526	Multimodal Learning	3
4	EE-529	Embedded Systems	4
5	GE-303	Modern Powertrains: Electric, Hybrid & IC/Aeroengines	3
6	IC-121	Mechanics of Particles and Waves	3
7	IC-253	Programming and Data Structures	3
8	ME-215	Manufacturing Engineering – I	3
9	ME-312P	Design Lab – II	1
<i>Compulsory project components — counted as DE, not selectable</i>			
10	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
11	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Communication Technology

Table 632: General Engineering - Communication Technology — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	34*
	Discipline Electives	32*
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

*EE-261 (Electrical System Around Us) and EE-201 (Electromechanics) are shown here at 3 credits each, as per the course catalogue; the programme brochure lists them at 5 and 4 credits respectively. If these are offered at the higher credits, the Discipline Core total rises accordingly and the Discipline Elective requirement falls by the same amount, so that $DC + DE = 66$ credits either way.

Table 633: B. Tech General Engineering - Communication Technology — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	ME-100	Reverse Engineering	0-0-2-1	1
2	EE-261	Electrical System Around Us	3-0-0-3	3*
3	EE-231	Measurement and Instrumentation	2-0-2-3	3
4	EE-316	Communication Systems	3-0-2-4	4
5	EE-201	Electromechanics	2.5-0.5-0-3	3*
6	EE-203	Network Theory	3-0-0-3	3
7	IC-253	Programming and Data Structures	2-0-2-3	3
8	EE-260	Signals and Systems	2.5-0.5-0-3	3
9	CS-313	Computer Networks	3-0-2-4	4
10	EE-314	Digital Signal Processing	3-0-2-4	4
11	EE-205	Electromagnetics and Wave Propagation	3-0-0-3	3
Total				34*

Table 634: B. Tech General Engineering - Communication Technology — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Computing and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 635: B. Tech General Engineering - Communication Technology — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 636: B. Tech General Engineering - Communication Technology — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CS-313	Computer Networks	3-0-2-4	4
4	DC	EE-203	Network Theory	3-0-0-3	3
5	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
6	DC	EE-261	Electrical System Around Us	3-0-0-3	3*
Total					19*

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 637: B. Tech General Engineering - Communication Technology — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3*
2	DC	EE-205	Electromagnetics and Wave Propagation	3-0-0-3	3
3	DC	EE-316	Communication Systems	3-0-2-4	4
4	DC	IC-253	Programming and Data Structures	2-0-2-3	3
5	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					14*

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 638: B. Tech General Engineering - Communication Technology — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	EE-314	Digital Signal Processing	3-0-2-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 639: B. Tech General Engineering - Communication Technology — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 640: B. Tech General Engineering - Communication Technology — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 641: B. Tech General Engineering - Communication Technology — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Beyond the core courses above, students choose 32 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Odd-semester electives are offered in Semesters 3, 5 and 7; even-semester electives in Semesters 4, 6 and 8.

Table 642: General Engineering - Communication Technology — Discipline Electives, Odd Semesters (3, 5, 7)

Sl. No	Course Code	Course Name	Credits
1	EE-503	Advanced Communication Theory	3
2	EE-517	Wireless Communications & Networks	3
3	EE-531	Estimation and Detection Theory	3
4	EE-553	Fundamentals of Intelligent Communication System-I	4
5	EE-641	Advanced Wireless Technologies	3
6	GE-303	Modern Powertrains: Electric, Hybrid & IC/Aeroengines	3
7	VL-402	RFIC Design	3

Table 643: General Engineering - Communication Technology — Discipline Electives, Even Semesters (4, 6, 8)

Sl. No	Course Code	Course Name	Credits
1	CS-549	Performance Analysis of Computer Networks	3
2	DS-314	Probability Theory and Statistics	4
3	EE-211	Analog Circuit Design	4
4	EE-518	Information Theory	3
5	EE-530	Applied Optimization	3
6	EE-533	Access Network	4
7	EE-536	IoT Systems	3
8	EE-541	Tensors: Applications for Signal Processing & ML	3
9	EE-574	Biomedical Signal & Image Processing	3
10	EE-581	Fundamentals of Intelligent Communication System-II	3
11	EE-607	Optical Communication System	3
12	EE-608	Digital Image Processing	4
13	IC-241	Material Science for Engineers	3
<i>Compulsory project components — counted as DE, not selectable</i>			
14	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
15	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Open Specialisation

Table 644: General Engineering - Open Specialisation — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	35*
	Discipline Electives (merged into Free Electives)	—
Electives	Free Electives (including merged DE pool)	53*
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

*EE-261 (Electrical System Around Us) and EE-201 (Electromechanics) are shown here at 3 credits each, as per the course catalogue; the programme brochure lists them at 5 and 4 credits respectively. If these are offered at the higher credits, the Discipline Core total rises accordingly and the Discipline Elective requirement falls by the same amount, so that $DC + DE = 66$ credits either way.

Table 645: B. Tech General Engineering - Open Specialisation — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	ME-100	Reverse Engineering	0-0-2-1	1
2	EE-261	Electrical System Around Us	3-0-0-3	3*
3	EE-231	Measurement and Instrumentation	2-0-2-3	3
4	ME-213	Thermodynamics	3-1-0-4	4
5	EE-201	Electromechanics	2.5-0.5-0-3	3*
6	EE-211	Analog Circuit Design	2-0-2-3	3
7	ME-212	Product Manufacturing Technology	2-0-3-3	3
8	IC-241	Material Science for Engineers	3-0-0-3	3
9	IC-253	Programming and Data Structures	2-0-2-3	3
10	HS-307	Macroeconomics	3-0-0-3	3
11	HS-541	Technical Communication	1-0-0-1	1
12	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total				33*

Table 646: B. Tech General Engineering - Open Specialisation — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Computing and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 647: B. Tech General Engineering - Open Specialisation — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 648: B. Tech General Engineering - Open Specialisation — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-261	Electrical System Around Us	3-0-0-3	3*
4	DC	IC-241	Material Science for Engineers	3-0-0-3	3
5	DC	ME-212	Product Manufacturing Technology	2-0-3-3	3
6	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					18*

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 649: B. Tech General Engineering - Open Specialisation — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3*
2	DC	EE-211	Analog Circuit Design	2-0-2-3	3
3	DC	HS-307	Macroeconomics	3-0-0-3	3
4	DC	IC-253	Programming and Data Structures	2-0-2-3	3
5	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					13*

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 650: B. Tech General Engineering - Open Specialisation — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-231	Measurement and Instrumentation	2-0-2-3	3
2	DC	ME-213	Thermodynamics	3-1-0-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 651: B. Tech General Engineering - Open Specialisation — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	HS-541	Technical Communication	1-0-0-1	1
2	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					5

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 652: B. Tech General Engineering - Open Specialisation — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 653: B. Tech General Engineering - Open Specialisation — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Beyond the core courses above, students choose 53 credits of **Free Electives** — for this major the Discipline Elective and Free Elective pools are merged into a single pool — from across the institute, subject to the elective policy below.*

Elective Policy. There is no prescribed Discipline Elective list for this major. Students may choose electives from any department, provided that no more than 40% of their elective credits come from the Discipline Core list of any single named major. The Discipline Elective and Free Elective pools are merged into one pool of 53 credits of Free Electives, giving students maximum flexibility in course selection.

The following compulsory project components count toward the DE credit requirement; they are not selectable DE options.

Table 654: General Engineering - Open Specialisation — Compulsory project components credited as DE

Sl. No	Course Code	Course Name	Credits
1	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
2	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — General Engineering - Data Analytics for Management and Finance

Table 655: General Engineering - Data Analytics for Management and Finance — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	41
	Discipline Electives	25
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 656: B. Tech General Engineering - Data Analytics for Management and Finance — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	DS-314*	Probability Theory and Statistics	3-1-0-4	4
2	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
3	HS-202	Principles of Economics	3-0-0-3	3
4	HS-307	Macroeconomics – I	3-0-0-3	3
5	IC-253	Programming and Data Structures	2-0-2-3	3
6	MA-222	Applied Mathematical Programming	3-1-0-4	4
7	MA-546	Introduction to Quantitative Finance		4
8	MA-653P	Computational Financial Modelling Lab	0-0-2-1	1
9	MA-654	Financial Engineering		3
10	MA-545	Technical Analysis of Financial Market		3
11	MA-579	Computational Methods in Financial Time Series Analysis		3
12	MB-311	Management Fundamentals for Engineers	3-0-0-3	3
13	MB-312	Economic Decision-Making for Engineers	3-0-0-3	3
Total				41

* Recoded w.e.f. batch 2025: DS-313 → DS-314 (Probability Theory and Statistics, 3-1-0-4 unchanged) and DS-413 → DS-416 (Fundamentals of Machine Learning, now 3-0-2-4).

Table 657: B. Tech General Engineering - Data Analytics for Management and Finance — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Computing and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 658: B. Tech General Engineering - Data Analytics for Management and Finance — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 659: B. Tech General Engineering - Data Analytics for Management and Finance — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	DS-314*	Probability Theory and Statistics	3-1-0-4	4
4	DC	HS-202	Principles of Economics	3-0-0-3	3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 660: B. Tech General Engineering - Data Analytics for Management and Finance — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	HS-307	Macroeconomics – I	3-0-0-3	3
2	DC	IC-253	Programming and Data Structures	2-0-2-3	3
3	DC	MA-222	Applied Mathematical Programming	3-1-0-4	4
4	DC	MA-546	Introduction to Quantitative Finance		4
5	DC	MA-653P	Computational Financial Modelling Lab	0-0-2-1	1
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 661: B. Tech General Engineering - Data Analytics for Management and Finance — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	DS-416*	Fundamentals of Machine Learning	3-0-2-4	4
2	DC	MA-654	Financial Engineering		3
3	DC	MB-311	Management Fundamentals for Engineers	3-0-0-3	3
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 662: B. Tech General Engineering - Data Analytics for Management and Finance — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	MB-312	Economic Decision-Making for Engineers	3-0-0-3	3
2	DC	MA-545	Technical Analysis of Financial Market		3
3	DC	MA-579	Computational Methods in Financial Time Series Analysis		3
4	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 663: B. Tech General Engineering - Data Analytics for Management and Finance — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 664: B. Tech General Engineering - Data Analytics for Management and Finance — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 25 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Odd-semester electives are offered in Semesters 3, 5 and 7; even-semester electives in Semesters 4, 6 and 8.

Table 665: General Engineering - Data Analytics for Management and Finance — Discipline Electives, Odd Semesters (3, 5, 7)

Sl. No	Course Code	Course Name	Credits
1	DS-412	Matrix Computations for Data Science	4
2	HS-510	Essentials of Entrepreneurship	3
3	MA-211	Ordinary Differential Equations	4
4	MA-311	Mathematical Modelling	3

Table 666: General Engineering - Data Analytics for Management and Finance — Discipline Electives, Even Semesters (4, 6, 8)

Sl. No	Course Code	Course Name	Credits
1	DS-312	Computing Systems for Data Processing	4
2	DS-415	Optimization for Machine Learning	4
3	HS-481	International Economics	3
4	HS-504	Personal Finance and Portfolio Management	3
5	HS-550	Statistical Methods	4
6	MA-220	Partial Differential Equations	4
7	MA-221	Numerical Analysis	4
8	MA-524	Probability and Statistics	4
9	MA-552	Number Theory	3
10	MA-604	Introduction to Game Theory	3
11	ME-100	Reverse Engineering	1

Table 667: General Engineering - Data Analytics for Management and Finance — Discipline Electives, Any Semester (Business)

Sl. No	Course Code	Course Name	Credits
1	MB-410	Marketing for Engineers	3
2	MB-411	Human Resource Management and Organizational Behaviour for Engineers	3
3	MB-412	Consumer Behaviour in Technology Markets	3
4	MB-413	Technology Entrepreneurship and Venture Creation	3
5	MB-414	Accounting Practices for Technology Businesses	3
6	MB-415	Financial Management for Technology Businesses	3
7	MB-416	Industrial Operations and Supply Chain Systems	3
<i>Compulsory project components — counted as DE, not selectable</i>			
8	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
9	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Materials Science and Engineering

Table 668: Materials Science and Engineering — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	45
	Discipline Electives	21
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 669: B. Tech Materials Science and Engineering — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
2	MT-201	Physics of Solids		3
3	MT-202	Applied Quantum Mechanics		3
4	MT-203	Materials Synthesis & Characterization		4
5	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
6	ME-100	Reverse Engineering	0-0-2-1	1
7	MT-204	Thermodynamics and Kinetics of Materials	3	3
8	MT-205	Functional Properties of Materials	4	4
9	MT-206	Extraction and Materials Processing	4	4
10	ME-212	Product Manufacturing Technology	2-0-3-3	3
11	MT-301	Phase Transformations		3
12	MT-302	Transport Phenomena		3
13	MT-303	Computational Materials Science		4
14	MT-304	Mechanical Behaviour of Materials		4
Total				45

Table 670: B. Tech Materials Science and Engineering — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 671: B. Tech Materials Science and Engineering — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-241	Material Science for Engineers	3-0-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 672: B. Tech Materials Science and Engineering — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	ME-212	Product Manufacturing Technology	2-0-3-3	3
4	DC	MT-201	Physics of Solids		3
5	DC	MT-202	Applied Quantum Mechanics		3
6	DC	MT-203	Materials Synthesis & Characterization		4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 673: B. Tech Materials Science and Engineering — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
2	DC	ME-100	Reverse Engineering	0-0-2-1	1
3	DC	MT-204	Thermodynamics and Kinetics of Materials	3	3
4	DC	MT-205	Functional Properties of Materials	4	4
5	DC	MT-206	Extraction and Materials Processing	4	4
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 674: B. Tech Materials Science and Engineering — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
2	DC	MT-301	Phase Transformations		3
3	DC	MT-302	Transport Phenomena		3
4	DC	MT-303	Computational Materials Science		4
5	DC	MT-304	Mechanical Behaviour of Materials		4
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 675: B. Tech Materials Science and Engineering — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 676: B. Tech Materials Science and Engineering — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	MS-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 677: B. Tech Materials Science and Engineering — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	MS-010	Internship		2
2	MTP	MS-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 21 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 678: Materials Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	ME-303	Heat Transfer	3
2	ME-308	Manufacturing Engineering	3
3	ME-353	Electronic Materials and Their Applications	3
4	ME-515	Carbon Materials and Technology	3
5	MT-501	Energy Conversion and Storage Technologies	3
6	MT-502	Recycling and Circular Economy	3
7	MT-503	Semiconductor Materials and Devices	3
8	MT-504	Powder metallurgical processing of Materials	3
9	MT-505	Thin Film Technology	3
10	MT-506	Biomaterials	3
11	MT-507	Modelling and Simulations in Material Science	3

Materials Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
12	MT-511	Sensor Materials and Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
13	MS-498P	Major Technical Project - I (compulsory; counted as DE)	4
14	MS-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Science — Chemical Sciences

Table 679: Chemical Sciences — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	31
	IC Basket	9
	IKS + HSS	12
Discipline Courses	Discipline Core	59
	Discipline Electives	23
Electives	Free Electives	15
	Research Lit Presentation (1+1)	2
	RP (equivalent as DE)	12
Total		163

Table 680: B. S. Chemical Sciences — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CY-201P	Physical Chemistry Laboratory	0-0-4-2	2
2	CY-301	Principles and Theories of Physical Chemistry	3-0-0-3	3
3	CY-302	Principles of Organic Chemistry	3-0-0-3	3
4	CY-303	Fundamentals of Inorganic Chemistry	3-0-0-3	3
5	CY-202P	Organic Chemistry Laboratory	0-0-4-2	2
6	CY-203P	Inorganic Chemistry Laboratory	0-0-4-2	2
7	CY-304	Fundamental Analytical Chemistry	3-0-0-3	3
8	CY-401	Introduction to Quantum Chemistry and Molecular Spectroscopy	3-0-0-3	3
9	CY-512	Advanced Quantum Chemistry	3-0-0-3	3
10	CY-512P	Physical Chemistry Laboratory		3
11	CY-531	Organic Reactions and Mechanisms		3
12	CY-533	Chemistry of Main Group Elements		3
13	CY-533P	Inorganic Chemistry Laboratory		3
14	CY-511	Group Theory and Spectroscopy	3-0-0-3	3
15	CY-531P	Organic Chemistry Laboratory		3
16	CY-532	Photochemistry and Pericyclic Reactions		3
17	CY-534	Chemistry of Transition Elements		3
18	CY-514	Chemical and Statistical Thermodynamics	3-0-0-3	3
19	CY-535	Introduction to Organometallic Chemistry		3
20	CY-504	Heterocyclic Chemistry	2-0-0-2	2
21	CY-513	Chemical Kinetics and Reaction Dynamics	3-0-0-3	3
Total				59

Table 681: B. S. Chemical Sciences — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 682: B. S. Chemical Sciences — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 683: B. S. Chemical Sciences — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC Basket	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
2	DC	CY-201P	Physical Chemistry Laboratory	0-0-4-2	2
3	DC	CY-301	Principles and Theories of Physical Chemistry	3-0-0-3	3
4	DC	CY-302	Principles of Organic Chemistry	3-0-0-3	3
5	DC	CY-303	Fundamentals of Inorganic Chemistry	3-0-0-3	3
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 684: B. S. Chemical Sciences — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-202P	Organic Chemistry Laboratory	0-0-4-2	2
2	DC	CY-203P	Inorganic Chemistry Laboratory	0-0-4-2	2
3	DC	CY-304	Fundamental Analytical Chemistry	3-0-0-3	3
4	DC	CY-401	Introduction to Quantum Chemistry and Molecular Spectroscopy	3-0-0-3	3
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 685: B. S. Chemical Sciences — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-512	Advanced Quantum Chemistry	3-0-0-3	3
2	DC	CY-512P	Physical Chemistry Laboratory		3
3	DC	CY-531	Organic Reactions and Mechanisms		3
4	DC	CY-533	Chemistry of Main Group Elements		3
5	DC	CY-533P	Inorganic Chemistry Laboratory		3
6	RP	DP-403P	Research Literature Presentation 1		1
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 686: B. S. Chemical Sciences — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-511	Group Theory and Spectroscopy	3-0-0-3	3
2	DC	CY-531P	Organic Chemistry Laboratory		3
3	DC	CY-532	Photochemistry and Pericyclic Reactions		3
4	DC	CY-534	Chemistry of Transition Elements		3
5	RP	DP-404P	Research Literature Presentation 2		1
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 687: B. S. Chemical Sciences — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-514	Chemical and Statistical Thermodynamics	3-0-0-3	3
2	DC	CY-535	Introduction to Organometallic Chemistry		3
3	RP	DP-551P	Undergraduate Research Project 1		6
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 688: B. S. Chemical Sciences — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-504	Heterocyclic Chemistry	2-0-0-2	2
2	DC	CY-513	Chemical Kinetics and Reaction Dynamics	3-0-0-3	3
3	RP	DP-552P	Undergraduate Research Project 2		6
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 23 credits of **Discipline Electives** and 15 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 689: Chemical Sciences — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BY-505	Nanobiotechnology	3
2	BY-514	Analytical Biotechniques	3
3	CE-519	Chemistry of Natural Waters	3
4	CH-101	General Chemistry	4
5	CY-001	Preparatory Chemistry - 1	4
6	CY-002	Preparatory Chemistry - 2	4
7	CY-100	Preparatory Chemistry 1	3
8	CY-101	Chemistry	5
9	CY-200	Foundations and applications of Chemistry	3
10	CY-241	Nano-scale Science and Technology	3
11	CY-242	Introduction to Molecular Spectroscopy	3
12	CY-243	Engineering Chemistry	3
13	CY-247	Introduction to Molecular Thermodynamics	3
14	CY-248	Molecular Physical Chemistry for Engineers	3
15	CY-342	Understanding Small Systems	3
16	CY-344	Food Chemistry: Processing, Preservation and Storage	3
17	CY-402	Applied Materials Chemistry	3
18	CY-403	Numerical Methods and Data Analysis in Chemistry	3
19	CY-404	Fundamentals of Soft Matter Science and Applications	3
20	CY-405	Introduction to structural chemistry	3
21	CY-501	Organic Reaction and Mechanism	4
22	CY-501P	Organic Chemistry Laboratory	4
23	CY-502	Photochemistry and pericyclic reactions	4
24	CY-506	Chemistry of Transition Elements	4
25	CY-506P	Inorganic Chemistry Lab	4
26	CY-507	Chemistry of Main Group Elements	4
27	CY-508	Organometallic Chemistry	2
28	CY-511P	Physical Chemistry Laboratory	4
29	CY-515	Advanced Inorganic Spectroscopy	3
30	CY-521	Mathematics for Chemists	3
31	CY-522	Computational Chemistry	3
32	CY-523	Colloids and Interface Science and Technology	3
33	CY-524	Basic and Applied Electrochemistry	3
34	CY-541	Fundamentals of Organic Chemistry	3
35	CY-545	Reagents in organic synthesis	3
36	CY-547	Chemical Crystallography	3
37	CY-550	Bioinspired Materials	3
38	CY-552	Hydrogen Generation and Storage	3
39	CY-553	Organic Inorganic Spectroscopy	3
40	CY-554	Science and Technology of Nanomaterials	3

Chemical Sciences — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
41	CY-555	Introduction to Polymer Science & Technology	3
42	CY-556	Organic Spectroscopy	3
43	CY-560	Introduction to Supramolecular Chemistry	3
44	CY-561	Natural Products Synthesis	3
45	CY-562	Stereochemistry of organic compounds	3
46	CY-562P	Design Practicum - II	1
47	CY-598P	Post Graduate Project-1 (CH)	4
48	CY-599P	Post Graduate Project-2	9
49	CY-641	Polymer Synthesis	3
50	CY-642	Molecular- and Bio-electronics	3
51	CY-643	Advanced Analytical Techniques	3
52	CY-644	Bioinorganic chemistry	3
53	CY-645	Reagents in organic synthesis	3
54	CY-646	Advanced NMR Spectroscopy - A Problem Based Approach	3
55	CY-660	Photocatalysis: Concepts and Applications	3
56	CY-670	Fluorescence spectroscopy, microscopy and applications	3
57	CY-701	Advances Physical Methods in Chemistry Theory and	3
58	CY-702	Advanced Inorganic Chemistry: Theory and Applications	3
59	CY-703	Advanced Organic Chemistry	3
60	CY-704	Introduction to Theoretical Chemistry	3
61	CY-705	Modern Methods in Organic Synthesis	3
62	CY-746	Self Assembly of Surfactants and Polymers in Solution	3
63	EN-503	Energy Storage Technologies	3
64	ME-353	Electronic Materials and Their Applications	3
65	ME-509	Nanomanufacturing	3
66	ME-515	Carbon Materials and Technology	3
67	ME-516	Polymer Technology for Engineers	3
68	MT-201	Physics of Solids	3
69	MT-501	Energy Conversion and Storage Technologies	3
70	MT-502	Recycling and Circular Economy	3
71	MT-503	Semiconductor Materials and Devices	3
72	MT-505	Thin Film Technology	3
73	MT-506	Biomaterials	3
74	MT-509	Hydrogen Energy	3
75	MT-510	Colloids and Interfaces	3
76	PH-504	Organic Optoelectronics	3
<i>Compulsory project components — counted as DE, not selectable</i>			
77	DP-551P	Undergraduate Research Project 1 (compulsory; counted as DE)	6

Chemical Sciences — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
78	DP-552P	Undergraduate Research Project 2 (compulsory; counted as DE)	6

Bachelor of Technology — Microelectronics and VLSI

Table 690: Microelectronics and VLSI — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	54
	Discipline Electives	12
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 691: B. Tech Microelectronics and VLSI — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EE-203	Network Theory	3-0-0-3	3
2	EE-212	Digital System Design	3-0-2-4	4
3	EE-260	Signals and Systems	2.5-0.5-0-3	3
4	EE-302	Control Systems	3-0-2-4	4
5	VL-201	Semiconductor Device for ICs	3-0-0-3	3
6	EE-202	Electromagnetic Theory and Transmission Lines	3-0-0-3	3
7	EE-211	Analog Circuit Design	2-0-2-3	4
8	VL-300	Reverse Engineering - E-waste management	0-0-2-1	1
9	VL-311	CMOS Processing and Practicum	3-0-2-4	4
10	VL-312	Electronic System Packaging	3-0-0-3	3
11	VL-326	Computer Organization	3-0-2-4	4
12	VL-401	RTL Design and Verification	2-0-2-3	3
13	VL-402	RF IC Design	3-0-0-3	3
14	VL-403	CMOS Digital IC Design	3-0-2-4	4
15	VL-404	CMOS Analog IC Design	3-0-2-4	4
16	VL-405	Design for Testability	3-0-2-4	4
Total				54

Table 692: B. Tech Microelectronics and VLSI — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket (choose 1)	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6		IC-136	Understanding Biotechnology and its Applications	3-0-0-3	
7		IC-230	Environmental Science	3-0-0-3	
8	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
9		IC-182	History of Science and Technology	3	
10	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
11		HS-112	Japanese Language I	3-0-0-3	
12		HS-342	German I	3-0-0-3	
13		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 693: B. Tech Microelectronics and VLSI — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket (choose 1)	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9		IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	
10		IC-241	Material Science for Engineers	3-0-0-3	
11		IC-253	Data Structures and Algorithms	2-0-2-3	
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 694: B. Tech Microelectronics and VLSI — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	EE-203	Network Theory	3-0-0-3	3
4	DC	EE-212	Digital System Design	3-0-2-4	4
5	DC	EE-260	Signals and Systems	2.5-0.5-0-3	3
6	DC	EE-302	Control Systems	3-0-2-4	4
7	DC	VL-201	Semiconductor Device for ICs	3-0-0-3	3
Total					23

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 695: B. Tech Microelectronics and VLSI — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-202	Electromagnetic Theory and Transmission Lines	3-0-0-3	3
2	DC	EE-211	Analog Circuit Design	2-0-2-3	4
3	DC	VL-300	Reverse Engineering - E-waste management	0-0-2-1	1
4	DC	VL-311	CMOS Processing and Practicum	3-0-2-4	4
5	DC	VL-312	Electronic System Packaging	3-0-0-3	3
6	DC	VL-326	Computer Organization	3-0-2-4	4
Total					19

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 696: B. Tech Microelectronics and VLSI — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	VL-401	RTL Design and Verification	2-0-2-3	3
2	DC	VL-402	RF IC Design	3-0-0-3	3
3	DC	VL-403	CMOS Digital IC Design	3-0-2-4	4
4	DC	VL-404	CMOS Analog IC Design	3-0-2-4	4
5	DC	VL-405	Design for Testability	3-0-2-4	4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 697: B. Tech Microelectronics and VLSI — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 698: B. Tech Microelectronics and VLSI — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	VL-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 699: B. Tech Microelectronics and VLSI — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	VL-010	Internship		2
2	MTP	VL-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 12 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 700: Microelectronics and VLSI — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	EE-210P	Digital Systems Design Lab	1
2	EE-507	Transmission Lines and Basic Microwave Engineering	4
3	EE-516	Biomedical Systems	4
4	EE-520	Microelectronic Devices and Modelling	3
5	EE-523	Digital VLSI Architecture Design	4
6	EE-529	Embedded Systems	4
7	EE-554	Low Power VLSI Design	3
8	EE-574	Biomedical Signal and Image Analysis	3
9	EE-585	Theory, Modelling and Characterization of Photo-voltaic Devices	3
10	EE-611	VLSI Technology	3

Microelectronics and VLSI — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
11	EE-615	Nanoelectronics and Micronanofabrication	3
12	EE-619	Mixed Signal VLSI Design	4
13	EE-621	Radiating systems	4
14	EE-622	Microwave Integrated Circuits	3
<i>Compulsory project components — counted as DE, not selectable</i>			
15	VL-498P	Major Technical Project - I (compulsory; counted as DE)	4
16	VL-499P	Major Technical Project - II (compulsory; counted as DE)	4

Bachelor of Technology — Chemical Engineering with Data Analytics

Table 701: Chemical Engineering with Data Analytics — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	56
	Discipline Electives	15
Electives	Free Electives	21
	MTP (4+4, equivalent as DE)	8
Total		160

A dash (—) in the Course Code column means that the source does not yet notify an official catalogue code.

Table 702: B. Tech Chemical Engineering with Data Analytics — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CED-101	Introduction to Chemical Engineering		3
2	—	Chemical Engineering Thermodynamics		3
3	—	Fluid Mechanics	2-1-2-4	4
4	—	Numerical Methods	3-1-0-4	4
5	—	Chemical Reaction Engineering	2-1-2-4	4
6	—	Heat Transfer	2-1-2-4	4
7	—	Mass Transfer I	2-1-2-4	4
8	—	Mass Transfer II		3
9	—	Data Science for Chemical Systems		3
10	—	Process Control and Instrumentation	2-1-2-4	4
11	—	Chemical Reaction Engineering II		3
12	—	Transport Phenomenon		3
13	—	Digital Process Equipment Design		3
14	—	Chemical Process Technology		3
15	—	Process Engineering and Economics		3
16	—	Chemical Engineering Lab (Remaining Experiments)		2
17	—	Climate Technology and Sustainable Process Engineering		3
Total				56

Table 703: B. Tech Chemical Engineering with Data Analytics — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
6		IC-182	History of Science and Technology	3	
7	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
8		HS-112	Japanese Language I	3-0-0-3	
9		HS-342	German I	3-0-0-3	
10		IK-101	Sanskrit language-Level 1		
11	DC	CED-101	Introduction to Chemical Engineering		3
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 704: B. Tech Chemical Engineering with Data Analytics — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE and Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket-I	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 705: B. Tech Chemical Engineering with Data Analytics — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	—	Chemical Engineering Thermodynamics		3
4	DC	—	Fluid Mechanics	2-1-2-4	4
5	DC	—	Numerical Methods	3-1-0-4	4
6	IC Basket-II	IC-XXX	Material and Energy Balance		3
Total					20

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 706: B. Tech Chemical Engineering with Data Analytics — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	—	Chemical Reaction Engineering	2-1-2-4	4
2	DC	—	Heat Transfer	2-1-2-4	4
3	DC	—	Mass Transfer I	2-1-2-4	4
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 707: B. Tech Chemical Engineering with Data Analytics — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	—	Mass Transfer II		3
2	DC	—	Data Science for Chemical Systems		3
3	DC	—	Process Control and Instrumentation	2-1-2-4	4
4	DC	—	Chemical Reaction Engineering II		3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 708: B. Tech Chemical Engineering with Data Analytics — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	—	Transport Phenomenon		3
2	DC	—	Digital Process Equipment Design		3
3	DC	—	Chemical Process Technology		3
4	DC	—	Process Engineering and Economics		3
5	DC	—	Chemical Engineering Lab (Remaining Experiments)		2
6	DC	—	Climate Technology and Sustainable Process Engineering		3
Total					17

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 709: B. Tech Chemical Engineering with Data Analytics — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	—	Undergraduate Research Project I	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 710: B. Tech Chemical Engineering with Data Analytics — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	—	Undergraduate Research Project II	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 15 credits of **Discipline Electives** and 21 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Bachelor of Technology — Quantum Science and Engineering

Table 711: Quantum Science and Engineering — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	40
	Discipline Electives	26
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

A dash (—) in the Course Code column means that the source does not yet notify an official catalogue code.

Table 712: B. Tech Quantum Science and Engineering — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	EP-301	Engineering Mathematics -2	3-1-0-4	4
2	EP-321	Foundation of Electrodynamics	3-0-0-3	3
3	QS-302 / PH-301	Foundation of Quantum Technology / Quantum Mechanics and Application	3-0-0-3	3
4	PH-302	Statistical Thermodynamics	3-0-0-3	3
5	PH-501	Solid State Physics	3-0-0-3	3
6	QS-407	Introduction to Quantum Sensing	3-0-0-3	3
7	—	Reverse Engineering	0-0-2-1	1
8	CS-212	Design of Algorithm	3-0-2-4	4
9	CS-315 [†]	Artificial Intelligence	3-0-2-4	4
10	QS-304P	Basic Laboratory Course on Quantum Technologies	1-0-4-3	3
11	PH-502	Photonics	3-0-0-3	3
12	QS-501P	Experiments in Quantum Optics	1-0-4-3	3
13	QS-509	Engineering Foundation of Quantum Technologies	3-0-0-3	3
Total				40

[†]CS-305 Artificial Intelligence (3-0-0-3) has been recoded as CS-315 (3-0-2-4) w.e.f. batch 2025; the extra credit is adjusted against Discipline Electives, keeping the total at 160.

Table 713: B. Tech Quantum Science and Engineering — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 714: B. Tech Quantum Science and Engineering — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE and Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 715: B. Tech Quantum Science and Engineering — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-272	Machine Learning	2-0-2-3	3
2	DC	EP-301	Engineering Mathematics -2	3-1-0-4	4
3	DC	EP-321	Foundation of Electrodynamics	3-0-0-3	3
4	DC	QS-302 / PH-301	Foundation of Quantum Technology / Quantum Mechanics and Application	3-0-0-3	3
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 716: B. Tech Quantum Science and Engineering — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-201P	Design Practicum	0-0-6-3	3
2	DC	PH-302	Statistical Thermodynamics	3-0-0-3	3
3	DC	PH-501	Solid State Physics	3-0-0-3	3
4	DC	QS-407	Introduction to Quantum Sensing	3-0-0-3	3
5	DC	—	Reverse Engineering	0-0-2-1	1
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 717: B. Tech Quantum Science and Engineering — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CS-212	Design of Algorithm	3-0-2-4	4
2	DC	CS-315 [†]	Artificial Intelligence	3-0-2-4	4
3	DC	QS-304P	Basic Laboratory Course on Quantum Technologies	1-0-4-3	3
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 718: B. Tech Quantum Science and Engineering — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	PH-502	Photonics	3-0-0-3	3
2	DC	QS-501P	Experiments in Quantum Optics	1-0-4-3	3
3	DC	QS-509	Engineering Foundation of Quantum Technologies	3-0-0-3	3
4	ISTP	DP-301P	ISTP or Equivalent	0-0-6-4	4
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 719: B. Tech Quantum Science and Engineering — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	DP-401P	MTP-1	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 720: B. Tech Quantum Science and Engineering — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	DP-402P	MTP-2	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 26 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 721: Quantum Science and Engineering — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	CS-520	Introduction to Quantum Computing	3
2	CS-671	Deep Learning and Applications	3
3	DS-416	Fundamentals of Machine Learning (CS-669)	3
4	EP-403	Physics of Atoms and Molecules	3
5	ME-353	Electronic Materials and their Applications	3
6	PH-559	Introduction to Quantum Optics	3
7	PH-579	Advanced Quantum Computation and Information	3
8	PH-601	Mesoscopic Physics and Quantum Transport	3

Quantum Science and Engineering — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
9	PH-603	Advanced Condensed Matter Physics	3
10	PH-605	Superconductivity	3
11	PH-613	Special Topics in Quantum Mechanics	3
12	QS-301	Survey of Quantum Technology	3
13	QS-406	Introduction to Quantum Communication	3
14	QS-510	Solid State Physics for Quantum Technologies	3
<i>Compulsory project components — counted as DE, not selectable</i>			
15	DP-401P	MTP-1 (compulsory; counted as DE)	4
16	DP-402P	MTP-2 (compulsory; counted as DE)	4

Bachelor of Technology — Agricultural Engineering with Data Analytics

Table 722: Agricultural Engineering with Data Analytics — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	36
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	49
	Discipline Electives	17
Electives	Free Electives	25
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

A dash (—) in the Course Code column means that the source does not yet notify an official catalogue code.

Table 723: B. Tech Agricultural Engineering with Data Analytics — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CE-310	Strength of Materials and Structures	3-0-0-3	3
2	CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
3	—	Crop Production Technology		3
4	—	Soil Science and Mechanics		4
5	—	Farm Machinery and Equipment		3
6	CE-201	Surveying: Traditional and Digital	2-0-4-4	4
7	CE-251	Hydraulics Engineering	3-0-0-3	3
8	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
9	EE-302	Control Systems	3-0-2-4	4
10	—	Theory and Design of Machine Elements		4
11	CE-303	Water Resources Engineering	3-0-0-3	3
12	CE-353P	Civil Engineering Drawing	0-0-2-1	1
13	CE-451	Irrigation Engineering and Hydraulic Structures	3-0-0-3	3
14	CS-541P	IoT Systems and Clouds	1-0-3-3	3
15	—	Food Science and Technology and Lab		4
16	—	Self-Sufficient Farming (Natural Farming)		2
17	—	Post-Harvest Engineering		3
Total				49

Table 724: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 725: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE and Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Rigid Body Dynamics	1.5-1.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 726: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CE-310	Strength of Materials and Structures	3-0-0-3	3
4	DC	CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
5	DC	—	Crop Production Technology		3
6	DC	—	Soil Science and Mechanics		4
7	DC	—	Farm Machinery and Equipment		3
Total					20

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 727: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	CE-201	Surveying: Traditional and Digital	2-0-4-4	4
3	DC	CE-251	Hydraulics Engineering	3-0-0-3	3
4	DC	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
5	DC	EE-302	Control Systems	3-0-2-4	4
6	DC	—	Theory and Design of Machine Elements		4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 728: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-303	Water Resources Engineering	3-0-0-3	3
2	DC	CE-353P	Civil Engineering Drawing	0-0-2-1	1
3	DC	CE-451	Irrigation Engineering and Hydraulic Structures	3-0-0-3	3
4	DC	CS-541P	IoT Systems and Clouds	1-0-3-3	3
5	DC	—	Food Science and Technology and Lab		4
6	DC	—	Self-Sufficient Farming (Natural Farming)		2
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 729: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	—	Post-Harvest Engineering		3
2	ISTP	DP-301P	ISTP or Equivalent	0-0-6-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 730: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	DP-401P	MTP-1	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 731: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	DP-402P	MTP-2	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 17 credits of **Discipline Electives** and 25 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

The following compulsory project components count toward the DE credit requirement; they are not selectable DE options.

Table 732: Agricultural Engineering with Data Analytics — Compulsory project components credited as DE

Sl. No	Course Code	Course Name	Credits
1	DP-401P	MTP-1 (compulsory; counted as DE)	4
2	DP-402P	MTP-2 (compulsory; counted as DE)	4